Reference-Point Fluency and Memorability*

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October 22, 2026

Abstract

Are all reference points equally influential? Round-number purchase prices are associated with a larger disposition effect, the tendency to sell winning investments while holding losing ones. The gain-loss gap in selling is 6.20 percentage points for regular-price positions and 8.60 percentage points for round-price positions, an amplification of 2.40 percentage points (about 39 percent of the regular-price baseline); in account-level regressions the round-versus-regular amplification remains about 1.2 to 2.0 percentage points, or roughly one-third of the baseline disposition effect on regular-price holdings. A stock-split design that rescales purchase bases after the trade addresses purchase-time selection: the amplification persists when the adjusted basis remains round and vanishes when the split breaks roundness. One interpretation is that a round purchase price is a more fluent and memorable reference point. Because it is easier to encode and to retrieve, the distinction between a gain and a loss is more salient when the investor decides whether to sell. The amplification is not explained by taxes, rank-based selling, rebalancing, portfolio concentration, or liquidity, so realization depends on the numerical form of the purchase price and not only on whether a position is at a gain or a loss. A shorter market-level exercise finds that round consensus earnings forecasts are associated with more announcement-window trading and a more complete reaction to positive surprises.

JEL Classification: G11, G12, G14, G41

Keywords: disposition effect; reference points; processing fluency; memorability; round numbers; investor attention; earnings announcements

*I especially thank my advisor, David Hirshleifer, for his invaluable advice and guidance. I also especially thank Cary Frydman for his suggestions and support throughout this research.

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1 Introduction

Prospect theory and the realization preference rely fundamentally on the comparison between a current outcome and a reference point (Kahneman and Tversky 1979). Perhaps the most robust empirical manifestation of this reference dependence in financial markets is the disposition effect, the tendency of investors to sell winning positions more readily than losing ones.[1] However, standard models of the disposition effect implicitly assume that the reference point (typically the purchase price) is perfectly encoded and effortlessly retrieved by the investor. In reality, human memory is constrained and noisy. A reference point is more likely to remain cognitively accessible when it was processed fluently at encoding and remains memorable at retrieval. Empirically, this implies that not all reference points are equally influential: stickier ones should amplify gain-loss trading more than opaque ones.

How does the cognitive accessibility of a reference point shape trading behavior? Prior literature has largely focused on external attention shocks, such as news events or extreme returns, to explain time-variation in the disposition effect. These studies suggest that external stimuli trigger the retrieval of past reference points thereby shaping trading decisions (Frazzini 2006; Frydman and Wang 2020; Andries, Bonelli and Sraer 2024).[2] Yet, this perspective overlooks an internal channel of attention: the ease with which a reference point was processed and encoded in the first place. Psychology refers to this ease of processing as fluency (Alter and Oppenheimer 2006; Green and Jame 2013). Fluency makes a reference point eas-

[1] This trading phenomenon has been documented among US retail investors (Andries, Bonelli and Sraer 2024), foreign retail investors (Grinblatt and Keloharju 2001; Frydman and Wang 2020), institutional investors (Shapira and Venezia 2001), homeowners (Genesove and Mayer 2001), corporate executives (Heath, Huddart and Lang 1999), and in experimental settings (Frydman, Hartzmark and Solomon 2018; Weber and Camerer 1998). Because of the high volume of potential citations, I refer readers to Kaustia (2010a)'s excellent overview of the literature.

[2] For example, Frydman and Wang (2020) finds that manipulating the presentation of a stock purchase price on a trading platform can affect the willingness of investors to sell winners versus losers in individual stocks. Birru (2015) finds that the disposition effect is absent on days when a stock undergoes a split. His interpretation is that investors fail to properly update their reference point, and thus become "confused" about a stock's gain or loss status.

ier to encode, which in turn makes that reference easier to retrieve when investors later evaluate outcomes relative to that reference. Consistent with this idea, recent advances in economics and psychology suggest that individuals allocate limited cognitive resources efficiently, encoding simple, salient information with greater precision (Frydman and Jin 2022; Charles, Frydman and Kilic 2024; Wachter and Kahana 2024; Frydman and Jin 2025). Fluent reference points are a natural instance of such simple stimuli.

In this paper, I study how the cognitive accessibility of a reference point shapes trading, mainly in account-level trading decisions and, more briefly, in market reactions around earnings announcements.

At the micro level, I propose that round number purchase prices (prices ending in zero and five) serve as a natural proxy for fluent reference points. Because round numbers are cognitively compressed and easier to retain than non-round numbers, they serve as sharper mental anchors. I hypothesize that when a stock is purchased at a round price, the reference point is more precisely defined, making the distinction between *gains* and *losses* more salient to the investor. Consequently, increasing the cognitive accessibility (roundness) of a stock purchase price will amplify the disposition effect.

To illustrate, consider an investor who bought one stock at $200.00 and another at $203.25. If round purchase prices are stickier anchors, the gain-loss gap in selling should be larger for the $200.00 position than for the $203.25 position.

Beyond this account-level setting, I ask, more briefly, whether a similar pattern appears for a shared market reference point. During earnings announcements, the analyst consensus earnings per share (EPS) forecast serves as the market's primary, shared reference point. While near-identical numerical values, such as $1.00 and $1.03, represent virtually the same financial target, they possess vastly different cognitive properties. A round consensus forecast may be easier to encode and retrieve. If so, firms with round targets might show a stronger trading response

to earnings surprises and a more complete incorporation of positive news at announcement, with less post-earnings drift. Because round consensus forecasts are not randomly assigned, I treat this evidence as secondary to the account-level results.

Estimating this channel in the field raises three identification challenges.

First, cognitive accessibility is hard to measure without confounding content. This body of literature often explores heterogeneities in scripts (Brewer and Treyens 1981; Graf and Mandler 1984; Hunt 1977; Heath and Heath 2007), stories (Graeber, Roth and Zimmermann 2024), emotions (Kensinger and Schacter 2008), mental imagery (Shepard and Cooper 1986; Standing 1973; Shepard 1967), and company name fluency (Green and Jame 2013), but sometimes faces an identification challenge: isolating fluency and memorability per se from intrinsic content differences. For example, the very features that make the content in certain structures easier to process and remember, such as salience, emotional intensity, concreteness, narrative structure, and diagnosticity, often unavoidably coincide with greater informational richness or perceived importance. As a result, comparisons across formats might risk conflating a cognitive accessibility channel with content-driven effects, making it difficult to attribute observed impacts to fluency features. In this paper, I propose that the round number is a useful proxy for cognitive accessibility. Humans are cognitively attuned to round numbers because they offer simple and more discretized representations of information, making them fluent to process and thus easier to remember, recall, and mentally manipulate. Owing to their less cognitively demanding nature, they are naturally convenient points of comparison (Schindler and Kirby 1997). For example, round numbers are frequently used as benchmarks for setting goals or measuring progress (Ranyard, Hinkley, Williamson and McHugh 2006), such as choosing loan amounts (Pursiainen 2024) and scheduling auto loan payments (Argyle, Nadauld and Palmer 2020). Unlike other proxies for memorability that carry different inherent content, round and non-round stock

prices are close to functionally equivalent economic objects: a purchase price of $200.00 and one of $203.25 convey nearly the same valuation. Using round numbers therefore addresses the content confound, because the two kinds of prices differ mainly in numerical form and not in the information they carry.

Second, in many empirical contexts, the cognitive accessibility of a stock is correlated with its external attention shock. Investors frequently respond to media coverage, abnormal trading volume, and extreme stock returns; these cues heighten external salience and correlate with inherent fluency and memorability, making it difficult to disentangle a fluency-induced trading channel as opposed to trading prompted by externally induced salience. To address this concern, I compare sell decisions across investors for the same stock on the same date, thereby holding stock-level external attention constant. This fixed effect approach exploits cross-investor variation in purchase-price fluency (e.g., round versus regular prices) that is orthogonal to same-day stock-level shocks. For example, consider two investors with unrealized gains in AAPL on the same day: one bought at $203.25 and the other at $200.00. I hypothesize that the $200 buyer is more likely to realize the gain, not because AAPL is more salient that day, but because the round purchase price provides a more fluent and memorable reference point that alleviates the cognitive load of gain-loss perception.

Third, a further identification challenge is purchase-time selection at the stock level. Whether a trade executes at a round price may be tied to the microstructure at the moment of purchase, such as order type, time of day, spread and depth, and liquidity tightness. These factors are time-varying and stock-specific, so they are not easily absorbed by investor fixed effects or by stock-by-day fixed effects defined at the sell date. As a result, a higher propensity to realize gains for round price stocks could reflect purchase time environments rather than memory encoded at purchase. I address this concern with a stock-split empirical design. A stock split rescales the original purchase basis after the transaction and mechanically deter-

mines the roundness of the split-adjusted purchase price for the same investor and stock. Because this change is ex post and fixed by the split ratio, the adjusted roundness of a given stock is orthogonal to the original microstructure environment in which the transaction was made. This setting, therefore, provides plausibly exogenous variation in whether a split-adjusted purchase price is round, allowing the observed differences in the subsequent disposition effect to be attributed to the fluency and memorability channel.

The main results of this paper are as follows. At the account level, round purchase prices amplify the disposition effect. Among accounts that ever purchased at a round price, the univariate gain-loss gap in selling is 6.20 percentage points for regular-price positions and 8.60 percentage points for round-price positions, an amplification of 2.40 percentage points (about 39% of the regular-price baseline). In investor-stock-day regressions, the $RoundBuyPrice \times Gain$ interaction remains positive and economically large , ranging from about 1.2 to 2.0 percentage points, or roughly one-third of the baseline disposition effect on regular-price holdings. The contribution of this evidence is to identify a source of the disposition effect. A round price differs from a regular price in numerical form and not in economic content, so the comparison isolates the accessibility of the purchase price. Round-price positions are about 4.6 percent of the sample and account for a modest part of the average disposition effect (see the discussion of Table 3), but the estimates show that realization depends on the numerical form of the purchase price and not only on whether a position is at a gain or a loss.

A stock-split design sharpens identification against purchase-time selection. When a split leaves the adjusted purchase basis round (Always), $Gain \times RoundBuyPriceAlways$ remains about 0.28 to 0.29. When the split breaks roundness (Breaks), the corresponding interaction is small and statistically indistinguishable from zero; a formal test rejects equality of the two interactions. The amplification is not explained by taxes, rank-based selling, rebalancing, concentration, or liquidity.

At the market level, round consensus EPS forecasts are associated with more announcement trading when surprises are large, and positive surprises are incorporated more thoroughly at announcement with less subsequent drift.

Taken together, the account- and market-level evidence points to the same conclusion. Reference dependence in financial markets may depend not only on what the reference point is, but on how fluently it is encoded and how easily it is retrieved.

Three strands of earlier work on salience and round numbers are closest to this paper, and each suggests a different reason why round purchase prices might matter. Frydman and Wang (2020) show that changing how prominently a trading platform displays the purchase price changes the gap in willingness to sell winners and losers, so the salience of the basis matters. Pope and Simonsohn (2011) show that people treat round numbers as goals and change behavior at the threshold, and Bhattacharya et al. (2012) show that buy–sell imbalances change on and around round transaction prices. These two papers concern where the current price stands relative to a round number. I ask a different question. For two positions in the same stock on the same day, or held by the same investor on the same day, is the gain–loss gap in selling larger when the investor's own purchase price is round? Within a stock-day every investor faces the same current price and the same distance to any round threshold, and within an investor-day every position is evaluated through the same platform and order process. The comparison therefore rests on the roundness of the investor's own purchase price rather than on where the current price stands or on how a platform displays it, and the stock-split design changes that roundness after the trade. This adds the form of the investor's own reference price to the known determinants of realization behavior. Throughout, accessibility refers to the ease with which the purchase price is encoded, retrieved, or compared. Internet Appendix Section I formalizes this idea in a simple model in which the investor recalls a noisy estimate of her cost basis, and derives the predic-

tions that the account-level tests examine.

These findings speak to several literatures. This paper contributes to a large literature in finance that explores how constraints on investor attention influence market outcomes (Barber and Odean 2008; Hirshleifer, Lim and Teoh 2009; Frydman and Wang 2020). Relative to work that emphasizes external salience, I find that round purchase prices amplify the disposition effect, consistent with the cognitive accessibility of the reference point itself being an economically meaningful internal channel of attention (Alter and Oppenheimer 2006; Green and Jame 2013).

An emerging literature across economics and psychology explores the nature of cognitive constraints in information processing (see Oprea (2020), for a review). This literature emphasizes that individuals should allocate their limited cognitive resources so that stimuli expected to occur more frequently are encoded with greater precision(Barlow et al. 1961). For example, models of rational inattention and resource rationality imply that individuals may preferentially encode stimuli that are frequent, simple, and easier to store and recall (Maćkowiak, Matějka and Wiederholt 2023). My findings contribute to this literature by suggesting that cognitive compression influences which reference points are precisely encoded and retrieved. This pattern is consistent with principles from efficient coding (Frydman and Jin 2022; Charles, Frydman and Kilic 2024; Frydman and Jin 2025), which predict heightened behavioral sensitivity to information that is either frequently encountered or easier to mentally represent. While previous empirical work on memory constraints has largely relied on lab settings, this paper offers field-based evidence consistent with efficient encoding of fluent, memorable stimuli shaping financial decision-making.

A growing literature explores the role of noisy or random behavior, which has been associated with patterns of global insensitivity to variation in choice parameters (Enke and Graeber 2023; Enke, Graeber, Oprea and Yang 2024). In contrast to this perspective, I propose that investor behavior is not uniformly noisy but selec-

tively sensitive: investors respond more strongly to certain reference points not because of their economic significance, but because they are more fluently processed and more easily recalled. This perspective complements the noise-based explanation by emphasizing cognitive accessibility and fluency-based and memory-based attention allocation as sources of behavioral heterogeneity.

The market-level evidence also speaks to limited attention around earnings announcements and to work on processing fluency in financial markets (Alter and Oppenheimer 2006; Green and Jame 2013). Prior research shows that attention constraints affect how earnings news is priced (Hirshleifer, Lim and Teoh 2009), and that markets respond asymmetrically to beats and misses of the consensus (Conrad, Cornell and Landsman 2002; Bhojraj, Hribar, Picconi and McInnis 2009; Johnson, Kim and So 2020). Relative to name-fluency studies that emphasize investor recognition of the firm itself, I study the cognitive accessibility of a shared forecast target. Round consensus EPS forecasts are associated with greater trading intensity when news is large, and a more thorough announcement reaction to positive surprises with less subsequent drift.

Lastly, my paper contributes to the literature on the disposition effect by documenting that its strength depends on the numerical form of the purchase price: the gain–loss gap in selling is larger when the purchase price is round, and the difference survives stock-by-day and investor-by-day comparisons and a stock-split design that changes roundness after the purchase. One strand of this literature has concentrated on the construction of formal models to better understand the mechanism driving the disposition effect (Barberis and Xiong 2009, 2012; Ingersoll and Jin 2013; Meng and Weng 2018). Another strand has tested these models empirically (Ben-David and Hirshleifer 2012; Frydman, Barberis, Camerer, Bossaerts and Rangel 2014; Birru 2015; Chang, Solomon and Westerfield 2016; Imas 2016; Heimer 2016).[3] Although the underlying cause of the disposition effect remains debated,

[3]My results complement Birru (2015), as I provide additional evidence that round purchase prices are associated with a stronger disposition effect, consistent with the fluency and memory of reference points

this paper provides new evidence that the effect is stronger when the purchase price is round, a pattern consistent with a more accessible reference point amplifying it.[4]

The rest of the paper is organized as follows. Section 2 describes the data and sample construction. Section 3 reports the main account-level evidence that round purchase prices amplify the disposition effect. Section 4 examines alternative mechanisms. Section 5 studies investor heterogeneity. Section 6 documents market-level price effects around earnings announcements. Section 7 concludes.

2 Data and Sample Construction

The empirical analysis draws on two main data sources. The account-level tests use retail brokerage transactions to study how round purchase prices shape the disposition effect. The market-level tests use an earnings-announcement panel built from IBES, Compustat, and CRSP to study how round consensus EPS forecasts relate to trading intensity and announcement returns.

The account-level data are provided by a nationwide discount brokerage house, which is the same as the data used by Barber and Odean (2000). The data are in three files: a trades file, a security number for the CUSIP file, and a positions file. Only the first two files are used in this study. The trades file includes stock transactions from more than 125,000 unique accounts over the period from January 1991 through December 1996. About 104,000 of these accounts hold common stock, and I use all of them. As a comparison, both Odean (1998) and Barber and Odean (2000) use a sample of 10,000 accounts from 1987 to 1996.

I follow several steps in cleaning and preparing the data. I begin with investor-

affecting it. See also D'Acunto, Prabhala and Rossi (2019), who find that the disposition effect decreases after the introduction of robo-advising.

[4]Memorable round number price makes the purchase price more salient. The interaction between salience and, for example, realization utility is suggested by Barberis and Xiong (2012), who writes that realization utility "is likely to play a larger role when the purchase price is more salient" (pg. 252).

transaction level observations and retain only those trades where the corresponding stock—identified via 8-character CUSIP—is covered in CRSP on every transaction date. I further exclude any investor-stock pairs that include at least one transaction with a negative commission, which may reflect a broker-initiated reversal or correction rather than a standard trade. To mitigate potential microstructure frictions, I exclude all stocks that exhibit at least one non-trading day within the prior 250 trading days, which ensures that all retained securities have a consistent record of trading activity. I compute returns as the percentage change from the purchase price to the closing price on the day before the sale. The cost basis is the most recent purchase price. For positions acquired through multiple purchases, Table IA.3 reports the results when the basis is instead the value-weighted average purchase price, the first purchase price, or the first-in-first-out (FIFO) price. Finally, I removed from the sample investor-stocks that include short-sale transactions or that have positions that were opened before 1991.01. I also keep only investor-days on which the investor holds at least two stocks.[5]

Next, for my main methodology, I use the transaction data set to construct a holding sample containing an observation for each investor-stock-day when sales take place.[6] I flag the days when a position is opened, when shares are sold (including a partial sale), or when additional shares of the same stock are purchased (increasing an existing position). I record purchase and selling prices (from the transaction data set), as well as daily closing prices from CRSP. When computing returns, I adjust for splits and dividends. In order to guard against regression re-

[5]I construct cumulative share positions for each investor-stock pair over time. If the cumulative position turns negative at any point—due to a prior holding established before the start of the sample or a short sale—I exclude the corresponding investor-stock from the sample. This rule primarily eliminates positions opened before 1991. An exception arises if an investor initiated a position before 1991, subsequently added shares post-1991, and retained some of the position beyond 1996. Such cases are exceedingly rare, given that investors seldom add to existing positions, and in over 90% of selling transactions, they liquidate their entire holding.

[6]Studies such as Frydman and Wang (2020), Barber and Odean (2013), and Feng and Seasholes (2005) also adopt a different method, incorporating all investor-day observations regardless of whether the investor sells at least one stock. Table IA.2 re-estimates the specifications of Table 3 on that broader panel; the $Gain \times RoundBuyPrice$ interaction remains positive and significant, so the main result is not an artifact of conditioning on sale days.

sults driven by outliers, I also winsorize independent variables at the 1st and 99th return percentiles within each prior holding period. Multiple buys or sells of the same stock in the same account on the same day, are aggregated.

I flag an investor-stock-day as a gain if the current price is strictly above the most recent purchase price. The current price is the selling price, the price of buying additional shares, or the end-of-day price. Symmetrically, the loss indicator is one if the current price is strictly below the most recent purchase price and zero otherwise. If the current stock price is equal to the most recent purchase price, the investor-stock-day is omitted. The interaction is similar when the basis is the first purchase price (1.960) or the FIFO price (0.906, marginally significant), and it does not depend on subsequent additional purchases (see Table IA.3 and Table IA.4). The value-weighted average purchase price is the one alternative for which the interaction is not positive (Column (2) of Table IA.3). Investors are less likely to track the average of several purchase prices as their reference point, so whether that average happens to be round does not appear to affect selling, and the estimate is indistinguishable from zero.

Insert Table 1 Panel A near here.

In Table 1 Panel A, I present descriptive information on the trading activity and portfolio characteristics of my sample. The dataset contains 1,393,400 investor–stock–day observations across 1,491 sell dates. On average, investors execute about 2.9 sales per year, hold 6.3 stocks per account–day, and maintain positions for roughly 222 days. The median holding period is 124 days, with substantial dispersion across the distribution. Average account–day total assets are about $82,792. The distribution of portfolio size is naturally right-skewed: the median is $31,125 and the 95th percentile is $305,375. These statistics highlight both the breadth of trading activity and the heterogeneity in portfolio size and holding horizons among individual investors.

Insert Table 1 Panel B near here.

In Table 1 Panel B, I provide descriptive statistics on stock purchase prices and capital gains. The average purchase price is about $31, and 64,627 observations have purchase prices at exact round-dollar amounts. Among these, purchase prices that end in 0 account for 31,158 observations, while those that end in 5 account for 33,469 observations. The distribution of round purchase prices is very similar to that of all purchases, suggesting that the occurrence of round numbers is not mechanically driven by extreme or atypical price levels, but rather reflects a broader tendency in how investors choose transaction prices.

Turning to capital gains, the mean return across all observations is 10.0%, with a standard deviation of 57.0%. About 54 percent of the observations have positive returns, with an average gain of 35.5%, while negative observations exhibit an average loss of 20.5%. For round-number returns, which total 64,627 observations, the mean is 9.7%. The distribution of round-number returns closely tracks that of overall returns, indicating that round returns do not disproportionately capture unusual price movements but instead mirror the general risk–return profile of the sample.

The brokerage records do not report order type, what the investor saw when deciding to trade, how much attention the investor paid to the stock, or what the investor recalled. The tests are built so that they do not require these variables. They need a contrast in the accessibility of the purchase price that does not come with a contrast in the decision environment, and the design obtains that contrast in three ways. First, the investor-by-day specification holds fixed the screen, the order process, and the investor's liquidity and tax circumstances, because all of an investor's positions are evaluated together, and the stock-by-day specification holds fixed the quote, the news, and the position of the current price relative to any round number, because all investors in a stock face the same price. Roundness

then varies through the purchase price of the individual position. Second, whatever the platform displayed about purchase prices applies to all of an investor's positions on a day, so display alone cannot generate a difference between round and non-round positions. It could do so only if roundness changes how readily the displayed number is used, which is the accessibility channel itself. Third, in the stock-split design both groups, Always and Breaks, bought at a round price, so any order-type or microstructure selection at purchase is shared by the two groups, and they differ only in whether the split-adjusted basis remains round. The data therefore do not identify whether roundness aids retrieval from memory or processing at the time of the sale. They do show that the gain–loss gap in selling varies with the roundness of the investor's own purchase price when the decision environment is held fixed.

For the market-level tests, I construct a firm-quarter earnings-announcement panel. For each firm-quarter, I take the last median consensus forecast observed strictly before the announcement date and classify the forecast target as round when that median consensus, rounded to two decimals, is a nonzero multiple of \$0.05. Percent earnings surprise is actual EPS minus the median consensus, scaled by the absolute value of the median consensus, and winsorized at the 1st and 99th percentiles. Cumulative abnormal returns are market-adjusted over trading days $[0, +1]$ and $[2, +75]$ relative to the announcement trading day and are winsorized at the 1st and 99th percentiles. Trading-intensity outcomes average retail order flow, identified following Boehmer, Jones, Zhang and Zhang (2021), and log share turnover over $[0, +1]$; retail flow is available for a shorter post-2015 subsample. The baseline return sample requires a usable consensus and actual EPS, a non-missing surprise, and at least two return days in $[0, +1]$.

Insert Table 2 near here.

Table 2 summarizes this market sample. It contains $N = 357{,}759$ firm-quarter

announcements for 11,108 firms over 1990–2024, of which 25.2% have round consensus targets ($N = 89{,}992$). Panel B reports means, with standard deviations in parentheses, for the median consensus, percent surprise, and announcement and post-announcement cumulative abnormal returns, for the full sample and separately for round and non-round targets.

3 Empirical Analysis

3.1 Univariate Results

To measure how cognitive accessibility influences the disposition effect, I start my empirical analysis with a univariate test.[7] To illustrate the idea, consider an investor with two stocks: X_R and X_{NR}, where X_R is bought at a round number price and X_{NR} is bought at a regular (non-round) price. If the disposition effect depends in part on the fluency and memorability of the holding stocks, then I would expect that $d_R > d_{NR} > 0$, where

$$d_i = Pr(X_i \text{ is sold} \mid X_i \text{ is at a gain}) \; - \; \Pr(X_i \text{ is sold} \mid X_i \text{ is at a loss}).$$

This implies that the gain-loss asymmetry in selling should be strictly larger when the purchase price is round than when it is not. This is precisely what is found in the data and can be illustrated with a simple figure shown below.[8]

Insert Figure 1 near here.

Figure 1 presents the probability that an investor sells a position (including partial sales) as a function of the position's current performance and the type of pur-

[7] I also replicate the Proportion of Gains Realized (PGR) and Proportion of Losses Realized (PLR) design of Odean (1998). Table IA.5 shows that PGR exceeds PLR in the full sample and that the PGR−PLR gap is significantly larger for round and whole-dollar purchase prices than for fractional prices, consistent with the sell-probability patterns in Figure 1.

[8] Figure 1 is constructed from the subset of investors who have purchased at least one round-number price at some point; accounts that never bought at a round price are excluded so that both the "RegularBuyPrice baseline" and "RoundBuyPrice" groups are drawn from the same investor pool.

chase price. Gains (losses) are defined as strictly positive (negative) paper returns relative to the most recent purchase price; ties are omitted. The left group, RegularBuyPrice Baseline, pools positions whose most recent purchase price is not a round number; the right group, RoundBuyPrice, pools positions purchased at a round number.

I first consider the probability that an investor sells a position purchased at a regular (non-round) price (left group). The disposition effect (blue bar) is the gap between the probability of selling when the position shows a paper gain (green bar) and the probability of selling when it shows a paper loss (red bar). In this baseline group, the sell probability is 19.60% in gains and 13.39% in losses, yielding a disposition effect of 6.20 percentage points. Put differently, investors are about 46.4% (19.60/13.39-1) more likely to sell after a gain than after a loss. This provides the benchmark against which I assess whether round purchase prices amplify the disposition effect.

Turning to positions purchased at a round number, the same univariate calculation shows a much larger gain-loss asymmetry (right group) in selling. The probability of selling when the position is at a gain is 24.99%, while the probability of selling at a loss is 16.39%, so the disposition effect equals 8.60 percentage points. In relative terms, investors are about 52.5% (24.99/16.39-1) more likely to sell after a gain than after a loss. Relative to the RegularBuyPrice baseline, both gain-state and loss-state selling probabilities are higher for round purchases, but the increase in gain-state selling is larger (+5.39 pp vs. +3.00 pp). As a result, the disposition effect amplifies by about 2.40 pp (8.60 - 6.20), which is a 38.7% (8.60/6.20-1) increase over the baseline. Taken together, the results in Figure 1 are consistent with the idea that round purchase prices are more fluent to process, leading investors to realize gains disproportionately more than losses.

3.2 Baseline Regression

My main methodology follows Feng and Seasholes (2005), Ben-David and Hirshleifer (2012), Barber and Odean (2013), and Frydman and Wang (2020). I estimate the disposition effect using a regression model, which takes the following form:

$$sell_{i,j,t} = \beta_0 + \beta_1 Gain_{i,j,t-1} + \epsilon_{i,j,t}, \tag{1}$$

where i, j, and t denote investor, stock, and day, respectively.[9] I only include days when a sale occurs within a portfolio comprising two or more stocks. $Sell_{i,j,t}$ is a dummy variable equal to one if investor i sells stock j (partially or fully) on the day t and $Gain_{i,j,t-1}$ is a dummy variable equal to one if stock j is in paper gain in investor i's portfolio, using the closing market price at $t-1$. With this structure, the coefficient β_1 measures the probability of selling a winning stock versus the probability of selling a losing stock. Many previous studies show that β_1 is positive and statistically significant, suggesting that investors are more prone to liquidate winning positions than losing positions.

My interest in this study is the relationship between the disposition effect and the cognitive accessibility of stock purchase prices. I analyze this relationship by estimating the following equation:

$$\begin{aligned} Sell_{i,j,t} = \alpha &+ \beta_1 Gain_{i,j,t-1} \\ &+ \beta_2 Gain_{i,j,t-1} \times RoundBuyPrice_{i,j,t} + \beta_3 RoundBuyPrice_{i,j,t} \\ &+ X\beta' + \epsilon_{i,j,t}. \end{aligned} \tag{2}$$

My additional variable, the proxy for fluency and memorability, $RoundBuyPrice_{i,j,t}$, is a dummy equal to one if investor i's purchase price for stock j is a round num-

[9] Feng and Seasholes (2005) and Seru et al. (2010) use a hazard model to estimate the disposition effect. However, this approach is limited in its ability to incorporate numerous high dimensional fixed effects due to the incidental parameters problem that affects maximum likelihood estimators. Additionally, this specification precludes the clustering of standard errors across multiple dimensions.

ber, such as \$10, \$15, \$25, \$100, \$1,000, etc. $X\beta'$ is a vector of stock-level control variables that can influence selling decisions. I follow Ben-David and Hirshleifer (2012) and include the natural logarithm of the most recent purchase price ($Log(BuyPrice)$), the stock volatility calculated using daily returns over the prior 252 days if the asset is in paper loss (gain) and zero otherwise ($Volatility(-)$ ($Volatility(+)$)), the number of days since purchase ($TimeOwned$), the return since purchase if the asset is in paper loss (gain) and zero otherwise ($Return(-)$ ($Return(+)$)). I also incorporate investor-by-day fixed effects, stock-by-day fixed effects, and investor-by-stock fixed effects in my main specification. These control for time-varying shocks to individual investors on a given day, common shocks to a stock on a given trading day, and persistent heterogeneity in investor–stock pairs, respectively. The regression in Equation 2 is estimated using linear regression, and standard errors are double clustered at the investor and day level.[10]

My main coefficient of interest in Equation 2 is β_2, the coefficient of the interaction term, which represents the difference in disposition effects for round price stocks and regular price stocks. In Equation 2, β_1 measures the disposition effect for regular price positions, and $\beta_1 + \beta_2$ measures the disposition effect for round price positions. A positive and significant β_2 would indicate that round-number purchase prices amplify the disposition effect, consistent with the idea that such prices are more fluent and memorable reference points. In other words, when investors anchor on a round number, the contrast between gains and losses becomes sharper, leading to a stronger asymmetry in realization behavior.

Insert Table 3 near here.

[10]Compared to the hazard model (See, for example, Feng and Seasholes (2005) and Seru et al. (2010)), which essentially measures the disposition effect as the ratio of the probability of selling winners and that of selling losers, the linear probability model estimates the disposition effect as the difference between these two. The benefit of this model is that it allows me to incorporate a number of fixed effects and won't suffer from the incidental parameter problem that the maximum likelihood estimator may have (Lancaster 2000). As a robustness check, I also estimate Cox hazard models later in the paper (see Table 9). The conclusion is the same: the $RoundBuyPrice \times Gain$ coefficient is positive and significant. The Cox coefficient is a log hazard ratio and is not comparable in magnitude to the linear-probability coefficients, which are scaled by 100.

Table 3 presents the estimation results of Equation 2. For presentation purposes, all coefficients are multiplied by 100. In all specifications, the coefficient β_1 on the *Gain* dummy is positive and statistically significant at the 1% confidence level and provides a baseline quantification for the disposition effect in my sample. The estimates of β_1 between 3.9 and 7.7 in Table 3 imply that a stock in a paper gain is 3.9 to 7.7 percentage points more likely to be sold than a stock in a paper loss. The estimates for β_2 on the interaction terms $RoundBuyPrice \times Gain$ are economically large and statistically significant: confirming that stocks purchased at a round number price exhibit a more substantial disposition effect than stocks purchased at regular numbers (with on average one-third of the total loading on the *Gain* dummy).[11][12]

The increase in the disposition effect among stocks purchased at round prices is economically large. In the univariate test results (Column 1), the estimate for the interaction terms is 1.156, which is roughly 15% ($\frac{1.156}{7.655}$) of the baseline disposition effect among stocks purchased at regular prices in the sample period. The interaction terms estimate increases to 1.445 with full controls (Column 2).

In Column (3), I re-estimate Equation 2, including stock-by-day fixed effects and investor fixed effects to address the "external salience" confounder. Stock-by-day fixed effects absorb any common, contemporaneous shocks that make a stock unusually salient on that date (e.g., earnings announcements, news, large returns, spikes in volume or volatility, or market-wide events), so the comparison is made within the same stock on the same day. Investor fixed effects remove time-invariant heterogeneity across investors (e.g., trading style, risk tolerance, tax status, default sell thresholds).

With these controls, identification comes from cross-sectional differences in reference prices across investors holding the same stock on the same day: positions

[11] Table IA.6 separates this amplification into $RoundBuyPrice \times Gain$ and $RoundBuyPrice \times Loss$: round purchase prices raise the propensity to sell winners and lower the propensity to sell losers, so the wider disposition gap is not driven by only one side of the gain/loss contrast.

[12] Table IA.7 shows that $RoundBuyPrice \times Gain$ remains large when excluding near-zero paper returns ($|r| \geq 1\%$ or 2%), whereas $RoundBuyPrice \times PaperReturn$ is insignificant, so the amplification reflects the gain/loss classification rather than tiny paper gains or a steeper continuous-return slope.

whose purchase prices happen to be round versus regular, and whether they are currently at a gain or a loss. The coefficient on the interaction term in Column (3) is 1.997, which remains positive and economically meaningful. This implies that the stronger realization asymmetry at round purchase prices cannot be attributed to contemporaneous stock-day salience; rather, it is consistent with the fluency and memorability of round reference points sharpening the gain-loss perception and amplifying the disposition effect. I use the specification in Column (3) as the baseline in the tests that follow, and the remaining columns vary which confounders the fixed effects absorb.

Column (4) changes the fixed effects in Column (3): it replaces the stock-by-day and investor fixed effects with investor-by-stock and day fixed effects, which targets a different set of confounders. Investor-by-stock fixed effects absorb any persistent heterogeneity specific to a given investor–stock pair—e.g., a stable trading style toward that stock (familiarity, default rules, etc) or a pair-specific tendency to transact at round prices that make the pair more (or less) likely to realize gains. Day fixed effects remove market-wide, time-varying shocks that affect selling across all stocks on a given date—macro news, market volatility spikes, calendar effects (e.g., month- or year-end), broad liquidity conditions—so the comparison is made net of common daily salience. By holding these pair-level traits fixed, the model avoids attributing long-run investor–stock matching or preferences to the fluency and memorability channel. The interaction is 1.233, smaller than in Column (3) but still significant at the 5% level.

Column (5) again changes the fixed effects: relative to Column (3), it replaces the investor fixed effects with investor-by-day fixed effects and the stock-by-day fixed effects with stock fixed effects, which addresses a different set of potential confounders. Investor-by-day fixed effects absorb all common factors that shape an investor's trading decisions on a given day, such as liquidity needs, portfolio rebalancing, tax or margin constraints, and shifts in attention or risk appetite. Stock

fixed effects, in turn, capture time-invariant characteristics of individual stocks, including their typical price levels and tick sizes, long-run liquidity or transaction costs, and any persistent tendency to trade at round numbers.

With these controls in place, identification comes from within-investor comparisons across different stocks held on the same day: some purchased at round prices and others at non-round prices. Under this demanding specification, the interaction coefficient (1.659) remains positive and significant at the 1% level. This result suggests the stronger realization asymmetry at round purchase prices cannot be attributed to investor-specific daily shocks or stock-specific static features, but is instead consistent with round numbers serving as more fluent and memorable reference points that amplify the disposition effect.[13]

The samples so far condition on days on which the investor sells at least one stock, which could introduce selection. Table IA.2 therefore re-estimates the specifications of Table 3 on the full investor-stock-day panel, including days with no sale. The $RoundBuyPrice \times Gain$ interaction remains positive and significant in all five columns (t-statistics from 2.2 to 3.4). Because the dependent variable is much less likely to equal one on this panel, the coefficients are smaller in absolute terms. Relative to the *Gain* coefficient, the interaction is 31 to 38 percent in Columns (3) to (5), comparable to the main sample.

The estimates above compare round-price positions with regular-price positions, and the comparison is designed to identify a source of the disposition effect. A round price and a regular price convey the same economic content, so the interaction measures how much the accessibility of the purchase price alone changes realization behavior. Where it applies, the effect is large: in Columns (2) to (5), the gain–loss gap of round-price positions is about 30 to 50 percent larger than that of regular-price positions. Round purchase prices are 64,627 of the 1,393,400 observa-

[13] Table IA.8 reassigns each investor's *RoundBuyPrice* labels across the stocks that investor holds: the $RoundBuyPrice \times Gain$ interaction collapses under these placebos (only about 1% of 100 reassignments match the baseline $|t|$), so the amplification is tied to the stock's own round purchase price rather than an investor-level tendency to buy round and sell winners.

tions, about 4.6 percent (Table 1), so they raise the sample-average gap by a smaller amount. In Column (3), $3.886 + 0.046 \times 1.997 \approx 3.98$, an increase of about 0.09 percentage points, or 2.4 percent. The aggregate figure is therefore not the right measure of the role of accessibility, for two reasons. First, round prices are one extreme of a continuum of accessibility, so if accessibility also varies among regular-price positions, the round versus regular contrast is conservative. Second, the test is informative about the mechanism regardless of how many positions are round: a small, clean contrast that shows realization depends on the numerical form of the basis is evidence that accessibility is one channel behind the disposition effect.

3.3 Roundness, Fluency, Memorability and Disposition Effect

One key assumption underlying my empirical test is that the price roundness serves as a useful proxy for stock fluency and memorability that affects the disposition effect. In this section, I exploit the prevalence of this premise against which other stimuli are evaluated to further highlight the role of fluency in shaping selling behavior.

A. Round Price and Whole Price

A potential concern with the baseline results in Table 3 is that they may capture a broader preference for whole number prices rather than a round-price effect specifically. That is, if investors systematically encode and respond differently to any purchase price without cents, then the stronger realization asymmetry at prices ending in 0 or 5 might simply be an artifact of whole-number pricing, not necessarily be of fluency and memorability. To address this concern, I estimate the following

regression model:

$$
\begin{aligned}
Sell_{i,j,t} = \alpha &+ \beta_1 Gain_{i,j,t-1} \\
&+ \beta_2 Gain_{i,j,t-1} \times RoundBuyPrice_{i,j,t} + \beta_3 RoundBuyPrice_{i,j,t} \\
&+ \beta_4 Gain_{i,j,t-1} \times WholeBuyPrice_{i,j,t} + \beta_5 WholeBuyPrice_{i,j,t} \\
&+ X\beta' + \epsilon_{i,j,t},
\end{aligned}
\tag{3}
$$

where the additional variable $WholeBuyPrice_{i,j,t}$ is the purchase prices without cents and whose last digit is 1–4 or 6–9.

If the effects documented earlier were merely reflections of a generic whole-number bias, then the coefficient on the interaction term for $RoundBuyPrice_{i,j,t}$ (β_2) should be indistinguishable from that for whole-but-non-round numbers (β_4). In that case, whether a purchase price ends in 0 or 5 versus 1–4 or 6–9 would make no difference in the disposition effect (e.g., $\beta_2 = \beta_4$).[14] By contrast, if fluency and memorability are the operative channel, then round prices provide a more fluent and cognitively accessible reference point than whole-but-non-round prices. This implies that β_2 should be larger in magnitude and more consistently statistically significant than β_4, reflecting the stronger asymmetry in the propensity to sell gains relative to losses when the purchase price is especially fluent to process and easy to recall.

Insert Table 4 near here.

In Table 4, I report the coefficient estimates and standard errors double clustered at the investor and day level from ordinary least squares (OLS) pooled regressions of various forms of Equation 3. For presentation purposes, all coefficients are multiplied by 100. Across all specifications, the interaction term for round purchase prices remains large, positive, and highly significant, with coefficients ranging from

[14] A visual inspection of Figure 2 also suggests rejecting the null $H_0 : \beta_2 = \beta_3$: the estimated effects and their 95% confidence Intervals mostly do not overlap.

1.25 to 2.23. However, it is worth noting that the coefficient on the interaction term $Gain_{i,j,t-1} \times WholeBuyPrice_{i,j,t}$ is also positive and statistically significant in most specifications, but considerably smaller in magnitude relative to its round-price counterpart. This finding suggests that investors may exhibit some tendency to treat whole numbers as salient anchors, possibly because the absence of the cents portion makes them more fluent to process and easier to encode and recall than fractional prices (the control group).

Taken together, the results in Table 4 suggest that the stronger realization of gains relative to losses is not a generic feature of whole number pricing, but is specifically tied to round purchase prices. In other words, investors are not simply reacting to prices without decimals; rather, they are disproportionately influenced by prices that end in 0 or 5, consistent with the idea that roundness provides a cognitively fluent and memorable reference point and amplifies the disposition effect.[15]

B. Falsification Test

While the previous section establishes that the baseline results are not driven by a broad preference for whole-number prices, an additional and more stringent question remains: whether the documented amplification reflects a smooth response to integer-valued prices, or instead hinges on the presence of specific numerical anchors.

If the effect were driven by general features of integer pricing, such as discreteness, execution conventions, or a coarse encoding of prices without cents, one would expect the realization asymmetry to vary smoothly across nearby integer values. In contrast, the fluency and memorability mechanism implies a knife-edge prediction: only prices that constitute salient numerical anchors should generate a

[15] Table IA.9 replaces the binary *RoundBuyPrice* indicator with continuous distance to, and proximity within \$0.05 of, the nearest whole-dollar and round (0/5) grids: the *Gain* amplification declines with distance and is larger near 0/5 than near other whole dollars, reinforcing that the effect is tied to round anchors rather than integer prices alone.

persistent amplification, while adjacent non-round integers should not.

To distinguish between these possibilities, I implement a local falsification design within the set of whole-number prices. Specifically, I replace purchase prices ending in 0 and 5 with adjacent non-round endings that differ by only one unit (1 and 9 for 0, and 4 and 6 for 5). These alternative endings are comparable to round prices in terms of integer status, local price level, and trading environment, but are not round numbers.

This design preserves mechanical aspects of integer pricing while introducing a minimal perturbation that breaks round-number salience. If the amplification documented above were driven by smooth integer effects, similar interaction patterns should emerge for these adjacent endings. Conversely, a sharp attenuation would indicate that the effect is tightly linked to round numerical anchors rather than to integer pricing more broadly.

Insert Table 5 near here.

The empirical results, reported in Table 5, provide consistent support for the fluency and memorability hypothesis. For the "1 and 6" group (Columns 1-5), the coefficient on the interaction term $BuyPriceEnd1or6 \times Gain$ is small and statistically insignificant across all specifications. For instance, in the most stringent specification with investor-by-day and stock fixed effects (Column 5), the coefficient is 0.710 ($p > 0.10$), which is less than half the magnitude of the 1.659 observed for 0- and 5-ending prices. Similarly, for the "9 and 4" group, the interaction effect $BuyPriceEnd9or4 \times Gain$ becomes negligible and loses all statistical significance (0.371, $p > 0.10$) once proper controls and fixed effects are included.

C. Last Digit and Fraction

To complement the regression-based falsification in Table 5, I plot the distribution of realization asymmetries by the last digit of all purchase price in Figure 2.

The goal is that the round–vs.–non-round dichotomy in the tables could hide richer heterogeneity across digits and fractions. I run the following regression to conduct the test:

$$\begin{aligned} Sell_{i,j,t} = \alpha &+ \beta_1 Gain_{i,j,t-1} \\ &+ \beta_2 Gain_{i,j,t-1} \times PriceNumCat_X_Y_{i,j,t} + \beta_3 PriceNumCat_X_Y_{i,j,t} \\ &+ X\beta' + FE_i + FE_{j\times t} + \epsilon_{i,j,t}, \end{aligned} \tag{4}$$

where $PriceNumCat_X_Y$, is a dummy indicating whether the investor's stock purchase price falls within one of the pricing categories. Specifically, I construct 20 sets of pricing categories based on the price's whole number portion and fractional portion. $PriceNumCat_X_Y$ equals 1 if the stock purchase price's right-most digit for the whole number portion is X ($X \in \{0,1,2,3,4,5,6,7,8,9\}$) and the fraction portion is Y ($Y \in \{[0,\frac{1}{2}), [\frac{1}{2},1)\}$).[16]

Insert Figure 2 near here.

Figure 2 illustrates the relationship between the propensity to sell gains and price roundness, following the model specified in Equation 4. This analysis incorporates all control variables from Ben-David and Hirshleifer (2012) and includes investor and stock-day fixed effects. The plot presents coefficient estimates for β_2 along with 95% confidence intervals, which are calculated using standard errors clustered by investor and day.

Overall, the estimates in Figure 2 reveal a sharp, discrete pattern. Purchase prices that end in 0 or 5 display clear positive spikes in the probability of selling, and the corresponding coefficients are above zero and statistically different from zero at conventional levels. By contrast, other whole-number endings cluster tightly around zero with only small, imprecise deviations, and fractional endings

[16] My sample overlaps the pre-decimalization period when U.S. equities traded at a \$1/8 tick. Table IA.10 reports purchase-price frequencies in eighth bins for four heavily traded names from April to May 1996: purchases cluster at whole and half dollars (0/8 and 4/8), illustrating the \$1/8 tick grid in the sample.

are generally at or below zero. The fact that the jumps occur exactly at 0 and 5, rather than tracing a smooth gradient across the last digits, is consistent with a fluency and memorability interpretation rather than a generic price-level effect.[17] In terms of economic magnitude, the 0/5 effects are between 1.5 and 2 percentage points, which is meaningful relative to baseline sell rates and dovetails with the stronger disposition effect for round purchases documented in Figure 1.

3.4 Stock Split and Memory Disruption

So far, the empirical evidence suggests that the disposition effect is significantly amplified when the purchase price is round, and this pattern is not explained by stock-specific external attention shocks or generic price-level effects. However, a remaining concern is purchase-time selection: executions at round prices may cluster in specific microstructure conditions (time-of-day, order type, spreads/liquidity) that fixed effects cannot fully absorb.[18] I address this concern in this section with a stock-split design. A split rescales the original purchase basis after the purchase and mechanically determines whether the split-adjusted purchase price for the same investor–stock is still round or not. The split does not change the conditions under which the purchase was made; it changes the number the investor must work with afterward. This setting, therefore, provides plausibly exogenous variation in the roundness of the basis as it applies to current shares, holding fixed the purchase-time conditions.

In the following test, the unit of observation is the investor–stock–day. I restrict the sample to investor–stock positions whose underlying stock experiences a split within the holding spell between January 1991 and December 1996.[19] Specifically,

[17] Table IA.11 interacts $RoundBuyPrice \times Gain$ with high-, mid-, and quartile purchase-price indicators and with demeaned $\log(BuyPrice)$: none of these triples is significant, so the amplification does not concentrate in high- or low-priced stocks.

[18] Table IA.12 shows that $RoundBuyPrice \times Gain$ remains large after controlling for buy-day relative spread, turnover, and price-in-range, and is not differentially larger when those buy-day conditions are high.

[19] The split must fall inside the holding spell: after the purchase that sets the basis, and before the spell

I estimate the following regression model:

$$\begin{aligned} Sell_{i,j,t} = \alpha &+ \beta_1 Gain_{i,j,t-1} \\ &+ \beta_2 Gain_{i,j,t-1} \times RoundBuyPriceAlways_{i,j} + \beta_3 RoundBuyPriceAlways_{i,j} \\ &+ \beta_4 Gain_{i,j,t-1} \times RoundBuyPriceBreaks_{i,j} + \beta_5 RoundBuyPriceBreaks_{i,j} \\ &+ X\beta' + \delta_{j,t} + \gamma_{i,j} + \epsilon_{i,j,t}, \end{aligned} \tag{5}$$

where $\delta_{j,t}$ is a stock-by-sell-date fixed effect and $\gamma_{i,j}$ is a stock-by-investor fixed effect. $RoundBuyPriceAlways$ = 1 if both the original purchase price and the split-adjusted purchase price have no fractional cents and the integer amount ends in 0 or 5; 0 otherwise. $RoundBuyPriceBreaks$ = 1 if the original purchase price is round pre-split but becomes non-round after applying the split ratio; 0 otherwise. Both indicators are fixed within a holding spell, so they take the same value on every day of that spell, including days before the split. The omitted category includes positions that are never round pre-/post-split.[20] Table IA.15 reports covariate means for Always, Breaks, and Never spells in this sample.

If the fluency and memorability hypothesis is operative, the gain-loss asymmetry in selling should remain sufficiently large only when the split leaves the purchase price round ("Always") and should be disrupted when the split breaks roundness ("Breaks"). In the specification in Equation 5, this implies a significantly positive β_2, an insignificant β_4, and $\beta_2 > \beta_4$. By contrast, a purchase-time selection hypothesis (where round executions merely proxy for particular microstructure environments) predicts a similar magnitude for the coefficients on the two interaction terms, because the ex post split does not alter the original buy-time conditions.[21]

ends. Splits before purchase, or only after the position is already closed, are dropped. If a spell has more than one split (rare here), I use the first split to classify the position. Unlike the baseline disposition regressions, I keep all holding days for these spells, not only sell days, to stay closer to Birru (2015) and to keep power when within-spell splits are scarce.

[20] Forward splits that would create a round split-adjusted purchase price are exceedingly rare in my sample. Consequently, the omitted/control group is effectively composed of positions that are not round either pre- or post-split.

[21] This Always-versus-Breaks contrast is robust in Table IA.16, which further interacts both groups with

Insert Table 6 near here.

Table 6 reports OLS estimates of various forms of Equation 5. The coefficients are scaled by 100, and standard errors are double-clustered by investor and stock. Across columns, the coefficient on *Gain* ranges from 0.05 to 0.11, but its magnitude and statistical significance are much smaller than in the baseline estimate in Column (3) of Table 3. This attenuation is consistent with Birru (2015): after a split, investors may fail to update the reference price, so for positions in the omitted group (those never round pre-/post-split) they become "confused" about gain versus loss status, dampening the measured disposition effect.

The main effect for the interaction $Gain \times RoundBuyPriceAlways$ is consistently large and statistically significant (ranging from 0.28 to 0.29). Thus, when a split preserves roundness, the gain-loss asymmetry in selling increases by approximately 0.28 percentage points relative to never-round positions. By contrast, the coefficient on the interaction term $Gain \times RoundBuyPriceBreaks$ is small and statistically indistinguishable from zero across specifications. In columns (4) and (5), where both interactions enter jointly, a formal test rejects equality: Always×Gain exceeds Breaks×Gain by 0.240 and 0.250 ($t = 2.36$ and $t = 2.43$; $N = 5{,}082{,}807$ and $N = 4{,}998{,}917$).[22] When a split converts a previously round purchase price into a non-round adjusted basis, the amplification disappears.[23]

This pattern is better described as reflecting the psychology of computational ac-

a 30-day post-split indicator: Always×Gain is differentially larger in that window, while the Breaks triple interaction remains insignificant.

[22]The treatment varies at the spell level, so the number of independent units is much smaller than the number of daily observations: the estimates rest on 144 Always and 301 Breaks spells drawn from 90 and 168 stocks (Table IA.18). Clustering at the spell or at the split-event level leaves the point estimates unchanged, and the Always–Breaks difference remains significant ($t = 2.41$ and $t = 2.17$; Table IA.19). Dropping each of the five stocks with the most Always and Breaks spells leaves the estimates essentially unchanged, and the difference remains significant at the 10 percent level when all five are dropped together (Table IA.20).

[23]Table IA.17, Column (1), keeps only days after the within-spell split, so Always versus Breaks is compared once the purchase basis has already been rescaled. Always×Gain is about 0.37 and Breaks×Gain remains near zero; the Always−Breaks gap stays significant. That cut also limits the concern that the main Always result is driven by selling between purchase and the split. Columns (2) and (3) instead drop the 30-day (or ±5-day) ex-date window and show the same pattern away from mechanical trading around the split.

cessibility and price-adjustment heuristics than memorability alone. A split leaves the encoding of the original price unchanged. What it changes is the number the investor must produce, by adjusting the remembered purchase price for the split ratio, in order to judge whether the position is at a gain. Birru (2015) reads the disappearance of the disposition effect on split days as a failure to adjust the reference point. The Always–Breaks contrast speaks to a related margin. If the number that mattered were the original purchase price, Always and Breaks positions would look alike, since both were round when bought; they do not, so the roundness of the adjusted basis matters. This reading does not move the account away from fluency and memory. The adjusted basis must be computed from a purchase price the investor can retrieve and a split ratio, and a round number is easier to retrieve and to manipulate mentally. A basis that is neither fluent nor memorable gives the investor little to compute from. The split design therefore shows that the accessibility of the basis as applied to current shares matters, which is consistent with a fluency and memory account, but it cannot separate retrieval of the original price from computation of the adjusted one. What it does argue against is purchase-time selection, which predicts similar interactions for the two groups.

Taken together, the amplification of the disposition effect appears only when the post-split basis remains a round number and vanishes when the split breaks roundness. This split-based evidence is difficult to explain by purchase-time microstructure selection, and it is consistent with the accessibility of the adjusted basis mattering for realization. Internet Appendix Figure IA.4 and Table IA.13 report how this amplification varies with holding time among positions that are still held.

4 Alternative Mechanisms

In this section, I examine whether the round-price amplification I document is simply a manifestation of prior empirical research.

4.1 Tax Incentives

Prior literature shows the disposition effect is time-varying due to tax incentives (e.g., Odean (1998)). To rule out a tax-based explanation for the round-price result, I re-estimate the baseline model separately for U.S. tax-exempt and taxable accounts. Panel A of Table 7 reports these estimates: Columns 1 and 2 (Tax Incentives) correspond to the tax-exempt and taxable subsamples, respectively. The interaction term is positive and statistically significant in both subsamples: about $+9.38$ pp in tax-exempt accounts ($p < 0.01$) and $+1.90$ pp in taxable accounts ($p < 0.01$). The results indicate that the round-price amplification of the disposition effect is present with and without tax incentives, implying that it is not driven by taxes.

Insert Table 7 Panel A near here.

4.2 The Rank Effect

Prior literature documents that investors are more likely to sell portfolio extremes (the best- and worst-performing positions; see Hartzmark (2015)). Since executions at round prices might be more frequent in heavily traded stocks, and such stocks could also be more likely to become portfolio winners (greater attention, momentum co-movement), leaving round-price purchases overrepresented among top-rank positions ex post. If investors systematically sell top-rank winners, the gain-state sell probability for round-price purchases will be mechanically higher, inflating the gain-loss gap even without a fluency and memorability channel.[24]

[24] While rank also predicts selling of bottom-rank losers, the empirical tilt of round-price purchases toward liquid winners implies a larger effect on gains, increasing the observed disposition effect for round prices.

To rule out that the main finding of this paper is driven by a preference to sell portfolio extremes rather than by fluency and memorability, I augment the baseline investor–stock–day regression with within-portfolio return ranks. On each investor–day, I compute for every open position its cumulative return since purchase and rank all contemporaneous holdings within that investor's portfolio. I then create indicator variables for (i) the top-ranked ("best") and bottom-ranked ("worst") position (Top1, Bottom1), following Hartzmark (2015), and (ii) in an alternative specification, for the top 15 and bottom 15 positions (Top15, Bottom15), following An et al. (2024). These rank indicators enter the regression as additional controls alongside the full set of baseline controls and fixed effects. The dependent variable and sample are identical to the baseline; standard errors are double-clustered by investor and day.

Columns 3 and 4 (Rank Effect) in Panel A of Table 7 show that in both specifications, the interaction remains positive and statistically significant and is very similar to the baseline regression in column 3 of Table 3. Thus, these results suggest that the main finding is not a mechanical consequence of rank-based selling.

4.3 Portfolio Rebalancing

It is possible that portfolio rebalancing causes the round-price amplification. Imagine a portfolio in which every position except one has declined. It is likely that the lone stock that is trading at a gain comprises a disproportionately large percentage of the investor's portfolio due to its gains and the rest of the stocks' losses. An investor aiming to restore target weights or reduce concentration may therefore sell (including partial sales) the winner. If stocks bought at round prices are systematically more liquid, or held in bigger positions, they are cheaper and more convenient to rebalance. Round-price holdings would then be disproportionately trimmed after gains, producing a larger gain-loss asymmetry that reflects rebalancing frictions rather than the fluency and memorability of the purchase price.

To address portfolio–rebalancing concerns, I redefine the dependent variable to capture only complete exits. Specifically, I construct an indicator, $FullLiquidation_{i,j,t}$ that equals 1 when investor i closes position j (shares fall to zero by the end of the day t) and 0 otherwise. I then re-estimate the baseline specification, replacing the dependent variable $Sell_{i,j,t}$ with $FullLiquidation_{i,j,t}$ while keeping the same set of controls, fixed effects, and double-clustering standard errors by investor and day. This test thus eliminates any variation from partial sales.

Column 5 (Rebalancing) in Panel A of Table 7 shows the results. The coefficient of interest is the interaction term, which remains positive and statistically significant (2.26 pp, $p<0.01$). Thus, the amplification of selling propensity after gains at round purchase prices is present even for complete exits, indicating that the main finding is not driven by portfolio rebalancing.

4.4 Portfolio Concentration

Investors who buy at round numbers may systematically take larger, more salient positions (e.g., in liquid, attention-grabbing stocks). Larger positions raise portfolio concentration, which creates independent incentives to sell gains: Big winners quickly dominate risk and trigger exposure trimming and closer monitoring, while losses are more likely to be retained to avoid realizing large losses. If round-price purchases coincide with higher portfolio concentration, the estimated disposition effect may reflect concentration-driven selling of winners.

To rule out that portfolio concentration explains the round-price results, I construct an investor-day portfolio Herfindahl–Hirschman Index (HHI). I then split the sample at the cross-sectional median of HHI into High-concentration and Low-concentration subsamples.

Columns 1–2 (HHI) in Panel B of Table 7 show the results. The coefficient of interest is the interaction term, which is positive in both specifications (1.711 pp for High HHI ($p<0.10$) and 1.340 pp for Low HHI ($p<0.05$)). The magnitudes

are comparable, with the effect, if anything, slightly larger for high-concentrated (higher HHI) portfolios. Taken together, these estimates indicate that portfolio concentration does not appear to drive the round-price amplification of the disposition effect.

Insert Table 7 Panel B near here.

4.5 Liquidity

Liquidity may cause the round-price amplification. Investors are mechanically more likely to realize gains in liquid stocks because trading costs are lower and partial sales are easier; if round-price purchases are concentrated in such stocks, the observed gain-loss asymmetry in selling could be inflated independent of reference-point fluency and memory. Therefore, the "round-price" indicator may partly proxy for liquidity rather than fluency and memorability, and failing to condition on it risks attributing liquidity-driven selling to the fluency and memorability channel.

To isolate the round-price amplification from liquidity, I proxy stock-day liquidity with a spread based on the CRSP daily high ask and low bid.[25] I merge this stock-day measure to the investor–stock–day panel and sort observations into high-, median-, and low-liquidity groups by the relative spread (tightest spreads = most liquid). In the estimation sample these groups contain 329,934, 698,434, and 365,001 observations, about 24%, 50%, and 26% of the sample, rather than equal thirds. I then re-estimate the baseline specification separately within each liquidity group, retaining the full set of controls and fixed effects and double-clustered standard errors by investor and day. If the main finding were driven by liquidity rather

[25] The proxy is $\text{RelSpread}_{j,t} \equiv \dfrac{\text{ASKHI}_{j,t} - \text{BIDLO}_{j,t}}{(\text{ASKHI}_{j,t} + \text{BIDLO}_{j,t})/2}$, where ASKHI and BIDLO are the highest ask and the lowest bid recorded for the stock that day. Smaller values indicate higher liquidity.

than fluency and memorability, the round-price amplification would concentrate in the high-liquidity group.

Columns 3–5 (Liquidity) in Panel B of Table 7 split the sample by this high-ask/low-bid proxy. The interaction term is positive and statistically significant for the high-liquidity and median-liquidity groups, which together account for 73.8% of the estimation sample. In contrast, while the coefficient remains positive in the low-liquidity group, it is estimated with less precision. Taken together, these results indicate that the round-price amplification of the disposition effect is not a liquidity artifact; if anything, it is more pronounced where spreads are tighter and trading frictions are lower.

5 Investor Heterogeneity

Another possibility is that the round-price amplification is explained by heterogeneity across investors and portfolio characteristics. In this section, I re-estimate the baseline specification in column 3 of Table 3 within subsamples defined by different characteristics. Because some demographic fields are missing, the subsample counts can be smaller than the full sample. On the investor side, I consider professional status, age, and trading frequency.

I follow Dhar and Zhu (2006) in defining an investor's level of sophistication using occupation groups. Specifically, I classify individuals as working in "professional" occupations if they report working in "professional/technical" or "managerial/administrative" positions. I classify individuals as working in "nonprofessional" occupations if they report working in "white collar/clerical," "blue collar/craftsman," or "service/sales." Dhar and Zhu (2006) document that investor sophistication is negatively correlated with the disposition effect, so I test whether the round-price amplification is consistent with this pattern.

In column 3 to 5 of Panel A, Age is split into three bins: 16–34, 35–54, and $\geq$55,

which roughly correspond to the first quartile, the middle two quartiles, and the top quartile of the age distribution. in column 1 to 3 of Panel B, Trading frequency is calculated as the conditional selling propensity of an investor over the whole sample period. Investors are partitioned into low-, median-, and high-frequency trading groups. On the portfolio side, I report two variations of the maximum holding period: 60 days, and 250 days.

Insert Table 8 near here.

In Table 8, there are several interesting observations. First, as reported in the first two columns of Panel A, while the round-price amplification remains pervasive, it is notably more pronounced among the less sophisticated investor group. This result is consistent with Dhar and Zhu (2006). Second, the round-price amplification is larger for the youngest (16–34) and middle age (35–54) investors, and smaller and even reversed for the oldest ($\geq$55) investors. Third, high- and median-frequency traders show a stronger round-price amplification than low-frequency traders (Panel B). Fourth, by horizon, the effect is consistent in both short and long horizons (Panel B).

6 Price Effects

The account-level evidence concerns private reference points. This section asks, more briefly, whether a shared reference point shows a similar pattern. At quarterly earnings announcements, the analyst consensus EPS forecast is a shared target against which earnings news is judged. Consensus values such as \$1.00 and \$1.03 are nearly the same target, but a round forecast may be easier to process and retain. If so, firms with round targets might trade more when earnings news arrives and react more fully to the surprise. Round consensus forecasts are not randomly

assigned, so I report this evidence as supporting rather than as a separate identification of the channel.

Throughout this section, *Round* equals one when the median IBES consensus EPS forecast, rounded to two decimal places, is a nonzero multiple of \$0.05 (for example, \$1.00 or \$1.25), and zero otherwise.[26] Percent earnings surprise is $(Actual - Consensus)/|Consensus|$, winsorized at the 1st and 99th percentiles and scaled to percent. Absolute and signed surprise components are standardized as described below. Cumulative abnormal returns are market-adjusted and winsorized at the 1st and 99th percentiles. Standard errors are clustered by announcement date.

6.1 Trading Intensity Around Earnings Announcements

A natural starting point is trading intensity: if a round forecast target is easier to process and remember, then a large miss or beat relative to that target should elicit a stronger market response, thereby generating higher trading volume. I begin with announcement-window trading intensity and estimate

$$\begin{aligned} Y_{i,t} = \alpha + \beta_1 Round_{i,t} + \beta_2 SR^{abs}_{i,t} \\ + \beta_3 Round_{i,t} \times SR^{abs}_{i,t} + X\gamma + \varepsilon_{i,t}, \end{aligned} \tag{6}$$

where i indexes firm-quarter announcements and t is the announcement date. The outcome $Y_{i,t}$ is either retail trade flow, $\log\big(1 + (mrbvol + mrsvol)/SHROUT \times 10^4\big)$ constructed from retail volume identified following Boehmer, Jones, Zhang and Zhang (2021), or $\log(\text{turnover})$, where turnover is daily share volume divided by shares outstanding. Each dependent variable is averaged over trading days $[0, +1]$ and winsorized at the 1% and 99% levels. SR^{abs} is the full-sample standardized absolute percent surprise. The vector X includes size and book-to-market deciles within calendar quarter; the log number of same-day announcements in the firm's

[26] Table IA.21 reports within-firm adjacent-quarter covariate differences when *Round* switches: size, analyst coverage, and institutional ownership are essentially unchanged, so round-target quarters are not accompanied by large within-firm shifts in these observables.

surprise decile and the log number of analyst EPS forecasts; reporting lag and its square and cube; institutional-ownership decile; earnings volatility and persistence; lagged share turnover; year, month, day-of-week, and Fama-French 10 industry indicators; and interactions of each control with the within-quarter surprise decile FE_q. My coefficient of interest is β_3: a positive β_3 means trading rises more steeply in surprise size when the forecast target is round.

Insert Table 10 near here.

Table 10 reports the estimates. Across both retail trade flow and log turnover, $Round \times SR^{abs}$ is positive and significant. With controls, the retail-flow interaction is 0.047 ($t = 3.69$; $N = 88{,}033$), and the turnover interaction is 0.045 ($t = 8.47$; $N = 346{,}313$). The main effect of *Round* is also positive once controls are included, so round-target firms are busier at announcement even after I hold surprise size fixed. In short, large earnings surprises draw more trading when the forecast target is round.

6.2 Announcement Reaction and Drift

If a round forecast target draws more attention to the earnings news, that attention may also be reflected in prices. Relative to other firms, firms with a round forecast target should therefore react more thoroughly on the announcement day, and leave less of the surprise to be resolved in subsequent drift. I study announcement returns $\text{CAR}(0, +1)$ and post-announcement returns $\text{CAR}(2, +75)$. Prior work shows that markets respond asymmetrically to good and bad earnings news and that just beating versus missing the consensus carries distinct return implications (Conrad, Cornell and Landsman 2002; Bhojraj, Hribar, Picconi and McInnis 2009; Johnson, Kim and So 2020). I therefore estimate the round-target effect separately for positive and non-positive surprises. Define $SR^{+} = \max(Surprise, 0)$ and $SR^{-} =$

$\min(Surprise, 0)$, each standardized to mean zero and unit variance within the corresponding surprise-sign subsample. For Surprise > 0, I estimate

$$\begin{aligned} CAR_{i,t} = \alpha + \beta_1 Round_{i,t} + \beta_2 SR^+_{i,t} \\ + \beta_3 Round_{i,t} \times SR^+_{i,t} + X\gamma + \varepsilon_{i,t}, \end{aligned} \tag{7}$$

and an analogous specification for Surprise ≤ 0 that replaces SR^+ with SR^- and $Round \times SR^+$ with $Round \times SR^-$. Returns are scaled to percent. Controls match those in Table 10.[27] A more complete announcement reaction to a round forecast target implies a positive β_3 on CAR$(0, +1)$. Less subsequent drift implies a smaller, or even negative, β_3 on CAR$(2, +75)$.[28]

Insert Table 11 near here.

Table 11 reports the results. The pattern shows up when the surprise is positive. In Panel A with controls, $Round \times SR^+$ equals 0.208 for CAR$(0, +1)$ and -0.346 for CAR$(2, +75)$. Because SR^+ is standardized within the positive-surprise sample, these estimates imply that a one-standard-deviation larger beat raises announcement CAR by an extra 0.21 percentage points for round-target firms relative to other firms, about one-fourth of the baseline SR^+ slope (0.838). Over the subsequent seventy-five trading days, that same one-standard-deviation increment is associated with 0.35 percentage points lower CAR for round-target firms. These estimates line up with the prediction above: relative to other firms, round-target firms react more thoroughly on the announcement day, and leave less of the positive surprise to be resolved in subsequent drift; if anything, some of the extra announcement reaction comes back later. The level coefficient on *Round* is negative in both windows, which fits the idea that a conspicuous round target raises the bar: merely meeting it looks less impressive.

[27] Relative to the trading-intensity specifications, I omit the FE_q interactions that are specific to the absolute-surprise trading regressions.

[28] Because SR^- is non-positive, a positive interaction in the miss sample means that round-target firms fall more as the miss gets larger.

Panel B (Surprise ≤ 0) does not deliver the same result for misses. $Round \times SR^{-}$ is small and insignificant in all four columns (for example, 0.050 with controls in the announcement window).

Together, the trading and return evidence is consistent with round consensus forecasts drawing more attention to earnings news when the news is large or positive. The tests do not isolate attention from other reasons a round target might matter, so I treat this evidence as secondary to the account-level results.

The Internet Appendix reports robustness checks. The trading-intensity result holds with firm and firm-by-year fixed effects (Table IA.22) and when announcement-window trading is measured relative to pre-announcement baselines (Table IA.23). It is weaker when surprises are scaled by price or by earnings volatility (Table IA.24), which is potentially consistent with investors responding to the surprise measures they actually see. The beat-sample return pattern holds under alternative round definitions, windows, and market-adjustment choices and under exact matching, propensity-score matching, and entropy balancing (Table IA.26, Table IA.27, Table IA.28). It is absent in pre-announcement windows (Table IA.25), survives controls for management guidance and just-beating the consensus by a penny (Table IA.29), and is not absorbed by interactions with analyst coverage and forecast dispersion (Table IA.31). The miss-sample null is not overturned by short-sale or attention moderators (Table IA.30). Internet Appendix Section VI also tests whether round-target announcers draw attention away from same-day non-round peers.

7 Conclusion

This paper studies how the cognitive accessibility of a reference point shapes trading behavior. Because memory is constrained, reference points matter only to the extent that investors can encode them fluently and retrieve them with ease. I use round numbers as a proxy for reference-point cognitive accessibility.

At the account level, round purchase prices amplify the disposition effect. The result survives rich fixed effects, falsification tests, and a stock-split design that disrupts post-split roundness, and it is not explained by taxes, rank-based selling, rebalancing, concentration, or liquidity. One interpretation is that when the cost basis is easier to process and remember, the gain-loss distinction becomes sharper and realization more asymmetric.

At the market level, round consensus EPS forecasts are associated with more trading when surprises are large, and positive surprises are incorporated more thoroughly at announcement with less subsequent drift.

The two settings point to the same conclusion. Reference dependence in financial markets may depend not only on what the reference point is, but on how easily it can be processed and remembered.

References

ALTER, A. L., AND D. M. OPPENHEIMER (2006): "Predicting Short-Term Stock Fluctuations by Using Processing Fluency," *Proceedings of the National Academy of Sciences*, 103, 9369–9372, 10.1073/pnas.0601071103.

AN, L., J. ENGELBERG, M. HENRIKSSON, B. WANG, AND J. WILLIAMS (2024): "The portfolio-driven disposition effect," *The Journal of Finance*, 79, 3459–3495.

ANDRIES, M., M. BONELLI, AND D. A. SRAER (2024): "Financial Advisors and Investors' Bias," *Available at SSRN*.

ARGYLE, B. S., T. D. NADAULD, AND C. J. PALMER (2020): "Monthly payment targeting and the demand for maturity," *The Review of Financial Studies*, 33, 5416–5462.

BARBER, B. M., AND T. ODEAN (2000): "Trading is hazardous to your wealth: The common stock investment performance of individual investors," *The journal of Finance*, 55, 773–806.

——— (2008): "All that glitters: The effect of attention and news on the buying behavior of individual and institutional investors," *The review of financial studies*, 21, 785–818.

——— (2013): "The behavior of individual investors," in *Handbook of the Economics of Finance*, Volume 2, 1533–1570: Elsevier.

BARBERIS, N., AND W. XIONG (2009): "What drives the disposition effect? An analysis of a long-standing preference-based explanation," *the Journal of Finance*, 64, 751–784.

——— (2012): "Realization utility," *Journal of Financial Economics*, 104, 251–271.

BARLOW, H. B. ET AL. (1961): "Possible principles underlying the transformation of sensory messages," *Sensory communication*, 1, 217–233.

BEN-DAVID, I., AND D. HIRSHLEIFER (2012): "Are investors really reluctant to realize their losses? Trading responses to past returns and the disposition effect," *The Review of Financial Studies*, 25, 2485–2532.

BHATTACHARYA, U., C. W. HOLDEN, AND S. JACOBSEN (2012): "Penny wise, dollar foolish: Buy–sell imbalances on and around round numbers," *Management Science*, 58, 413–431.

BHOJRAJ, S., P. HRIBAR, M. PICCONI, AND J. MCINNIS (2009): "Making Sense of Cents: An Examination of Firms That Marginally Miss or Beat Analyst Forecasts," *Journal of Finance*, 64, 2361–2388, 10.1111/j.1540-6261.2009.01503.x.

BIRRU, J. (2015): "Confusion of confusions: A test of the disposition effect and momentum," *The Review of Financial Studies*, 28, 1849–1873.

BOEHMER, E., C. M. JONES, X. ZHANG, AND X. ZHANG (2021): "Tracking Retail Investor Activity," *Journal of Finance*, 76, 2249–2305, 10.1111/jofi.13033.

BREWER, W. F., AND J. C. TREYENS (1981): "Role of schemata in memory for places," *Cognitive psychology*, 13, 207–230.

CHANG, T. Y., D. H. SOLOMON, AND M. M. WESTERFIELD (2016): "Looking for someone to blame: Delegation, cognitive dissonance, and the disposition effect," *The Journal of Finance*, 71, 267–302.

CHARLES, C., C. FRYDMAN, AND M. KILIC (2024): "Insensitive investors," *The Journal of Finance*, 79, 2473–2503.

CONRAD, J., B. CORNELL, AND W. R. LANDSMAN (2002): "When Is Bad News Really Bad News?" *Journal of Finance*, 57, 2507–2532, 10.1111/1540-6261.00504.

DHAR, R., AND N. ZHU (2006): "Up close and personal: Investor sophistication and the disposition effect," *Management science*, 52, 726–740.

D'ACUNTO, F., N. PRABHALA, AND A. G. ROSSI (2019): "The promises and pitfalls of robo-advising," *The Review of Financial Studies*, 32, 1983–2020.

ENKE, B., AND T. GRAEBER (2023): "Cognitive uncertainty," *The Quarterly Journal of Economics*, 138, 2021–2067.

ENKE, B., T. GRAEBER, R. OPREA, AND J. YANG (2024): "Behavioral attenuation," technical report, National Bureau of Economic Research.

FENG, L., AND M. S. SEASHOLES (2005): "Do investor sophistication and trading experience eliminate behavioral biases in financial markets?" *Review of finance*, 9, 305–351.

FRAZZINI, A. (2006): "The disposition effect and underreaction to news," *The Journal of Finance*, 61, 2017–2046.

FRYDMAN, C., N. BARBERIS, C. CAMERER, P. BOSSAERTS, AND A. RANGEL (2014): "Using neural data to test a theory of investor behavior: An application to realization utility," *The Journal of finance*, 69, 907–946.

FRYDMAN, C., S. M. HARTZMARK, AND D. H. SOLOMON (2018): "Rolling mental accounts," *The Review of Financial Studies*, 31, 362–397.

FRYDMAN, C., AND L. J. JIN (2022): "Efficient coding and risky choice," *The Quarterly Journal of Economics*, 137, 161–213.

——— (2025): "Adaptation through Experience or Description? Evidence from an Experiment on Efficient Coding," *Evidence from an Experiment on Efficient Coding (August 16, 2025)*.

FRYDMAN, C., AND B. WANG (2020): "The impact of salience on investor behavior: Evidence from a natural experiment," *The Journal of Finance*, 75, 229–276.

GENESOVE, D., AND C. MAYER (2001): "Loss aversion and seller behavior: Evidence from the housing market," *The quarterly journal of economics*, 116, 1233–1260.

GRAEBER, T., C. ROTH, AND F. ZIMMERMANN (2024): "Stories, statistics, and memory," *The Quarterly Journal of Economics*, qjae020.

GRAF, P., AND G. MANDLER (1984): "Activation makes words more accessible, but not necessarily more retrievable," *Journal of verbal learning and verbal behavior*, 23, 553–568.

GREEN, T. C., AND R. JAME (2013): "Company Name Fluency, Investor Recognition, and Firm Value," *Journal of Financial Economics*, 109, 813–834, 10.1016/j.jfineco.2013.04.007.

GRINBLATT, M., AND M. KELOHARJU (2001): "What makes investors trade?" *The journal of Finance*, 56, 589–616.

HARTZMARK, S. M. (2015): "The worst, the best, ignoring all the rest: The rank effect and trading behavior," *The Review of Financial Studies*, 28, 1024–1059.

HEATH, C., AND D. HEATH (2007): *Made to stick: Why some ideas survive and others die*, Random House.

HEATH, C., S. HUDDART, AND M. LANG (1999): "Psychological factors and stock option exercise," *The Quarterly Journal of Economics*, 114, 601–627.

HEIMER, R. Z. (2016): "Peer pressure: Social interaction and the disposition effect," *The Review of Financial Studies*, 29, 3177–3209.

HIRSHLEIFER, D., S. S. LIM, AND S. H. TEOH (2009): "Driven to distraction: Extraneous events and underreaction to earnings news," *The journal of finance*, 64, 2289–2325.

HUNT, E. (1977): "Scripts, plans, goals, and understanding: An inquiry into human knowledge structures."

IMAS, A. (2016): "The realization effect: Risk-taking after realized versus paper losses," *American Economic Review*, 106, 2086–2109.

INGERSOLL, J. E., AND L. J. JIN (2013): "Realization utility with reference-dependent preferences," *The Review of Financial Studies*, 26, 723–767.

JOHNSON, T. L., J. KIM, AND E. C. SO (2020): "Expectations Management and Stock Returns," *Review of Financial Studies*, 33, 4580–4626, 10.1093/rfs/hhz141.

KAHNEMAN, D., AND A. TVERSKY (1979): "Prospect theory: An analysis of decision under risk," *Econometrica*, 47, 363–391.

KAUSTIA (2010a): *Behavioral finance: investors, corporations, and markets*, Volume 6: John Wiley & Sons.

KENSINGER, E. A., AND D. L. SCHACTER (2008): "Memory and emotion," *Handbook of emotions*, 3, 601–617.

LANCASTER, T. (2000): "The incidental parameter problem since 1948," *Journal of econometrics*, 95, 391–413.

MAĆKOWIAK, B., F. MATĚJKA, AND M. WIEDERHOLT (2023): "Rational inattention: A review," *Journal of Economic Literature*, 61, 226–273.

MENG, J., AND X. WENG (2018): "Can prospect theory explain the disposition effect? A new perspective on reference points," *Management Science*, 64, 3331–3351.

ODEAN, T. (1998): "Are investors reluctant to realize their losses?" *The Journal of finance*, 53, 1775–1798.

OPREA, R. (2020): "What makes a rule complex?" *American economic review*, 110, 3913–3951.

POPE, D., AND U. SIMONSOHN (2011): "Round numbers as goals: Evidence from baseball, SAT takers, and the lab," *Psychological science*, 22, 71–79.

PURSIAINEN, V. (2024): "Inaccurate borrower information and credit risk: Evidence from marketplace loans," *The Review of Corporate Finance Studies*, cfae025.

RANYARD, R., L. HINKLEY, J. WILLIAMSON, AND S. MCHUGH (2006): "The role of mental accounting in consumer credit decision processes," *Journal of Economic Psychology*, 27, 571–588.

SCHINDLER, R. M., AND P. N. KIRBY (1997): "Patterns of rightmost digits used in advertised prices: implications for nine-ending effects," *Journal of Consumer Research*, 24, 192–201.

SERU, A., T. SHUMWAY, AND N. STOFFMAN (2010): "Learning by trading," *The Review of Financial Studies*, 23, 705–739.

SHAPIRA, Z., AND I. VENEZIA (2001): "Patterns of behavior of professionally managed and independent investors," *Journal of Banking & Finance*, 25, 1573–1587.

SHEPARD, R. N. (1967): "Recognition memory for words, sentences, and pictures," *Journal of verbal Learning and verbal Behavior*, 6, 156–163.

SHEPARD, R. N., AND L. A. COOPER (1986): *Mental images and their transformations.*, The MIT Press.

STANDING, L. (1973): "Learning 10000 pictures," *Quarterly Journal of Experimental Psychology*, 25, 207–222.

WACHTER, J. A., AND M. J. KAHANA (2024): "A retrieved-context theory of financial decisions," *The Quarterly Journal of Economics*, 139, 1095–1147.

WEBER, M., AND C. F. CAMERER (1998): "The disposition effect in securities trading: An experimental analysis," *Journal of Economic Behavior & Organization*, 33, 167–184.

Figures and Tables

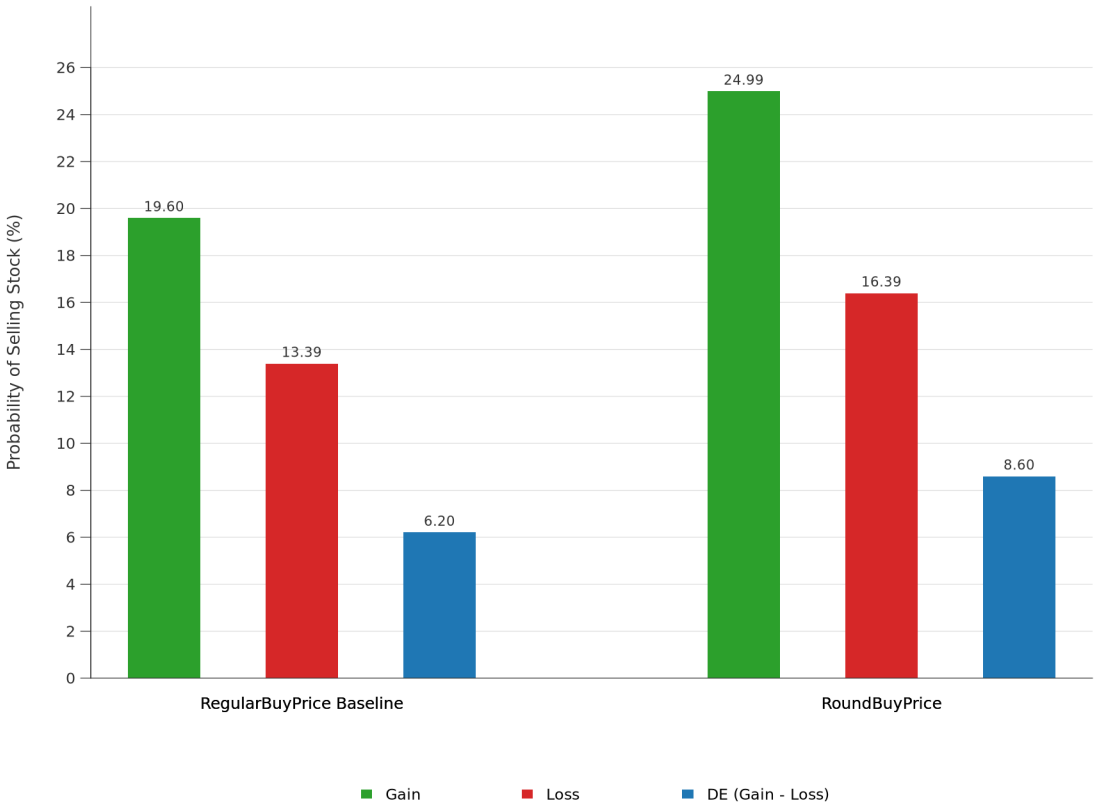


Figure 1: **Propensity to Sell Gains and Buy Price** Figure 1 presents the probability that an investor sells a position (including partial sales) as a function of the position's current performance and the type of purchase price. The figure is built from the subset of accounts that have purchased at least one round-number price at some point; accounts that never bought at a round price are excluded so that both groups are drawn from the same investor pool. The left group, RegularBuyPrice Baseline, pools positions whose most recent purchase price is not a round number; the right group, RoundBuyPrice, pools positions purchased at a round number. Within each group, the green bar gives Pr[sell | gain] and the red bar gives Pr[sell | loss], where gains (losses) are defined as strictly positive (negative) paper returns relative to the most recent purchase price; ties are omitted. The blue bar labeled DE (Gain−Loss) is my univariate measure of the disposition effect, Pr[sell | gain] − Pr[sell | loss], reported in percentage points.

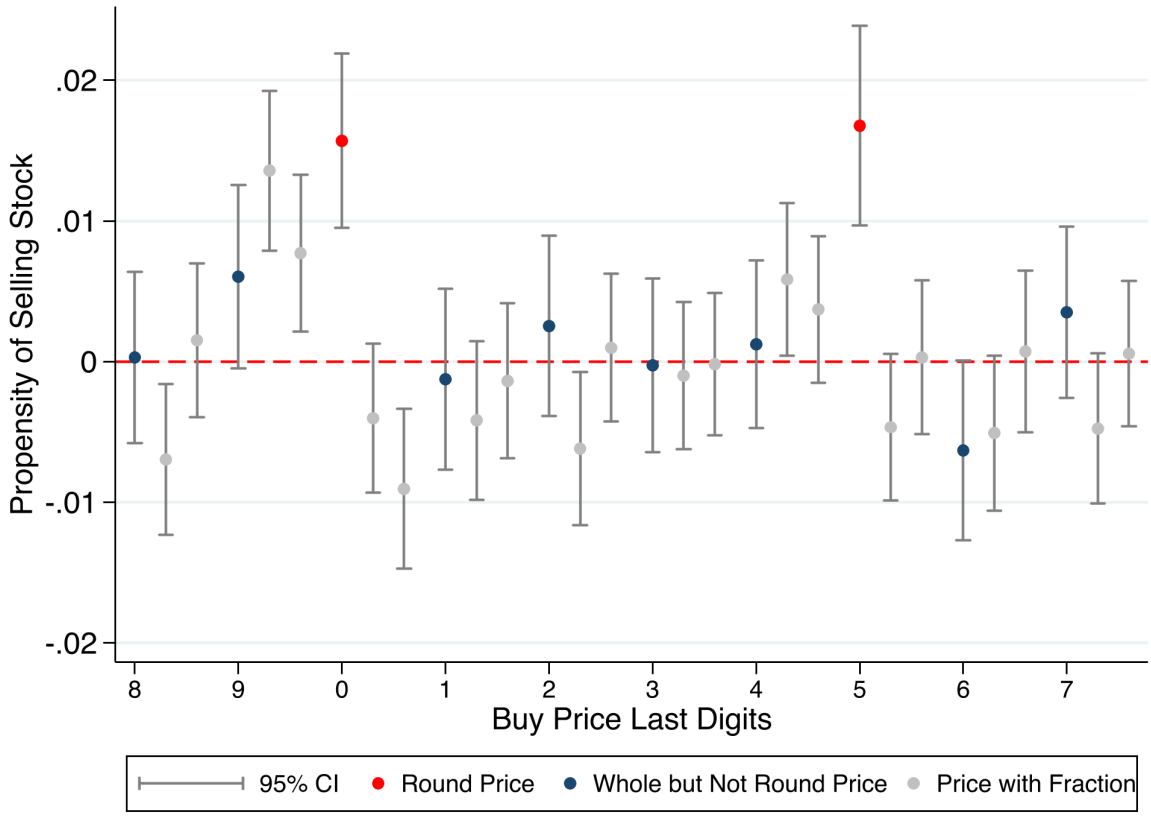


Figure 2: **Propensity to Sell Gains and Buy Price Roundness**

This figure plots coefficient estimates for β_2 from

$$Sell_{i,j,t} = \alpha + \beta_1 Gain_{i,j,t-1} + \beta_2 Gain_{i,j,t-1} \times PriceNumCat_X_Y_{i,j,t} + \beta_3 PriceNumCat_X_Y_{i,j,t} + X\beta' + FE_i + FE_{j\times t} + \varepsilon_{i,j,t},$$

where $PriceNumCat_X_Y$ equals one when the purchase price's last whole-number digit is X ($X \in \{0, 1, \ldots, 9\}$) and the fractional part falls in bin Y ($Y \in \{[0, \frac{1}{2}), [\frac{1}{2}, 1)\}$). Controls X follow Ben-David and Hirshleifer (2012): $\log(BuyPrice)$, $Return(+)$, $Return(-)$, $TimeOwned$, $Volatility(+)$, and $Volatility(-)$. Vertical bars are 95% confidence intervals; standard errors are clustered by investor and day. See Section 3.2 and Equation (4) for variable construction.

Table 1: **Summary Statistics**

Panel A: Sample Overview								
	N	Mean	SD	5%	25%	50%	75%	95%
Observations	1,393,400	.	.	.	.	.	.	.
Number of Sell Dates	1,491	.	.	.	.	.	.	.
Number of Sells Per Year	83,517	2.865	4.059	1	1	2	3	9
Number of Stocks Held	239,245	6.332	8.262	2	3	4	7	17
Holding Period (Days)	1,393,400	222.180	255.566	7	42	124	306	789
Total Assets (USD)	239,245	82,792.37	224,080.9	4,729.852	13,671.5	31,125.00	77,294	305,375

Number of Sell Dates is the count of distinct trading dates on which there is at least one sale. *Number of Stocks Held* is computed at the account–day level: for each sell date, I count the number of unique stocks in the portfolio. *Holding Period* is measured by taking the differences between the purchase date and the sell date. *Total Assets (USD)* is the account–day market value of the portfolio. *Number of Sells Per Year* is computed at the account–year level. Reported quantiles are the 5th, 25th, 50th, 75th, and 95th percentiles.

Table 1: **Summary Statistics (Cont.)**

Panel B: Purchase Price ($) and Capital Gain (%)								
	N	Mean	SD	5%	25%	50%	75%	95%
Purchase Price ($)								
Purchase Price (All)	1,393,400	30.993	133.238	4.125	13.125	24	40	72.375
Round Purchase Price	64,627	31.708	21.065	5	15	25	45	75
Round Purchase Price (Ends in 0)	31,158	33.337	21.092	10	20	30	50	80
Round Purchase Price (Ends in 5)	33,469	30.192	20.926	5	15	25	45	75
Capital Gain (%)								
Return (All)	1,393,400	9.985	57.011	-47.143	-12.214	2.183	19.792	86.173
Positive Return	758,147	35.512	65.094	1.434	6.967	17.424	39.610	124.390
Negative Return	633,568	-20.534	19.144	-61.111	-29.808	-14.286	-5.806	-1.250
Round Return	64,627	9.696	56.162	-48.333	-14	1.655	19.988	90.833
Positive Round Return	34,135	37.286	63.493	1.500	7.500	18.250	41	132.500
Negative Round Return	30,407	-21.249	19.126	-60.909	-31	-15.250	-6.250	-1.250

Round Purchase Price (Ends in 0) is the purchase price that has no fractional part (cents = 00) and the integer amount ends in 0 (i.e., the price is an integer multiple of 10). *Round Purchase Price (Ends in 5)* is the purchase price has no fractional part (cents = 00) and the integer amount ends in 5. *Round Purchase Price* is the combination of these 2 above-mentioned round prices. Capital gain is computed as the difference between the current market price and the purchase price, divided by the purchase price. *Positive Return* (*Negative Return*) refers to returns when the capital gain is positive (negative). *Positive Round Return* (*Negative Round Return*) refers to positive (negative) capital gains for which the purchase price has no fractional part (cents = 00) and the integer amount ends in 0 or 5. Reported quantiles are the 5th, 25th, 50th, 75th, and 95th percentiles.

Table 2: Market-Level Earnings Announcement Sample

Panel A: Sample Overview

	Value
Firm-quarter announcements (N)	357,759
Distinct PERMNOs	11,108
Sample period	1990–2024
Round-target share	25.2% ($N = 89,992$)
Positive-surprise share	55.2% ($N = 197,596$)

Panel B: Key Variables

	All	Round	Non-round
N	357,759	89,992	267,767
Median consensus EPS	0.13 (4.23)	-0.44 (7.72)	0.32 (1.93)
Percent surprise	-5.58 (113.67)	-8.72 (105.11)	-4.53 (116.39)
CAR$(0,+1)$ (%)	0.011 (7.810)	-0.198 (7.897)	0.082 (7.779)
CAR$(2,+75)$ (%)	1.397 (23.247)	0.492 (24.833)	1.701 (22.680)

The unit of observation is a firm-quarter earnings announcement. The sample requires a valid pre-announcement IBES median consensus (`medest`), actual EPS, percent surprise, roundness eligibility, and at least two return days in CAR$(0,+1)$. *Round* $=1$ when `medest`, rounded to two decimals, is a nonzero multiple of \$0.05 (with $|\texttt{medest}| \leq 100$). Percent surprise is $(Actual - \texttt{medest})/|\texttt{medest}|$, winsorized at the 1st and 99th percentiles. CARs are market-adjusted and winsorized at the 1st and 99th percentiles. Panel B reports means with standard deviations in parentheses. Overall $N = 357,759$.

Table 3: **Propensity to Sell Gains and Round Buy Price**

The table presents estimation results from an OLS regression, where the dependent variable equals 1 if a stock is sold (fully or partially) on a given day and 0 otherwise. *Gain* is a dummy variable equal to 1 if the return since purchase is positive and 0 otherwise. *RoundBuyPrice* is a dummy variable equal to 1 if the purchase price has no fractional part (cents = 00) and the integer amount ends in 0 or 5 (i.e., the price is an integer multiple of 5); otherwise, it is 0. $Log(BuyPrice)$ is the natural logarithm of the most recent purchase price. $Volatility(-)$ ($Volatility(+)$) is the stock volatility calculated using daily returns over the prior 252 days if the return since purchase is negative (positive) and 0 otherwise. $\sqrt{TimeOwned}$ is the square root of the number of days since purchase. $Return(-)$ ($Return(+)$) is the return since purchase if the return since purchase is negative (positive) and 0 otherwise. All coefficients are multiplied by 100. t-statistics in parentheses; standard errors are double-clustered at the investor and day level. ***, ** and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

	(1)	(2)	(3)	(4)	(5)
RoundBuyPrice	-0.379*	-0.603*	-1.143***	-1.018*	-0.907***
	(-1.68)	(-1.70)	(-3.30)	(-1.90)	(-3.03)
Gain	7.655***	3.960***	3.886***	4.143***	4.787***
	(109.70)	(16.88)	(16.38)	(15.56)	(15.57)
RoundBuyPrice × Gain	1.156***	1.445***	1.997***	1.233**	1.659***
	(3.57)	(2.84)	(3.90)	(2.29)	(3.58)
Log(BuyPrice)		0.761***	-0.046	5.495***	1.224***
		(3.58)	(-0.12)	(12.06)	(5.73)
Return(+)		-0.044	-0.401***	0.358	0.237
		(-0.19)	(-2.88)	(1.50)	(1.52)
Return(−)		-7.861***	-7.176***	-0.635	-3.176***
		(-7.11)	(-5.87)	(-0.59)	(-3.03)
$\sqrt{TimeOwned}$		-0.308***	-0.039***	2.116***	-0.104***
		(-16.79)	(-2.71)	(30.56)	(-6.92)
Volatility(+)		203.258***	62.337	147.506***	193.198***
		(14.38)	(1.24)	(14.32)	(20.71)
Volatility(−)		-16.043	-121.542**	-21.692**	-24.124***
		(-1.56)	(-2.41)	(-2.52)	(-3.15)
Constant	16.701***	16.653***			
	(337.88)	(15.33)			
Investor FE	No	No	Yes	No	No
SellDate FE	No	No	No	Yes	No
Stock FE	No	No	No	No	Yes
Investor × SellDate FE	No	No	No	No	Yes
Investor × Stock FE	No	No	No	Yes	No
Stock × SellDate FE	No	No	Yes	No	No
Observations	1,393,400	1,393,400	1,393,400	1,393,400	1,393,400
Adjusted R^2	0.009	0.017	0.190	0.297	0.077

t-statistics in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 4: **Whole-but-Non-Round Buy Prices**

This table investigates whether whole-number purchase prices that do not end in 0 or 5 are associated with the same selling behavior as round purchase prices. *RoundBuyPrice* is a dummy variable equal to 1 if the purchase price has no fractional part (cents = 00) and the integer amount ends in 0 or 5 (i.e., the price is an integer multiple of 5); 0 otherwise. *WholeBuyPrice* is a dummy variable equal to 1 if the purchase price has no fractional part (cents = 00) and the integer amount does not end in 0 or 5 (i.e., the last digit is 1–4 or 6–9); 0 otherwise. Other variable definitions are provided in the caption of Table 3. All coefficients are multiplied by 100. t-statistics in parentheses; standard errors are double-clustered at the investor and day level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)
RoundBuyPrice	-0.380*	-0.645*	-1.249***	-1.070**	-1.021***
	(-1.68)	(-1.75)	(-3.60)	(-1.97)	(-3.40)
Gain	7.559***	3.814***	3.657***	4.092***	4.579***
	(98.44)	(15.74)	(14.83)	(15.17)	(14.76)
RoundBuyPrice × Gain	1.252***	1.585***	2.228***	1.283**	1.867***
	(3.84)	(3.11)	(4.34)	(2.35)	(4.00)
WholeBuyPrice	-0.007	-0.231	-0.579***	-0.263	-0.613***
	(-0.05)	(-1.09)	(-2.89)	(-0.88)	(-3.83)
WholeBuyPrice × Gain	0.567***	0.815***	1.269***	0.281	1.152***
	(3.08)	(2.88)	(4.28)	(0.99)	(4.76)
Log(BuyPrice)		0.757***	-0.029	5.506***	1.234***
		(3.52)	(-0.08)	(12.06)	(5.77)
Return(+)		-0.045	-0.400***	0.357	0.237
		(-0.19)	(-2.88)	(1.50)	(1.52)
Return(−)		-7.870***	-7.199***	-0.638	-3.190***
		(-7.12)	(-5.89)	(-0.59)	(-3.05)
$\sqrt{TimeOwned}$		-0.308***	-0.038***	2.116***	-0.104***
		(-16.81)	(-2.69)	(30.57)	(-6.90)
Volatility(+)		203.325***	62.146	147.606***	193.687***
		(14.34)	(1.23)	(14.33)	(20.75)
Volatility(−)		-16.246	-122.138**	-21.689**	-24.083***
		(-1.58)	(-2.42)	(-2.52)	(-3.14)
Constant	16.702***	16.710***			
	(306.14)	(15.32)			
Investor FE	No	No	Yes	No	No
SellDate FE	No	No	No	Yes	No
Stock FE	No	No	No	No	Yes
Investor × SellDate FE	No	No	No	No	Yes
Investor × Stock FE	No	No	No	Yes	No
Stock × SellDate FE	No	No	Yes	No	No
Observations	1,393,400	1,393,400	1,393,400	1,393,400	1,393,400
Adjusted R^2	0.009	0.017	0.190	0.297	0.077

t-statistics in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 5: **Falsification Tests**

Panel A: Buy Price Ends in 1 or 6

This table reports results from a falsification test that replaces round-number purchase prices with adjacent non-round integer prices. Specifically, *BuyPriceEnd*1*or*6 and *BuyPriceEnd*9*or*4 are dummy variables constructed using purchase prices ending in 1 or 6 (Panel A), and 9 or 4 (Panel B), respectively. Other variable definitions are provided in the caption of Table 3. All coefficients are multiplied by 100. t-statistics in parentheses; standard errors are double-clustered at the investor and day level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)
BuyPriceEnd1or6	0.101	0.021	-0.176	0.059	-0.388
	(0.39)	(0.06)	(-0.44)	(0.10)	(-1.23)
Gain	7.705***	4.021***	3.967***	4.226***	4.838***
	(110.97)	(17.10)	(16.63)	(15.79)	(15.73)
BuyPriceEnd1or6 × Gain	0.054	0.185	0.453	-0.688	0.710
	(0.15)	(0.37)	(0.80)	(-1.29)	(1.54)
Log(BuyPrice)		0.761***	-0.046	5.492***	1.221***
		(3.59)	(-0.12)	(12.08)	(5.72)
Return(+)		-0.044	-0.402***	0.359	0.237
		(-0.18)	(-2.90)	(1.50)	(1.52)
Return(−)		-7.842***	-7.113***	-0.621	-3.151***
		(-7.09)	(-5.81)	(-0.57)	(-3.01)
$\sqrt{TimeOwned}$		-0.308***	-0.039***	2.116***	-0.104***
		(-16.79)	(-2.70)	(30.57)	(-6.91)
Volatility(+)		203.242***	62.210	147.428***	193.143***
		(14.39)	(1.24)	(14.31)	(20.70)
Volatility(−)		-15.879	-121.485**	-21.654**	-24.008***
		(-1.55)	(-2.41)	(-2.52)	(-3.13)
Constant	16.679***	16.620***			
	(339.50)	(15.26)			
Investor FE	No	No	Yes	No	No
SellDate FE	No	No	No	Yes	No
Stock FE	No	No	No	No	Yes
Investor × SellDate FE	No	No	No	No	Yes
Investor × Stock FE	No	No	No	Yes	No
Stock × SellDate FE	No	No	Yes	No	No
Observations	1,393,400	1,393,400	1,393,400	1,393,400	1,393,400
Adjusted R^2	0.009	0.017	0.190	0.297	0.077

t-statistics in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Panel B: Buy Price Ends in 9 or 4

	(1)	(2)	(3)	(4)	(5)
BuyPriceEnd9or4	-0.134	-0.173	-0.117	1.153**	0.285
	(-0.55)	(-0.48)	(-0.31)	(2.20)	(0.99)
Gain	7.675***	3.992***	3.962***	4.199***	4.850***
	(110.41)	(17.00)	(16.62)	(15.62)	(15.72)
BuyPriceEnd9or4 × Gain	0.855**	0.925*	0.578	0.017	0.371
	(2.43)	(1.87)	(1.02)	(0.03)	(0.83)
Log(BuyPrice)		0.761***	-0.045	5.488***	1.222***
		(3.59)	(-0.12)	(12.07)	(5.72)
Return(+)		-0.043	-0.401***	0.359	0.239
		(-0.18)	(-2.89)	(1.50)	(1.53)
Return(−)		-7.845***	-7.115***	-0.614	-3.152***
		(-7.09)	(-5.81)	(-0.57)	(-3.01)
$\sqrt{TimeOwned}$		-0.308***	-0.039***	2.116***	-0.104***
		(-16.79)	(-2.70)	(30.58)	(-6.91)
Volatility(+)		203.234***	62.111	147.554***	193.173***
		(14.39)	(1.23)	(14.32)	(20.71)
Volatility(−)		-15.911	-121.588**	-21.544**	-23.990***
		(-1.55)	(-2.41)	(-2.51)	(-3.13)
Constant	16.688***	16.629***			
	(339.03)	(15.30)			
Investor FE	No	No	Yes	No	No
SellDate FE	No	No	No	Yes	No
Stock FE	No	No	No	No	Yes
Investor × SellDate FE	No	No	No	No	Yes
Investor × Stock FE	No	No	No	Yes	No
Stock × SellDate FE	No	No	Yes	No	No
Observations	1,393,400	1,393,400	1,393,400	1,393,400	1,393,400
Adjusted R^2	0.009	0.017	0.190	0.297	0.077

t-statistics in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 6: **Stock Split and Memory Disruption**

Regressions are estimated at the investor-stock-day level where the underlying security experienced at least one stock split from January 1991 to December 1996. *RoundBuyPrice_Always* is a dummy variable equal to 1 if both (i) the raw purchase price and (ii) the split-adjusted purchase price (expressed on the post-split share basis) have no fractional part (cents = 00) and the integer amount ends in 0 or 5; 0 otherwise. *RoundBuyPrice_Breaks* is a dummy variable equal to 1 if (i) the raw purchase price has no fractional part (cents = 00) and the integer amount ends in 0 or 5, but (ii) the split-adjusted purchase price (expressed on the post-split share basis) becomes non-round; 0 otherwise. Other variable definitions are provided in the caption of Table 3. All coefficients are multiplied by 100. t-statistics in parentheses; standard errors are double-clustered at the investor and stock level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)
RoundBuyPrice_Always	0.107	-0.426		0.106	-0.426
	(0.18)	(-1.01)		(0.18)	(-1.01)
Gain	0.109***	0.053**	0.110***	0.107***	0.052*
	(6.16)	(1.99)	(6.14)	(5.97)	(1.93)
RoundBuyPrice_Always × Gain	0.279***	0.292***		0.281***	0.294***
	(2.78)	(2.88)		(2.80)	(2.90)
RoundBuyPrice_Breaks			0.596	0.606	0.521*
			(1.46)	(1.51)	(1.72)
RoundBuyPrice_Breaks × Gain			0.038	0.041	0.044
			(0.96)	(1.03)	(1.08)
Log(BuyPrice)		0.049			0.051
		(0.74)			(0.76)
Return(+)		0.007			0.007
		(0.52)			(0.52)
Return(−)		0.012			0.011
		(0.12)			(0.10)
$\sqrt{TimeOwned}$		0.047***			0.047***
		(9.67)			(9.65)
Volatility(+)		1.721			1.718
		(1.63)			(1.63)
Volatility(−)		0.978			0.975
		(1.63)			(1.62)
Stock × SellDate FE	Yes	Yes	Yes	Yes	Yes
Stock × Investor FE	Yes	Yes	Yes	Yes	Yes
Observations	5,082,807	4,998,917	5,082,807	5,082,807	4,998,917
Adjusted R^2	0.023	0.024	0.023	0.023	0.024

t-statistics in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 7: **Alternative Mechanisms**

This table examines whether alternative mechanisms can explain the main finding. In Panel A columns 1 and 2, I run the test within tax-exempt accounts and taxable accounts, respectively. Following Hartzmark (2015), in column 3, I add indicator variables for the best and worst performing stocks in an investor's portfolio. Following An et al. (2024), in column 4, I include rank indicator variables for the 15 best and 15 worst performing stocks. In column 5, the dependent variable is defined differently: it is equal to 1 for full liquidations and 0 otherwise. For Panel B, columns 1 and 2 test whether the results are explained by portfolio concentration (HHI), and HHI is the HHI of portfolio holdings. Columns 3–5 examine liquidity by splitting holdings into high, median, and low-liquidity groups using a proxy based on the daily high ask and low bid. All coefficients are multiplied by 100. t-statistics in parentheses; standard errors are double-clustered at the investor and day level. ***, ** and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

Panel A: Tax Incentives, Rank Effect and Rebalancing

	Tax Incentives?		Rank Effect?		Rebalancing?
	Tax-exempt	Taxable	1 best worse	15 best worse	Full Liquidation
RoundBuyPrice	-4.485**	-1.046***	-1.072***	-0.968***	-1.154***
	(-2.12)	(-2.85)	(-3.19)	(-2.88)	(-3.68)
Gain	3.329***	3.740***	3.622***	3.619***	2.544***
	(3.13)	(16.37)	(17.59)	(17.62)	(15.22)
RoundBuyPrice × Gain	9.376***	1.896***	1.845***	1.806***	2.263***
	(3.14)	(3.38)	(3.73)	(3.68)	(4.82)
Controls	Yes	Yes	Yes	Yes	Yes
Investor FE	Yes	Yes	Yes	Yes	Yes
Stock × SellDate FE	Yes	Yes	Yes	Yes	Yes
Observations	140,362	1,253,038	1,393,400	1,393,400	1,393,400
Adjusted R^2	0.223	0.183	0.205	0.215	0.194

t-statistics in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Panel B: HHI and Liquidity

	HHI		Liquidity		
	High	Low	High	Median	Low
RoundBuyPrice	-0.847	-0.987**	-1.413*	-0.896**	-1.485**
	(-1.31)	(-2.36)	(-1.91)	(-1.97)	(-2.12)
Gain	6.634***	1.414***	2.029***	4.176***	4.960***
	(17.99)	(5.81)	(5.84)	(15.41)	(10.65)
RoundBuyPrice × Gain	1.711*	1.340**	3.392***	1.455**	1.475
	(1.94)	(2.05)	(3.39)	(2.16)	(1.29)
Controls	Yes	Yes	Yes	Yes	Yes
Investor FE	Yes	Yes	Yes	Yes	Yes
Stock × SellDate FE	Yes	Yes	Yes	Yes	Yes
Observations	695,808	697,583	329,934	698,434	365,001
Adjusted R^2	0.176	0.107	0.202	0.193	0.192

t-statistics in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 8: **Subsample Analysis**

This table reports the results from the Table 3 Column 3 regression using different subsamples based on the investor's professional status (follow the specifications in Dhar and Zhu (2006)), age, trading frequency (High/Med/Low terciles), and stock holding period (in days), respectively. The dependent variable equals 1 if a stock is sold (fully or partially) on a given day and 0 otherwise. *Gain* is a dummy variable equal to 1 if the return since purchase is positive and 0 otherwise. *RoundBuyPrice* is a dummy variable equal to 1 if the purchase price has no fractional part (cents = 00) and the integer amount ends in 0 or 5 (i.e., the price is an integer multiple of 5); otherwise, it is 0. All coefficients are multiplied by 100. t-statistics in parentheses; standard errors are double-clustered at the investor and day level. ***, ** and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

Panel A: Professional Status and Age

	Professional		Age		
	Yes	No	16–34	35–54	≥55
RoundBuyPrice	-2.803**	-7.574***	-7.826**	-4.594***	2.541
	(-2.01)	(-2.88)	(-2.44)	(-3.22)	(1.19)
Gain	4.330***	2.619**	4.213***	6.493***	4.450***
	(6.44)	(2.28)	(2.91)	(7.72)	(3.89)
RoundBuyPrice × Gain	2.681	11.197***	13.934***	5.803***	-1.086
	(1.33)	(3.14)	(3.20)	(2.64)	(-0.34)
Controls	Yes	Yes	Yes	Yes	Yes
Investor FE	Yes	Yes	Yes	Yes	Yes
Stock × SellDate FE	Yes	Yes	Yes	Yes	Yes
Observations	222,854	120,473	96,180	188,352	126,748
R^2	0.890	0.936	0.945	0.904	0.931

t-statistics in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Panel B: Trading Frequency and Holding Period

	Trading Frequency			Holding Period	
	High	Med	Low	1 to 60	1 to 250
RoundBuyPrice	-0.876***	-1.232*	0.037	-1.498**	-1.608***
	(-3.35)	(-1.71)	(0.03)	(-2.06)	(-3.83)
Gain	3.838***	5.956***	7.584***	4.652***	4.514***
	(27.03)	(16.86)	(11.37)	(9.86)	(18.92)
RoundBuyPrice × Gain	1.602***	2.577**	0.752	3.571***	2.910***
	(3.85)	(2.56)	(0.39)	(3.19)	(4.83)
Controls	Yes	Yes	Yes	Yes	Yes
Stock FE	Yes	Yes	Yes	Yes	Yes
Investor × SellDate FE	Yes	Yes	Yes	Yes	Yes
Observations	1,018,176	252,614	95,013	449,542	970,688
R^2	0.214	0.241	0.311	0.806	0.723

t-statistics in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 9: **Cox Hazard Model**

The table provides estimation results from a Cox proportional hazard model regression. The dependent variable is a dummy which is equal to 1 if a stock is sold (fully or partially) on a given day, and 0 otherwise. *Gain* is a dummy variable equal to 1 if the return since purchase is positive and 0 otherwise. *RoundBuyPrice* is a dummy variable equal to 1 if the purchase price has no fractional part (cents = 00) and the integer amount ends in 0 or 5 (i.e., the price is an integer multiple of 5); otherwise, it is 0. $Log(BuyPrice)$ is the natural logarithm of the most recent purchase price. $Volatility(-)$ ($Volatility(+)$) is the stock volatility calculated using daily returns over the prior 252 days if the return since purchase is negative (positive) and 0 otherwise. *StockReturn* is the return since purchase. *MarketReturn* is the market return calculated based on the CRSP value-weighted index. t-statistics in parentheses; standard errors are clustered at the investor level. ***, ** and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

	(1)	(2)	(3)	(4)	(5)
Gain	0.423***	0.435***	0.436***	0.389***	0.475***
	(25.98)	(35.59)	(29.67)	(99.81)	(26.26)
RoundBuyPrice	-0.053*	-0.093***	-0.035	-0.017	-0.035
	(-1.89)	(-3.25)	(-1.17)	(-1.24)	(-1.15)
RoundBuyPrice × Gain	0.105***	0.083***	0.066*	0.037**	0.063*
	(3.20)	(2.63)	(1.93)	(2.04)	(1.81)
Log(BuyPrice)	0.272***	0.160***	0.113***		0.090***
	(18.25)	(6.98)	(7.77)		(5.63)
Volatility(+)	17.942***	17.646***	10.486***		9.956***
	(12.38)	(31.51)	(9.26)		(7.98)
Volatility(−)	1.405*	-2.268***	-0.052		-1.495*
	(1.81)	(-3.82)	(-0.07)		(-1.76)
Stock Return	-0.506***	-0.863***	-0.507***		-0.570***
	(-32.55)	(-58.87)	(-33.41)		(-36.40)
Market Return	-10.427***	-15.397***			-13.630***
	(-11.63)	(-13.67)			(-9.59)
Stratified by account	Yes	No	No	No	No
Stratified by stock	No	Yes	No	No	No
Stratified by date	No	No	Yes	No	No
Observations	1,393,400	1,393,400	1,393,400	1,393,400	1,393,400
Pseudo-R^2	0.019	0.014	0.010	0.001	0.005

t-statistics in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 10: **Retail Order Flow and Turnover Around Earnings Announcements**

The table reports OLS estimates for quarterly earnings announcements with valid roundness, consensus, and $[0,+1]$ return data. Cols. (1)–(2) use retail trade flow, $\log(1+(mrbvol+mrsvol)/SHROUT\times 10^4)$ (Boehmer, Jones, Zhang and Zhang 2021); cols. (3)–(4) use $\log(\text{turnover})$, where turnover is daily share volume divided by shares outstanding. Each dependent variable is averaged over trading days $[0,+1]$ relative to the announcement date and winsorized at the 1% and 99% levels. Cols. (1) and (3) include *Round*, SR^{abs}, and $Round\times SR^{abs}$; cols. (2) and (4) add size and book-to-market deciles (within calendar quarter); $\log(1+N_{FEk,t})$; $\log(1+N_{est})$; reporting lag, its square, and its cube; institutional-ownership decile (within calendar quarter, lagged one quarter); earnings volatility and persistence; lagged share turnover; year, month, day-of-week, and Fama–French 10 industry indicators; and each control interacted with FE_q. *Round* equals one when median IBES consensus EPS, rounded to two decimals, is a nonzero multiple of \$0.05. SR^{abs} is the full-sample standardized absolute percent earnings surprise (medest). FE_q is the within-calendar-quarter decile of percent earnings surprise. $N_{FEk,t}$ is the number of $FE_q = k$ earnings announcements on date t; N_{est} is the number of IBES EPS forecasts. t-statistics in parentheses; standard errors clustered by announcement date. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	Retail trade flow		log(Turnover)	
	(1)	(2)	(3)	(4)
Round	0.1906***	0.0566***	-0.0114**	0.0217***
	(16.15)	(7.20)	(-2.14)	(6.39)
SR^{abs}	0.1273***	0.0690***	-0.0044	0.0369***
	(18.78)	(15.27)	(-1.55)	(19.57)
$Round\times SR^{abs}$	0.0812***	0.0471***	0.0252***	0.0454***
	(4.84)	(3.69)	(3.67)	(8.47)
Constant	8.7059***	8.3768***	1.9674***	-0.4718**
	(633.50)	(93.23)	(222.02)	(-2.34)
Controls	No	Yes	No	Yes
Observations	90,696	88,033	361,296	346,313
Adj. R^2	0.012	0.517	0.000	0.595

Table 11: **Announcement Reaction and Drift: Round Forecast Targets and Signed Surprise**

OLS estimates for roundness-eligible earnings announcements with valid CAR and return data. Panel A restricts to Surprise > 0 and Panel B to Surprise ≤ 0. Each panel includes *Round*, the sign-specific standardized surprise ($SR^+ = \max(Surprise, 0)$ in Panel A and $SR^- = \min(Surprise, 0)$ in Panel B, standardized to mean zero and unit variance within the subsample), their interaction, and a constant. Dependent variables are CAR$(0, +1)$ in cols. (1)–(2) and CAR$(2, +75)$ in cols. (3)–(4), market-adjusted, winsorized at the 1% and 99% levels, and scaled to percent. Percent surprise is $(Actual - Consensus)/|Consensus|$ from IBES, winsorized at the 1% and 99% levels. $Round = 1$ when the median IBES consensus EPS, rounded to two decimals, is a nonzero multiple of \$0.05. Cols. (2) and (4) add the controls described in Table 10. t-statistics in parentheses; standard errors clustered by announcement date. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

	CAR$(0, +1)$		CAR$(2, +75)$	
	(1)	(2)	(3)	(4)
Panel A: Surprise > 0				
Round	-0.1873***	-0.0732*	-1.0484***	-0.9469***
	(-4.69)	(-1.82)	(-8.20)	(-7.71)
SR^+	0.8583***	0.8380***	0.7000***	0.4303***
	(35.11)	(33.36)	(9.48)	(6.11)
Round $\times SR^+$	0.1421**	0.2082***	-0.3719**	-0.3458**
	(2.55)	(3.61)	(-2.16)	(-2.07)
Constant	1.7625***	-4.0458	2.2612***	-13.8699
	(67.65)	(-0.50)	(21.11)	(-0.82)
Controls	No	Yes	No	Yes
Observations	197,596	191,166	197,596	191,166
Adj. R^2	0.014	0.025	0.001	0.027
Panel B: Surprise ≤ 0				
Round	-0.1240***	-0.1542***	-1.2908***	-1.1010***
	(-2.76)	(-3.40)	(-8.33)	(-7.18)
SR^-	0.6519***	0.5943***	-0.1507	0.1561
	(22.47)	(19.73)	(-1.46)	(1.60)
Round $\times SR^-$	0.0222	0.0498	0.0594	0.0995
	(0.36)	(0.78)	(0.30)	(0.48)
Constant	-2.0576***	0.4478	0.9760***	3.2554
	(-64.59)	(0.22)	(7.43)	(0.39)
Controls	No	Yes	No	Yes
Observations	160,163	151,829	160,163	151,829
Adj. R^2	0.007	0.029	0.001	0.030

Internet Appendix for

"Reference-Point Fluency and Memorability"

Litong Zhang

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I A Simple Model of Noisy Cost-Basis Recall

This section develops a deliberately minimal model of how the roundness of a purchase price could amplify the disposition effect. The investor does not use her cost basis without error; she recalls a noisy estimate of it. Recall is more precise when the basis is round and less precise as time passes since the purchase. Selling yields realization utility that depends on the gain or loss the investor believes she is realizing. The model is not meant to be estimated. Its purpose is to make explicit which assumptions are needed for roundness to amplify the disposition effect, and to derive testable predictions that separate noisy recall from other ways in which roundness could matter. Table IA.1 summarizes the predictions and the corresponding evidence.

I.1 Setup

Consider a position with log cost basis b and log price p on the decision date, so that the log paper return is $r = p - b$ and the position is at a gain when $r > 0$. Let $k \in \{\mathrm{R}, \mathrm{N}\}$ indicate whether the purchase price is round or non-round, and let $\tau > 0$ denote the time since purchase.

Noisy recall. When deciding whether to sell, the investor uses a recalled basis $\hat{b} = b + \varepsilon$, so her perceived return is $\hat{r} = r - \varepsilon$. The recall error $\varepsilon \sim N\big(0, \sigma_k^2(\tau)\big)$ is independent of r, with

$$\sigma_k^2(\tau) = s_k^2\,\tau, \qquad 0 < s_{\mathrm{R}} < s_{\mathrm{N}}. \tag{8}$$

Recall precision, $1/\sigma_k^2(\tau)$, is therefore higher for a round basis and falls with time since purchase. Variance that grows linearly in time is the simplest form with these two properties. Below I state which results depend on it and which hold for any $\sigma_{\mathrm{R}}(\tau) < \sigma_{\mathrm{N}}(\tau)$.

Realization utility. Building on Barberis and Xiong (2012), selling generates a burst of utility from the gain or loss the investor believes she is realizing. The burst has a linear part, with possibly different slopes for gains and losses, and a jump at the reference point:

$$u(x) = \alpha_+ \, x \, \mathbf{1}\{x \geq 0\} + \alpha_- \, x \, \mathbf{1}\{x < 0\} + j \, \mathbf{1}\{x > 0\}, \qquad \alpha_+, \alpha_- > 0, \; j > 0. \quad (9)$$

The jump j captures the idea that realizing an outcome classified as a gain is pleasant, and realizing one classified as a loss is painful, beyond the size of the amount.

Sale rule. The investor sells with probability

$$S = a + c \, \mathbb{E}\big[u(\tilde{r})\big], \qquad a, c > 0, \quad (10)$$

where $\tilde{r}$ is the return the investor believes she would realize and the expectation is over recall noise. This linear form arises when the investor sells whenever the realization payoff plus an independent, uniformly distributed sale shock, such as a liquidity need, exceeds a threshold, and the resulting probability is interior.

I consider two ways the investor can use the recalled basis. A naive investor treats $\hat{r}$ as the true return, so $\tilde{r} = \hat{r}$. A Bayesian investor with a flat prior over b knows that recall is noisy; her posterior is $r \mid \hat{r} \sim N(\hat{r}, \sigma_k^2)$, and she evaluates u under this posterior.

Lemma 1: *For both the naive and the Bayesian investor, the sale probability of a position with return r, averaged over recall noise, is*

$$S_k(r, \tau) = a + c \, \mathbb{E}\big[u\big(r + \kappa \, \sigma_k(\tau) \, Z\big)\big], \qquad Z \sim N(0, 1),$$

with $\kappa = 1$ for the naive investor and $\kappa = \sqrt{2}$ for the Bayesian investor.

Proof: *For the naive investor, $\tilde{r} = r - \varepsilon$, and ε is symmetric. For the Bayesian investor, a draw from the posterior is $\hat{r} + \eta = r - \varepsilon + \eta$, where $\eta \sim N(0, \sigma_k^2)$ is independent of ε, so the draw is distributed $N(r, 2\sigma_k^2)$. By the law of iterated expectations, averaging her posterior expected utility over ε gives the expectation of u under this distribution.*

Because κ only rescales s_k, I set $\kappa = 1$ below. Every result holds for the Bayesian investor with s_k replaced by $\sqrt{2}\, s_k$.

I.2 The Disposition Effect Under Noisy Recall

For a return magnitude $x > 0$, define the gain–loss gap in selling as $D_k(x, \tau) = S_k(x, \tau) - S_k(-x, \tau)$.

Lemma 2: *The gain–loss gap is*

$$D_k(x, \tau) = c\,(\alpha_+ + \alpha_-)\,x + c\,j\left[2\,\Phi\left(\tfrac{x}{\sigma_k(\tau)}\right) - 1\right],$$

where Φ is the standard normal distribution function. The linear part of u contributes a term that does not depend on recall precision; only the jump contributes a term that does.

Proof: *Write the linear part of u as $\ell(y) = \alpha_+ y - (\alpha_+ - \alpha_-)\min(y, 0)$. For $Y \sim N(\mu, \sigma^2)$, $\mathbb{E}[\min(Y, 0)] = \mu\,\Phi(-\mu/\sigma) - \sigma\,\phi(\mu/\sigma)$, where ϕ is the standard normal density. Hence $\mathbb{E}[\min(x + \sigma Z, 0)] - \mathbb{E}[\min(-x + \sigma Z, 0)] = x\,[\Phi(-x/\sigma) + \Phi(x/\sigma)] = x$, and the linear part contributes $2\alpha_+ x - (\alpha_+ - \alpha_-)x = (\alpha_+ + \alpha_-)x$ for every σ. The jump contributes $j\,[\Pr(x + \sigma Z > 0) - \Pr(-x + \sigma Z > 0)] = j\,[2\Phi(x/\sigma) - 1]$.*

Lemma 2 has two implications. First, recall noise changes the gain–loss gap only through the jump. If realization utility were linear in the realized amount, even with different slopes for gains and losses, mean-zero recall noise would leave the gap unchanged, and recall precision could not make the disposition effect depend on roundness. The model therefore needs realization utility to depend on whether a position is classified as a gain or a loss, and not only on the amount. Curvature

that is concave over gains and convex over losses, as in the S-shaped realization utility of Ingersoll and Jin (2013), pushes in the same direction for returns away from the reference point. I use the jump because it is the simplest case and delivers closed forms. Second, the linear part is the same for round and non-round positions. In the main-text regressions it corresponds to the $Return(+)$ and $Return(-)$ controls, which are not interacted with roundness, so $RoundBuyPrice \times Gain$ loads on the jump component.

I.3 Predictions

Define the round-number amplification of the disposition effect as

$$A(x,\tau) = D_{\mathrm{R}}(x,\tau) - D_{\mathrm{N}}(x,\tau) = 2\,c\,j\left[\Phi\left(\tfrac{x}{s_{\mathrm{R}}\sqrt{\tau}}\right) - \Phi\left(\tfrac{x}{s_{\mathrm{N}}\sqrt{\tau}}\right)\right]. \tag{11}$$

Proposition 1 (Round-number amplification): *$A(x,\tau) > 0$ for all $x > 0$ and $\tau > 0$.*

Proof: *$s_{\mathrm{R}} < s_{\mathrm{N}}$ implies $x/(s_{\mathrm{R}}\sqrt{\tau}) > x/(s_{\mathrm{N}}\sqrt{\tau})$, and Φ is strictly increasing.*

Proposition 1 holds for any $\sigma_{\mathrm{R}}(\tau) < \sigma_{\mathrm{N}}(\tau)$, not only for the form in Equation 8. It corresponds to the main result, a positive $RoundBuyPrice \times Gain$ interaction (Table 3).

Proposition 2 (A single index): *A depends on (x,τ) only through $z = x/\sqrt{\tau}$:*

$$A(x,\tau) = 2cj\,H(z), \qquad H(z) = \Phi(z/s_{\mathrm{R}}) - \Phi(z/s_{\mathrm{N}}).$$

H is positive, tends to zero as $z \to 0$ and as $z \to \infty$, and is single-peaked with its maximum at

$$z^* = s_{\mathrm{R}}\,s_{\mathrm{N}}\sqrt{\frac{2\ln(s_{\mathrm{N}}/s_{\mathrm{R}})}{s_{\mathrm{N}}^2 - s_{\mathrm{R}}^2}}.$$

Proof: *The limits follow from $\Phi(0) = 1/2$ and $\Phi(\infty) = 1$. The derivative is $H'(z) = \phi(z/s_{\mathrm{R}})/s_{\mathrm{R}} - \phi(z/s_{\mathrm{N}})/s_{\mathrm{N}}$, which is positive if and only if $\exp\left[-\frac{z^2}{2}\left(s_{\mathrm{R}}^{-2} - s_{\mathrm{N}}^{-2}\right)\right] > s_{\mathrm{R}}/s_{\mathrm{N}}$, that is, if and only if $z < z^*$.*

Proposition 2 yields three predictions.

Corollary 1 (Paper return): *For a given holding time, the amplification is hump-shaped in the absolute paper return. It is small when $|r|$ is close to zero, largest at $|r| = z^*\sqrt{\tau}$, and vanishes as $|r|$ grows.*

The intuition is that roundness matters only when it changes how the position is classified. When the return is small relative to even the round-basis recall noise, neither investor can reliably tell a gain from a loss. When it is large relative to even the non-round-basis noise, both can. Roundness matters in between. This shape holds for any $\sigma_R < \sigma_N$ at a given holding time.

Corollary 2 (Holding time): *For a given absolute paper return x, the amplification is hump-shaped in holding time. It vanishes as $\tau \to 0$, peaks at $\tau^* = (x/z^*)^2$, and vanishes as $\tau \to \infty$.*

The decline at long horizons is robust: it holds whenever recall noise for both kinds of basis grows without bound. The rise at short horizons is not. It requires that both round and non-round bases be recalled almost exactly right after purchase. If roundness also improves encoding, so that $\sigma_k^2(\tau) = e_k^2 + s_k^2\tau$ with $e_R < e_N$, then $A(x, 0) = 2cj\,[\Phi(x/e_R) - \Phi(x/e_N)] > 0$, and the amplification need not rise at short horizons. The model therefore makes a firm prediction about long horizons and a conditional one about short horizons.

Corollary 3 (Joint pattern): *The holding time at which the amplification peaks, $\tau^* = (x/z^*)^2$, increases with the absolute paper return. Positions with equal $|r|/\sqrt{\tau}$ have equal amplification.*

Corollary 3 links the paper-return and holding-time patterns: under noisy recall they are two views of the same function.

The next result shows what the paper-return pattern can distinguish. Suppose, instead, that roundness does not change recall precision but makes the gain–loss classification more rewarding, for example because a gain measured against

a round price feels more like reaching a goal (Pope and Simonsohn 2011). In the model, this means a larger jump for round positions, $j_R > j_N$, with common recall noise $\sigma(\tau)$.

Proposition 3 (A preference-based alternative): *If $j_R > j_N$ and $\sigma_R(\tau) = \sigma_N(\tau) = \sigma(\tau)$, then*

$$A(x,\tau) = c\,(j_R - j_N)\left[2\,\Phi\left(\tfrac{x}{\sigma(\tau)}\right) - 1\right],$$

which increases in x and converges to $c\,(j_R - j_N) > 0$ as $x \to \infty$.

Proof: *Apply Lemma 2 with j_k in place of j and a common σ.*

Under noisy recall, the amplification vanishes for large paper returns; under a preference-based bonus, it is largest there. The two accounts therefore make opposite predictions for large gains and losses.

I.4 Stock Splits

A forward split with ratio $m > 1$ rescales both the basis and the price by $\ln m$ in logs, so the true return r is unchanged. Before the split, the investor recalls b as above. After the split, she must work with the adjusted basis $b - \ln m$. Her recalled adjusted basis is $b - \ln m + \varepsilon + \xi$, where ε is the recall error of the original basis, with variance $s_{k_0}^2 \tau$ determined by the roundness k_0 of the original price, and $\xi \sim N(0, \omega_{k_1}^2)$ is an independent computation error that depends on the roundness k_1 of the adjusted basis, with $0 \le \omega_R < \omega_N$. The assumption is that an adjusted basis is easier to compute and hold when it is itself round: a \$100 basis after a two-for-one split becomes \$50, whereas a \$25 basis becomes \$12.50. In the notation of the main text, Always positions have $(k_0, k_1) = (R, R)$, Breaks positions have (R, N), and Never positions have (N, N).

Proposition 4 (Splits): *Holding x and τ fixed:*

(i) Before the split, $D_{\text{Always}} = D_{\text{Breaks}} > D_{\text{Never}}$.

(ii) After the split, $D_{\text{Always}} > D_{\text{Breaks}} > D_{\text{Never}}$.

(iii) For each group with $\omega_{k_1} > 0$*, the gain–loss gap is smaller after the split than before.*

(iv) If instead the investor does not adjust and compares the post-split price with the unadjusted basis b*, Always and Breaks positions have the same gain–loss gap after the split as well as before.*

Proof: *By Lemma 2,* D *is strictly decreasing in the variance of the error in the basis the investor uses. Before the split, this variance is* $s_{k_0}^2\tau$*, which is equal for Always and Breaks and larger for Never, giving (i). After the split, it is* $s_{k_0}^2\tau + \omega_{k_1}^2$*, which equals* $s_{\text{R}}^2\tau + \omega_{\text{R}}^2$*,* $s_{\text{R}}^2\tau + \omega_{\text{N}}^2$*, and* $s_{\text{N}}^2\tau + \omega_{\text{N}}^2$ *for the three groups, in increasing order, giving (ii). Adding* $\omega_{k_1}^2 > 0$ *raises the variance, giving (iii). Without adjustment, the perceived return is* $r - \ln m - \varepsilon$*, which depends on* k_0 *but not on* k_1*, giving (iv).*

Proposition 4(iii) gives a recall-based reading of Birru (2015), who finds that the disposition effect is absent on days when a stock undergoes a split. Proposition 4(iv) shows that a pure failure to adjust, by itself, cannot generate a difference between Always and Breaks positions, because the two groups differ only in the roundness of the adjusted basis.

I.5 Evidence

Proposition 1 corresponds to the baseline result: *RoundBuyPrice* × *Gain* is positive, about 1.2 to 2.0 percentage points across specifications (Table 3).

Corollary 1 and Proposition 3 concern how the amplification varies with the absolute paper return. In Figure IA.3, the implied *RoundBuyPrice* × *Gain* effect is about 1.1 percentage points for $|r| < 5\%$, with a confidence interval that includes zero; about 3.0 percentage points for returns between 5% and 15%; about 2.1 to 2.3 percentage points between 15% and 50%; and about 0.8 percentage points, again with a confidence interval that includes zero, above 50%. In the subsample estimates of Table IA.7, the interaction falls from 2.0 percentage points in the full

sample to 1.3 ($t = 1.84$) for $|r| \geq 10\%$ and 0.3 ($t = 0.30$) for $|r| \geq 20\%$. Both sets of estimates decline at large returns, as noisy recall predicts and the preference-based alternative does not. The bin-level estimates are imprecise and their confidence intervals overlap, so this evidence is suggestive rather than decisive.

Corollary 2 concerns holding time. In Figure IA.4 and Table IA.13, the implied effect is about 2.7 and 3.8 percentage points at 15 to 90 and 91 to 180 days, and about 0.8 and 0.4 percentage points, both insignificant, at 181 to 365 days and beyond one year. This decline matches the robust long-horizon prediction. The estimate for the first two weeks, about 1.8 percentage points with a confidence interval from about -1.5 to 5.1, is too imprecise to tell whether the amplification rises at short horizons, which is the part of the prediction that depends on whether roundness also aids encoding. Two caveats apply. The model's comparative statics hold a position fixed as it ages, whereas the estimates compare positions still held at different horizons, which are selected on earlier sale decisions. And holding time is not randomly assigned.

Proposition 4 concerns splits. Consistent with part (ii), on post-split days the Always×Gain interaction is 0.372 ($t = 2.66$), the Breaks×Gain interaction is 0.047 ($t = 0.81$), and their difference is 0.324 ($t = 2.29$) (Table IA.17, Column (1)). In the model, a Breaks interaction close to zero means that the computation error for a non-round adjusted basis is large relative to the recall advantage of a round original price. Consistent with part (iii), the coefficient on *Gain*, which measures the gain–loss gap for Never positions, is 0.006 ($t = 0.13$) on post-split days, compared with 0.052 to 0.110 over full holding spells (Table 6). The estimates reject part (iv)'s implication that Always and Breaks positions behave alike, so investors appear to work with the adjusted basis rather than ignore the split.

I.6 Limitations

The model takes recall precision as a primitive. It does not explain why a round basis is recalled more precisely, which the main text attributes to fluency and ease of encoding (Frydman and Jin 2022). The closed forms rely on normal recall errors and on a sale probability that is linear in expected realization utility; I do not claim that the exact hump shapes survive arbitrary functional forms. Proposition 1 and the long-horizon decline in Corollary 2 require only that a round basis be recalled more precisely and that recall noise grow with time. Finally, the model describes a single position. The estimates come from cross-sectional comparisons, so the predictions concern the sign and shape of the estimated interactions rather than their magnitudes.

Table IA.1: **Predictions of the Noisy-Recall Model and Corresponding Evidence**

This table summarizes the predictions of the model in Internet Appendix Section I. "Noisy recall" gives the prediction when a round cost basis is recalled more precisely and recall precision falls with holding time. "Alternative" gives the prediction of the alternative indicated in the row. "Evidence" lists the corresponding estimates.

Outcome	Noisy recall	Alternative	Evidence
RoundBuyPrice × *Gain*	Positive (Proposition 1).	–	Positive, about 1.2 to 2.0 pp (Table 3).
Variation with $\|r\|$	Hump-shaped; vanishes for large $\|r\|$ (Corollary 1).	Roundness raises the gain–loss reward: increases with $\|r\|$ (Proposition 3).	Largest at 5% to 15%, smaller above 50% (Figure IA.3); 1.3 pp for $\|r\| \geq 10\%$, 0.3 pp for $\|r\| \geq 20\%$ (Table IA.7). Imprecise.
Variation with holding time	Vanishes at long horizons; the short-horizon pattern depends on whether roundness also aids encoding (Corollary 2).	–	Smaller and insignificant beyond 180 days; short horizon imprecise (Figure IA.4, Table IA.13). Subject to survivor selection.
Splits, post-split days	Always > Breaks > Never (Proposition 4(ii)).	Failure to adjust: Always = Breaks (Proposition 4(iv)).	Always 0.372, Breaks 0.047, difference 0.324 ($t = 2.29$) (Table IA.17).
Gain–loss gap after a split	Smaller (Proposition 4(iii)).	–	*Gain* is 0.006 on post-split days versus 0.052 to 0.110 over full spells (Table IA.17, Table 6).

II Account-Level Additional Evidence

II.1 Purchase-Price Distributions

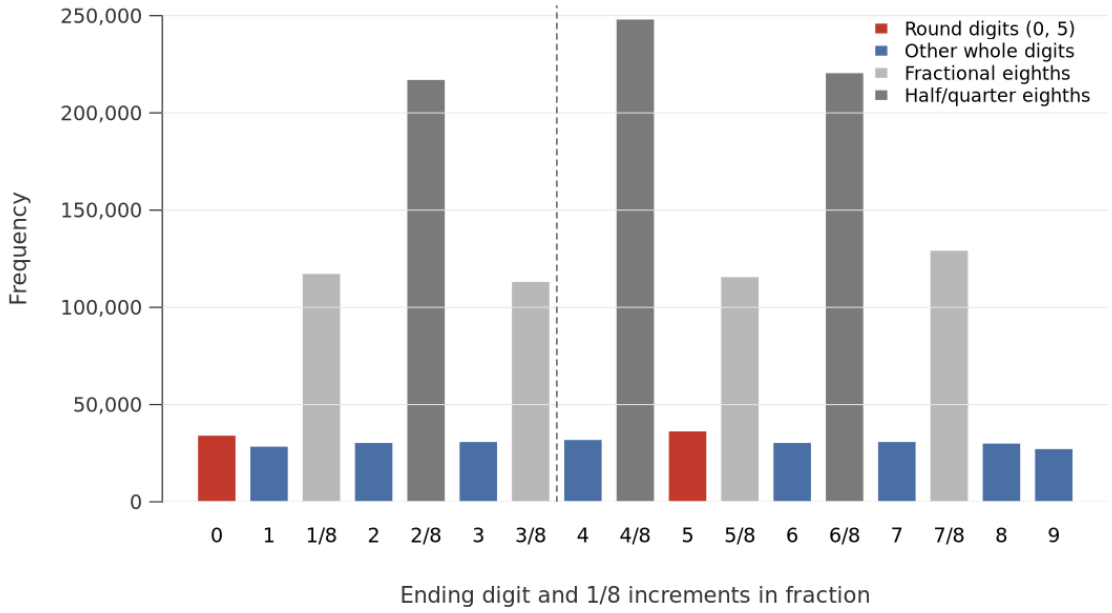


Figure IA.1: **Distribution of Stock Purchase Prices by Ending Digit and Fraction**

Household-level purchase prices from the brokerage sample (1991–1996). The left block pools fractional prices into eighth-of-a-dollar bins ($\frac{1}{8}$–$\frac{7}{8}$); the right block reports whole-dollar prices by the ending digit (1–9 and 0). Bars in red highlight round digits (0 and 5); darker gray bars mark half- and quarter-dollar fractions ($\frac{2}{8}$, $\frac{4}{8}$, $\frac{6}{8}$).

II.2 Distance and Paper-Return Figures

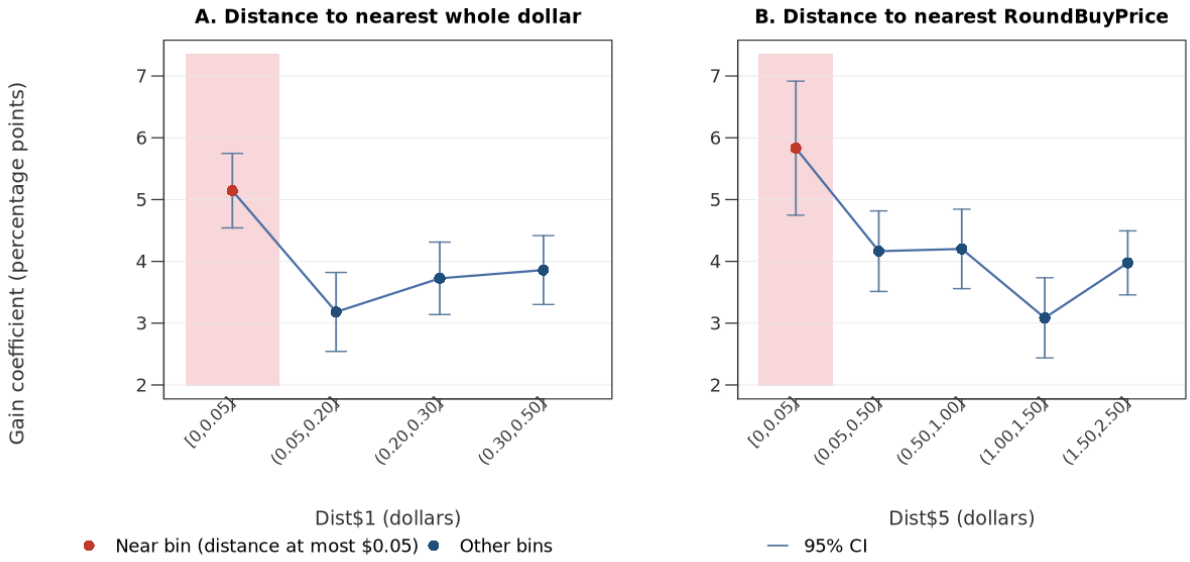


Figure IA.2: **Disposition Effect by Distance to Round Purchase Prices**

The dependent variable is the sell indicator in percentage points. Panel A estimates

$$Sell100_{ijt} = \beta_1 Gain_{ijt} + \sum_{k \in \mathcal{B}_A \setminus \{r_A\}} \left(\alpha_k D^k_{ijt} + \gamma_k \, Gain_{ijt} \times D^k_{ijt} \right) + X'_{ijt}\lambda + \mu_i + \delta_{j \times t} + \varepsilon_{ijt},$$

where $\mathcal{B}_A = \{[0, 0.05], (0.05, 0.20], (0.20, 0.30], (0.30, 0.50]\}$ are $Dist\$1$ bins and the omitted reference is $r_A = (0.30, 0.50]$. Panel B uses the same specification with $Dist\$5$ bins $\mathcal{B}_B = \{[0, 0.05], (0.05, 0.50], (0.50, 1.00], (1.00, 1.50], (1.50, 2.50]\}$ and omitted reference $r_B = (1.50, 2.50]$. X_{ijt} includes $\log(BuyPrice)$, $Return(+)$, $Return(-)$, $TimeOwned$, $Volatility(+)$, and $Volatility(-)$. Each point is the implied $Gain$ effect in that bin (β_1 in the reference bin; $\beta_1 + \gamma_k$ otherwise). Vertical bars are 95% confidence intervals; standard errors are double-clustered by investor and day. The shaded bin is distance at most \$0.05. $N = 1{,}393{,}400$ investor-stock-days.

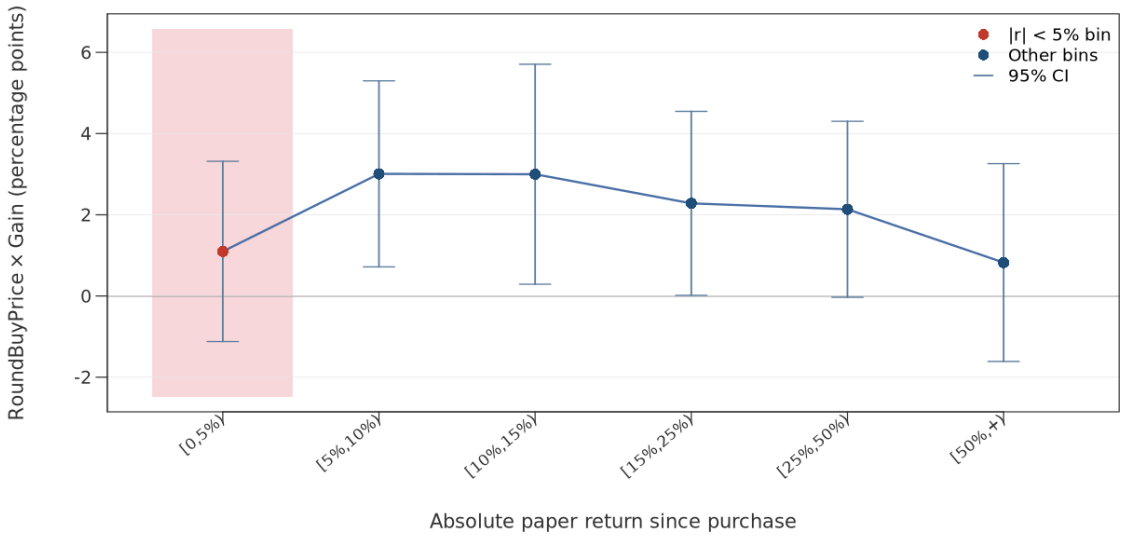


Figure IA.3: **Disposition Effect and Round Purchase Prices by Absolute Paper Return**

The dependent variable is the sell indicator in percentage points. Let $R_{ijt} \equiv RoundBuyPrice_{ijt}$ and $G_{ijt} \equiv Gain_{ijt}$. Absolute $PaperReturn$ is partitioned into mutually exclusive bins. The figure estimates

$$\begin{aligned} Sell100_{ijt} = {} & \beta_R R_{ijt} + \beta_G G_{ijt} + \beta_{RG}\, R_{ijt} \times G_{ijt} \\ & + \sum_{k \in \mathcal{B} \setminus \{r\}} \left(\alpha_k B^k_{ijt} + \beta_{Rk}\, R_{ijt} \times B^k_{ijt} + \beta_{Gk}\, G_{ijt} \times B^k_{ijt} + \gamma_k\, R_{ijt} \times G_{ijt} \times B^k_{ijt} \right) \\ & + X'_{ijt}\lambda + \mu_i + \delta_{j \times t} + \varepsilon_{ijt}, \end{aligned}$$

where

$$\mathcal{B} = \{ [0, 5\%),\ [5\%, 10\%),\ [10\%, 15\%),\ [15\%, 25\%),\ [25\%, 50\%),\ [50\%, +) \}$$

and the omitted reference is $r = [50\%, +)$. X_{ijt} includes $\log(BuyPrice)$, $Return(+)$, $Return(-)$, $TimeOwned$, $Volatility(+)$, and $Volatility(-)$. Each plotted point is the implied $R \times G$ effect in that bin (β_{RG} in the reference bin; $\beta_{RG} + \gamma_k$ otherwise). The shaded bin is $|PaperReturn| < 5\%$. Vertical bars are 95% confidence intervals; standard errors are double-clustered by investor and day. $N = 1{,}393{,}400$ investor-stock-days.

II.3 Baseline Sample and Measurement

Table IA.2: **Disposition Effect and Round Purchase Prices: All Investor-Day Sample**

This table reports estimation results from the full investor-stock-day holding panel, including days with no sale. The dependent variable equals 1 if a stock is sold (fully or partially) on a given day and 0 otherwise (scaled to percentage points). *Gain* equals one if the return since purchase is positive. *RoundBuyPrice* equals one if the most recent purchase price has no fractional part and the integer amount ends in 0 or 5. Other variable definitions follow Table 3. All coefficients are multiplied by 100. t-statistics in parentheses; standard errors are double-clustered at the investor and day level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)
RoundBuyPrice	-0.006**	-0.015***	-0.014***	-0.048***	-0.013***
	(-2.04)	(-4.29)	(-2.95)	(-3.49)	(-2.93)
Gain	0.114***	0.034***	0.061***	0.071***	0.064***
	(114.93)	(8.78)	(15.12)	(16.10)	(16.28)
RoundBuyPrice × Gain	0.010**	0.017***	0.019***	0.027***	0.024***
	(2.22)	(2.91)	(2.65)	(3.03)	(3.44)
Log(BuyPrice)		0.044***	0.036***	0.141***	0.041***
		(13.76)	(6.12)	(13.46)	(10.01)
Return(+)		-0.000	-0.019***	0.020***	-0.009***
		(-0.19)	(-7.69)	(8.44)	(-5.47)
Return(−)		-0.093***	-0.144***	-0.202***	-0.121***
		(-8.27)	(-8.76)	(-13.22)	(-9.22)
$\sqrt{TimeOwned}$		-0.018***	-0.002***	0.059***	-0.003***
		(-47.44)	(-8.25)	(62.81)	(-10.43)
Volatility(+)		8.152***	-1.589	4.949***	4.724***
		(29.57)	(-1.47)	(26.17)	(25.91)
Volatility(−)		1.999***	-6.418***	-1.318***	-0.531***
		(11.78)	(-5.90)	(-10.79)	(-4.25)
Investor FE	No	No	Yes	No	No
SellDate FE	No	No	No	Yes	No
Stock FE	No	No	No	No	Yes
Investor × SellDate FE	No	No	No	No	Yes
Investor × Stock FE	No	No	No	Yes	No
Stock × SellDate FE	No	No	Yes	No	No
Observations	110,790,367	110,790,367	110,790,367	110,790,367	110,790,367
Adjusted R^2	0.000	0.001	0.024	0.030	0.127

t-statistics in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table IA.3: **Disposition Effect and Round Purchase Prices: Alternative Cost-Basis Definitions**

This table reports estimation results under alternative executed purchase-price cost bases. The dependent variable equals 1 if a stock is sold (fully or partially) on a given day and 0 otherwise (scaled to percentage points). Each column defines the gain indicator and *RoundBuyPrice* from the same cost basis shown in the column heading. Paper gain equals one when the lagged CRSP closing price exceeds that cost basis. Column (1) repeats the baseline specification in Table 3, Column (3), and uses that sample. Column (2) retains investor-stock-day observations with at least one additional purchase after the initial buy; columns (3) and (4) use the full sample. In columns (2) to (4), return controls use $(\text{lagged closing price}/\text{cost basis}) - 1$. All specifications include log purchase price, return components, $\sqrt{\text{TimeOwned}}$, and volatility controls, investor fixed effects, and stock-by-sell-date fixed effects. All coefficients are multiplied by 100. t-statistics in parentheses; standard errors are double-clustered at the investor and sell-date level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	Recent purchase (1)	Vol.-weighted avg. (2)	First purchase (3)	FIFO (4)
RoundBuyPrice	-1.143***	3.397	-1.061***	-0.933***
	(-3.30)	(1.00)	(-3.10)	(-2.90)
Gain	3.886***	1.236	4.592***	3.116***
	(16.38)	(0.50)	(12.21)	(9.43)
RoundBuyPrice × Gain	1.997***	-3.981	1.960***	0.906*
	(3.90)	(-0.51)	(3.79)	(1.82)
Log(BuyPrice)	-0.046	-3.831	-6.580***	-7.515***
	(-0.12)	(-0.58)	(-7.49)	(-8.69)
Return(+)	-0.401***	-3.061	-0.995*	-2.041***
	(-2.88)	(-0.74)	(-1.95)	(-3.86)
Return(−)	-7.176***	-18.619*	-20.649***	-19.735***
	(-5.87)	(-1.68)	(-13.17)	(-12.87)
$\sqrt{TimeOwned}$	-0.039***	0.109	-0.093***	0.339***
	(-2.71)	(1.04)	(-6.73)	(21.75)
Volatility(+)	62.337	-41.253	88.666	75.854***
	(1.24)	(-0.00)	(0.00)	(6.92)
Volatility(−)	-121.542**	-109.920	-26.672	
	(-2.41)	(-0.00)	(-0.00)	
Investor FE	Yes	Yes	Yes	Yes
Stock × SellDate FE	Yes	Yes	Yes	Yes
Observations	1,393,400	97,681	1,397,507	1,267,139
R^2	0.679	0.909	0.679	0.667

t-statistics in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table IA.4: **Disposition Effect and Round Purchase Prices: Additional Purchases**

This table examines whether additional purchases after the initial buy affect the relationship between round purchase prices and the disposition effect. The dependent variable equals 1 if a stock is sold (fully or partially) on a given day and 0 otherwise (scaled to percentage points). Gain and *RoundBuyPrice* are defined from the most recent executed purchase price, as in the baseline. Column (1) is estimated on the baseline sample and interacts *RoundBuyPrice* and Gain with an indicator equal to one if the position includes at least one additional purchase after the initial buy. Column (2) retains single-purchase positions in that sample. All specifications include log purchase price, return components, $\sqrt{\text{TimeOwned}}$, and volatility controls, investor fixed effects, and stock-by-sell-date fixed effects. All coefficients are multiplied by 100. t-statistics in parentheses; standard errors are double-clustered at the investor and sell-date level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	Interaction (1)	Single purchase (2)
RoundBuyPrice	-1.146***	-1.167***
	(-3.23)	(-3.20)
Gain	4.100***	3.713***
	(17.45)	(14.97)
RoundBuyPrice × Gain	1.961***	2.042***
	(3.71)	(3.79)
AdditionalPurchase	-2.196***	
	(-5.22)	
RoundBuyPrice × Gain × AdditionalPurchase	-0.488	
	(-0.25)	
Log(BuyPrice)	-0.111	-0.561
	(-0.30)	(-1.45)
Return(+)	-0.494***	-0.352**
	(-3.50)	(-2.52)
Return(−)	-7.008***	-8.489***
	(-5.69)	(-6.71)
$\sqrt{TimeOwned}$	-0.018	-0.039***
	(-1.24)	(-2.68)
Volatility(+)	49.884	-100.688*
	(1.00)	(-1.68)
Volatility(−)	-129.795***	-294.426***
	(-2.58)	(-4.84)
Investor FE	Yes	Yes
Stock × SellDate FE	Yes	Yes
Observations	1,393,400	1,297,969
Adjusted R^2	0.191	0.198

t-statistics in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table IA.5: **PGR and PLR**

	All Prices	Round & Whole Prices	Fractional Prices
PGR	0.231	0.236	0.230
PLR	0.158	0.156	0.159
Difference	0.073***	0.080***	0.071***
t-statistic	(112.8)	(56.0)	(98.1)

	Round & Whole minus Fractional	
	Difference	t-statistic
PGR	0.006***	(5.1)
PLR	-0.003***	(-2.8)
Difference	0.009***	(5.6)

This table compares the Proportion of Gains Realized (PGR) to the Proportion of Losses Realized (PLR), where

$$\text{Proportion of Gains Realized (PGR)}^p = \frac{\text{Realized Gains}^p}{\text{Realized Gains}^p + \text{Paper Gains}^p}$$

$$\text{Proportion of Losses Realized (PLR)}^p = \frac{\text{Realized Losses}^p}{\text{Realized Losses}^p + \text{Paper Losses}^p}$$

and $p \in \{\text{all-price}, \text{round/whole-price}, \text{fractional-price}\}$.
Realized gains, paper gains, realized losses, and paper losses are aggregated over time for the whole sample period (1991–1996). PGR and PLR are reported for all purchase prices, for round and whole-dollar purchase prices (integer prices ending in 0–9), and for fractional purchase prices only. The within-column t-statistic tests H_0 : PGR $=$ PLR; the bottom panel reports Round & Whole minus Fractional differences with t-statistics computed under the same independence assumption as Odean (1998). ***, ** and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

II.4 Gains, Losses, and Paper Returns

Table IA.6: **Disposition Effect and Round Purchase Prices: Gains versus Losses**

This table reports estimation results that separate the association between round purchase prices and selling propensities for positions in paper gains and paper losses. The dependent variable equals 1 if a stock is sold (fully or partially) on a given day and 0 otherwise (scaled to percentage points). *Gain* equals one if the return since purchase is strictly positive. *Loss* equals one if the return since purchase is strictly negative. Observations with a zero paper return are excluded. *RoundBuyPrice* equals one if the most recent purchase price has no fractional part and the integer amount ends in 0 or 5. RoundBuyPrice×Gain − RoundBuyPrice×Loss is the difference between the two interaction coefficients. Other variable definitions follow Table 3. All coefficients are multiplied by 100. t-statistics in parentheses; standard errors are double-clustered at the investor and day level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)
RoundBuyPrice × Gain	0.777***	0.842**	0.854**	0.215	0.752**
	(3.33)	(2.00)	(2.14)	(0.42)	(2.22)
RoundBuyPrice × Loss	-0.379*	-0.603*	-1.143***	-1.018*	-0.907***
	(-1.68)	(-1.70)	(-3.30)	(-1.90)	(-3.03)
Gain	7.655***	3.960***	3.886***	4.143***	4.787***
	(109.70)	(16.88)	(16.38)	(15.56)	(15.57)
RoundBuyPrice×Gain − RoundBuyPrice×Loss	1.156***	1.445***	1.997***	1.233**	1.659***
	(3.57)	(2.84)	(3.90)	(2.29)	(3.58)
Log(BuyPrice)		0.761***	-0.046	5.495***	1.224***
		(3.58)	(-0.12)	(12.06)	(5.73)
Return(+)		-0.044	-0.401***	0.358	0.237
		(-0.19)	(-2.88)	(1.50)	(1.52)
Return(−)		-7.861***	-7.176***	-0.635	-3.176***
		(-7.11)	(-5.87)	(-0.59)	(-3.03)
$\sqrt{TimeOwned}$		-0.308***	-0.039***	2.116***	-0.104***
		(-16.79)	(-2.71)	(30.56)	(-6.92)
Volatility(+)		203.258***	62.337	147.506***	193.198***
		(14.38)	(1.24)	(14.32)	(20.71)
Volatility(−)		-16.043	-121.542**	-21.692**	-24.124***
		(-1.56)	(-2.41)	(-2.52)	(-3.15)
Investor FE	No	No	Yes	No	No
SellDate FE	No	No	No	Yes	No
Stock FE	No	No	No	No	Yes
Investor × SellDate FE	No	No	No	No	Yes
Investor × Stock FE	No	No	No	Yes	No
Stock × SellDate FE	No	No	Yes	No	No
Observations	1,393,400	1,393,400	1,393,400	1,393,400	1,393,400
Adjusted R^2	0.009	0.017	0.190	0.297	0.077

t-statistics in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table IA.7: **Disposition Effect and Round Purchase Prices: Paper Returns**

The dependent variable equals one if the stock is sold on day t and zero otherwise (percentage points). *RoundBuyPrice* equals one if the most recent purchase price is a whole dollar ending in 0 or 5. *Gain* equals one if the return since purchase is positive. *PaperReturn* is the continuous return since purchase, winsorized at the 1st and 99th percentiles. Purchase prices are winsorized at the 1st and 99th percentiles. Column (1) reports *RoundBuyPrice* × *Gain* in the baseline specification, Column (3) of Table 3, on the full estimation sample. Column (2) reports the same specification on observations with $|PaperReturn| \geq 0.01$. Column (3) reports the same specification on observations with $|PaperReturn| \geq 0.02$. Column (4) reports the same specification on observations with $|PaperReturn| \geq 0.10$. Column (5) reports the same specification on observations with $|PaperReturn| \geq 0.20$. Column (6) reports the same specification on observations with $|PaperReturn| < 0.01$. Column (7) reports *RoundBuyPrice* × *PaperReturn* on the full estimation sample. Controls, fixed effects, and clustering follow Column (3) of Table 3. Coefficients are multiplied by 100. t-statistics are in parentheses. Standard errors are double-clustered by investor and day. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. N is reported by column.

	Baseline	$\|r\| \geq 1\%$	$\|r\| \geq 2\%$	$\|r\| \geq 10\%$	$\|r\| \geq 20\%$	$\|r\| < 1\%$	Cont. Ret
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
RoundBuyPrice	-1.143***	-1.150***	-1.112***	-0.525	-0.210	-1.033	-0.124
	(-3.30)	(-3.21)	(-2.98)	(-1.09)	(-0.35)	(-0.14)	(-0.46)
Gain	3.886***	4.133***	4.346***	4.367***	3.613***	0.542	
	(16.38)	(16.54)	(16.72)	(12.40)	(8.15)	(0.19)	
RoundBuyPrice × Gain	1.997***	1.992***	2.114***	1.280*	0.265	-5.515	
	(3.90)	(3.79)	(3.89)	(1.84)	(0.30)	(-0.55)	
PaperReturn							-2.377***
							(-8.97)
RoundBuyPrice × PaperReturn							0.184
							(0.32)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Investor FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Stock × SellDate FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,393,400	1,343,611	1,280,752	888,834	588,711	49,789	1,393,400
R^2	0.679	0.688	0.697	0.756	0.808	0.988	0.679

t-statistics in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

II.5 Placebos, Distance, and Price Level

Table IA.8: **Disposition Effect and Round Purchase Prices: Placebo RoundBuyPrice Reassignment**

This table reports estimates from a placebo reassignment of *RoundBuyPrice* across stocks within investor. The dependent variable equals one when a stock is sold on a given day and zero otherwise (percentage points). *RoundBuyPrice* equals one if the most recent purchase price is a whole dollar ending in 0 or 5. Column (1) uses observed *RoundBuyPrice* in the baseline specification, Column (3) of Table 3. Columns (2)–(4) report three independent within-investor reassignments of each investor-stock *RoundBuyPrice* label across stocks held by the same investor, holding Gain and other regressors fixed. Other variables follow Table 3. Coefficients are multiplied by 100. *t*-statistics are in parentheses; standard errors are double-clustered by investor and day. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	Baseline (1)	Placebo 1 (2)	Placebo 2 (3)	Placebo 3 (4)
RoundBuyPrice	-1.143***	-0.557	-0.107	-0.162
	(-3.30)	(-1.64)	(-0.31)	(-0.47)
Gain	3.886***	3.945***	3.975***	3.959***
	(16.38)	(16.63)	(16.62)	(16.63)
RoundBuyPrice × Gain	1.997***	0.819	0.196	0.528
	(3.90)	(1.59)	(0.40)	(1.03)
Log(BuyPrice)	-0.046	-0.048	-0.046	-0.044
	(-0.12)	(-0.13)	(-0.12)	(-0.12)
Return(+)	-0.401***	-0.403***	-0.402***	-0.402***
	(-2.88)	(-2.90)	(-2.89)	(-2.89)
Return(−)	-7.176***	-7.129***	-7.117***	-7.114***
	(-5.87)	(-5.82)	(-5.81)	(-5.81)
$\sqrt{TimeOwned}$	-0.039***	-0.039***	-0.039***	-0.039***
	(-2.71)	(-2.70)	(-2.70)	(-2.70)
Volatility(+)	62.337	62.158	62.277	62.218
	(1.24)	(1.23)	(1.24)	(1.24)
Volatility(−)	-121.542**	-121.589**	-121.419**	-121.459**
	(-2.41)	(-2.41)	(-2.41)	(-2.41)
Investor FE	Yes	Yes	Yes	Yes
Stock × SellDate FE	Yes	Yes	Yes	Yes
Observations	1,393,400	1,393,400	1,393,400	1,393,400
Adjusted R^2	0.190	0.190	0.190	0.190

t-statistics in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table IA.9: **Disposition Effect and Distance to Round Purchase Prices**

The dependent variable equals one if the stock is sold on day t and zero otherwise (percentage points). *RoundBuyPrice* equals one if the most recent purchase price is a whole dollar ending in 0 or 5. *Gain* equals one if the return since purchase is positive. Purchase prices are winsorized at the 1st and 99th percentiles. *Dist*\$1 is the absolute distance from the purchase price to the nearest whole-dollar price. *Dist*\$5 is the absolute distance to the nearest *RoundBuyPrice* grid point (a whole dollar ending in 0 or 5). *Near*\$1 equals one if *Dist*\$1 $\leq$ \$0.05. *Near*\$5 equals one if *Dist*\$5 $\leq$ \$0.05. Column (1) reports *RoundBuyPrice* $\times$ *Gain*. Column (2) reports *Dist*\$1 $\times$ *Gain*. Column (3) reports *Dist*\$5 $\times$ *Gain*. Column (4) reports *Near*\$1 $\times$ *Gain*. Column (5) reports *Near*\$5 $\times$ *Gain*. Controls, fixed effects, and clustering follow Table 3. Coefficients are multiplied by 100. t-statistics are in parentheses. Standard errors are double-clustered by investor and day. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. N = 1,393,400 investor-stock-days.

	Baseline (1)	Dist\$1 (2)	Dist\$5 (3)	Near\$1 (4)	Near\$5 (5)
RoundBuyPrice	-1.143***				
	(-3.30)				
Gain	3.886***	4.369***	4.361***	3.655***	3.888***
	(16.38)	(15.87)	(13.82)	(14.83)	(16.40)
RoundBuyPrice $\times$ Gain	1.997***				
	(3.90)				
Dist\$1		0.852*			
		(1.93)			
Dist\$1 $\times$ Gain		-1.661***			
		(-2.59)			
Dist\$5			0.140		
			(1.34)		
Dist\$5 $\times$ Gain			-0.301**		
			(-2.01)		
Near\$1				-0.724***	
				(-4.07)	
Near\$1 $\times$ Gain				1.485***	
				(5.64)	
Near\$5					-1.085***
					(-3.14)
Near\$5 $\times$ Gain					1.936***
					(3.79)
Controls	Yes	Yes	Yes	Yes	Yes
Investor FE	Yes	Yes	Yes	Yes	Yes
Stock $\times$ SellDate FE	Yes	Yes	Yes	Yes	Yes
Observations	1,393,400	1,393,400	1,393,400	1,393,400	1,393,400
Adjusted R^2	0.190	0.190	0.190	0.190	0.190

t-statistics in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table IA.10: **Price Clustering Frequency Distributions in the $1/8th Tick-Size Era for Four Frequently Traded Stocks** (April 10–May 30, 1996)

The table reports the distribution of household stock *purchase* prices in each eighth-of-a-dollar bin during the final $1/8 tick-size regime. Sample firms are Intel, Cisco Systems, Micron Technology, and Sun Microsystems, selected as frequently traded names in the brokerage data over April 10–May 30, 1996. N is the number of purchase-price observations for each firm in that window. Price range reports the minimum and maximum purchase price (in dollars). Column $\frac{k}{8}$ gives the percentage of observations whose fractional part rounds to $k/8$ ($k = 0, \ldots, 7$).

Security	Price range	N	$\frac{0}{8}$	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{3}{8}$	$\frac{4}{8}$	$\frac{5}{8}$	$\frac{6}{8}$	$\frac{7}{8}$
CISCO SYSTEMS INC	20–95	685	18.1%	13.6%	10.4%	8.6%	16.6%	7.7%	16.1%	8.9%
SUN MICROSYSTEMS INC	20–102	510	17.3%	13.5%	13.3%	6.9%	9.6%	7.5%	17.3%	14.7%
MICRON TECHNOLOGY INC	28–92	934	20.3%	10.4%	13.6%	8.8%	13.9%	10.2%	11.2%	11.6%
INTEL CORP	50–118	1,231	25.3%	7.3%	10.9%	7.9%	14.5%	8.4%	13.9%	11.9%

Table IA.11: **Disposition Effect and Round Purchase Prices: Purchase-Price Level**

The dependent variable equals one if the stock is sold on day t and zero otherwise (percentage points). *RoundBuyPrice* equals one if the most recent purchase price is a whole dollar ending in 0 or 5. *Gain* equals one if the return since purchase is positive. Purchase prices are winsorized at the 1st and 99th percentiles. *BuyPrice* denotes that winsorized purchase price. *HighBuyPrice* equals one if *BuyPrice* is at or above the sample median. *MidBuyPrice* equals one if *BuyPrice* lies between the 25th and 75th percentiles (inclusive). *PriceQ*2, *PriceQ*3, and *PriceQ*4 are indicators for the second, third, and fourth *BuyPrice* quartiles; the first quartile is the omitted reference group. *LogBuyPrice* is the natural logarithm of *BuyPrice*, demeaned in the estimation sample. Column (1) reports *RoundBuyPrice* × *Gain*. Column (2) interacts *RoundBuyPrice* × *Gain* with *HighBuyPrice*. Column (3) interacts *RoundBuyPrice* × *Gain* with *MidBuyPrice*. Column (4) interacts *RoundBuyPrice* × *Gain* with *PriceQ*2, *PriceQ*3, and *PriceQ*4. Column (5) interacts *RoundBuyPrice* × *Gain* with *LogBuyPrice*. All columns use the full estimation sample. Controls, fixed effects, and clustering follow Table 3. Coefficients are multiplied by 100. t-statistics are in parentheses. Standard errors are double-clustered by investor and day. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. N = 1,393,400 investor-stock-days.

	Baseline (1)	HighPrice (2)	MidPrice (3)	Quartiles (4)	Cont. Price (5)
RoundBuyPrice	-1.143***	-1.405**	-1.134**	0.512	-1.114***
	(-3.30)	(-2.33)	(-2.36)	(0.48)	(-2.85)
Gain	3.886***	3.916***	4.171***	4.222***	3.860***
	(16.38)	(12.24)	(15.21)	(9.49)	(15.68)
RoundBuyPrice × Gain	1.997***	2.035**	1.651**	0.182	1.935***
	(3.90)	(2.08)	(2.25)	(0.11)	(3.21)
HighBuyPrice		0.163			
		(0.52)			
RoundBuyPrice × Gain × HighBuyPrice		-0.055			
		(-0.05)			
MidBuyPrice			-0.229		
			(-1.03)		
RoundBuyPrice × Gain × MidBuyPrice			0.697		
			(0.70)		
PriceQ2				-0.103	
				(-0.25)	
RoundBuyPrice × Gain × PriceQ2				2.674	
				(1.36)	
PriceQ3				-0.149	
				(-0.29)	
RoundBuyPrice × Gain × PriceQ3				1.760	
				(0.98)	
PriceQ4				0.109	
				(0.17)	
RoundBuyPrice × Gain × PriceQ4				1.809	
				(1.00)	
LogBuyPrice					-0.075
					(-0.19)
RoundBuyPrice × Gain × LogBuyPrice					0.155
					(0.21)
Controls	Yes	Yes	Yes	Yes	Yes
Investor FE	Yes	Yes	Yes	Yes	Yes
Stock × SellDate FE	Yes	Yes	Yes	Yes	Yes
Observations	1,393,400	1,393,400	1,393,400	1,393,400	1,393,400
Adjusted R^2	0.190	0.190	0.190	0.190	0.190

t-statistics in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

II.6 Buy-Day Market Conditions

Table IA.12: **Disposition Effect and Round Purchase Prices: Buy-Day Market Conditions**

This table reports estimation results that relate round purchase prices and the disposition effect to buy-day market conditions. The dependent variable equals 1 if a stock is sold (fully or partially) on a given day and 0 otherwise (scaled to percentage points). *RoundBuyPrice* equals one if the most recent purchase price has no fractional part and the integer amount ends in 0 or 5. Buy-day covariates are measured on the most recent purchase date: relative spread equals $(\text{ASKHI} - \text{BIDLO})/((\text{ASKHI} + \text{BIDLO})/2)$; turnover equals share volume divided by shares outstanding; price-in-range equals $(\text{purchase price} - \text{BIDLO})/(\text{ASKHI} - \text{BIDLO})$. High indicators equal one when the buy-day covariate is at or above its sample median. Column (1) is the baseline specification, Column (3) of Table 3. Column (2) adds continuous buy-day relative spread, turnover, and price-in-range. Columns (3)–(4) interact *RoundBuyPrice* × *Gain* with the high buy-day spread and high buy-day price-in-range indicators, respectively. Other variable definitions follow Table 3. All coefficients are multiplied by 100. t-statistics in parentheses; standard errors are double-clustered at the investor and day level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	Baseline	Buy-day controls	× High spread	× High price-in-range
	(1)	(2)	(3)	(4)
RoundBuyPrice	-1.143***	-1.120***	-0.761	-0.725
	(-3.30)	(-3.22)	(-1.62)	(-1.42)
Gain	3.886***	3.843***	3.501***	4.052***
	(16.38)	(16.24)	(13.82)	(15.31)
RoundBuyPrice × Gain	1.997***	1.966***	2.208***	1.541**
	(3.90)	(3.83)	(3.22)	(2.22)
HighBuySpread			-0.403**	
			(-2.10)	
RoundBuyPrice × Gain × HighBuySpread			-0.568	
			(-0.58)	
HighBuyPriceInRange				-0.094
				(-0.56)
RoundBuyPrice × Gain × HighBuyPriceInRange				0.802
				(0.79)
BuyRelSpread		-0.336		
		(-0.16)		
BuyTurnover		-7.608***		
		(-2.98)		
BuyPriceInRange		-0.629***		
		(-3.91)		
Controls	Yes	Yes	Yes	Yes
Investor FE	Yes	Yes	Yes	Yes
Stock × SellDate FE	Yes	Yes	Yes	Yes
Observations	1,393,400	1,383,059	1,383,059	1,383,059
Adjusted R^2	0.190	0.191	0.191	0.191

t-statistics in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

II.7 Holding Time

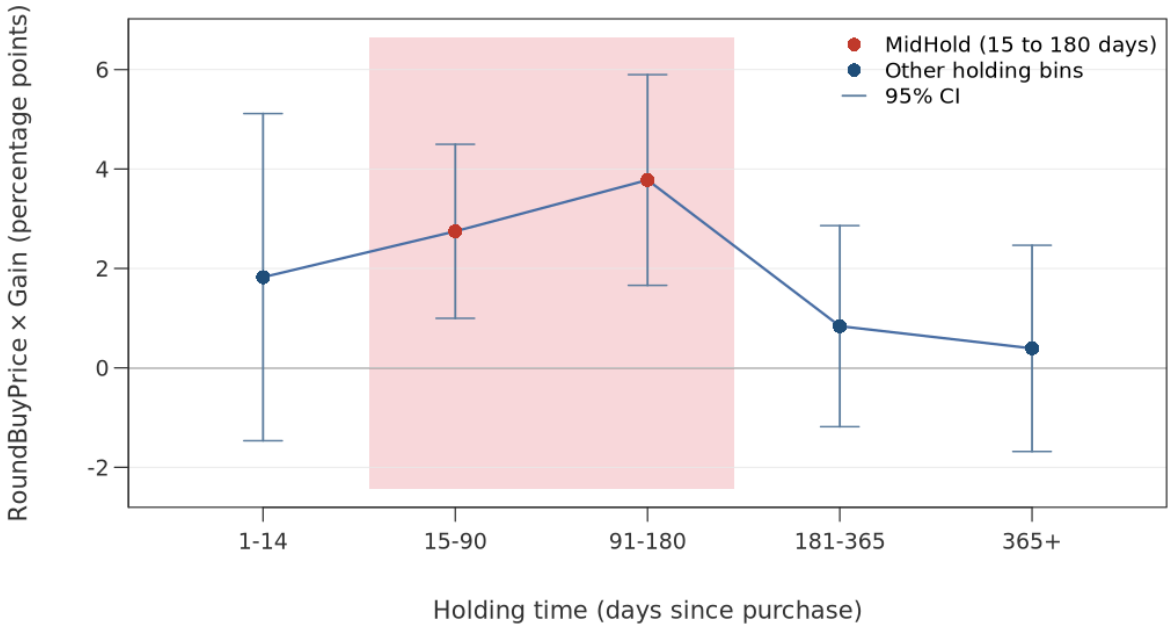


Figure IA.4: **Round Purchase Prices and the Disposition Effect by Holding Time**

The dependent variable is the sell indicator in percentage points. Let $R_{ijt} \equiv RoundBuyPrice_{ijt}$ and $G_{ijt} \equiv Gain_{ijt}$. The figure estimates

$$\begin{aligned} Sell100_{ijt} = {} & \beta_R R_{ijt} + \beta_G G_{ijt} + \beta_{RG}\, R_{ijt} \times G_{ijt} \\ & + \sum_{k \in \mathcal{B} \setminus \{r\}} \left(\alpha_k H^k_{ijt} + \beta_{Rk}\, R_{ijt} \times H^k_{ijt} + \beta_{Gk}\, G_{ijt} \times H^k_{ijt} + \gamma_k\, R_{ijt} \times G_{ijt} \times H^k_{ijt} \right) \\ & + X'_{ijt}\lambda + \mu_i + \delta_{j \times t} + \varepsilon_{ijt}, \end{aligned}$$

where $\mathcal{B} = \{1 \text{ to } 14,\ 15 \text{ to } 90,\ 91 \text{ to } 180,\ 181 \text{ to } 365,\ 365+\}$ are holding-time bins in calendar days since purchase and the omitted reference is $r = 365+$. X_{ijt} includes $\log(BuyPrice)$, $Return(+)$, $Return(-)$, $TimeOwned$, $Volatility(+)$, and $Volatility(-)$. Each plotted point is the implied $R \times G$ effect in that bin (β_{RG} in the reference bin; $\beta_{RG} + \gamma_k$ otherwise). The shaded region is $MidHold$ (15 to 180 days). Vertical bars are 95% confidence intervals; standard errors are double-clustered by investor and day. $N = 1{,}393{,}400$ investor-stock-days.

Table IA.13: **Disposition Effect and Round Purchase Prices: Holding Time**

This table reports estimates that relate round purchase prices and the disposition effect to holding time. The dependent variable equals one when a stock is sold on a given day and zero otherwise (percentage points). *RoundBuyPrice* equals one if the most recent purchase price is a whole dollar ending in 0 or 5. *Gain* equals one if the return since purchase is positive. Holding time is calendar days since purchase. Holding-time windows follow Figure IA.4: 1 to 14, 15 to 90, 91 to 180, 181 to 365, and more than 365 days. *MidHold* equals one when holding time is between 15 and 180 days. *EarlyHold* equals one when holding time is below 15 days. *LateHold* equals one when holding time exceeds 180 days. *Hold*[15, 90] and *Hold*[91, 180] are the two MidHold sub-bins in that figure. Column (1) is the baseline specification, Column (3) of Table 3. Column (2) interacts *RoundBuyPrice* × *Gain* with *MidHold*. Column (3) interacts *RoundBuyPrice* × *Gain* with *EarlyHold* and *LateHold*, omitting the 15 to 180 day interval as the reference group. Column (4) interacts *RoundBuyPrice* × *Gain* with *Hold*[15, 90] and *Hold*[91, 180]. Other variables follow Table 3. Coefficients are multiplied by 100. *t*-statistics are in parentheses; standard errors are double-clustered by investor and day. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	Baseline	× MidHold	Early / Late vs Mid	× MidHold sub-bins
	(1)	(2)	(3)	(4)
RoundBuyPrice	-1.143***	-1.004**	-1.273***	-1.008**
	(-3.30)	(-2.14)	(-2.71)	(-2.15)
Gain	3.886***	3.024***	4.814***	3.013***
	(16.38)	(11.61)	(17.19)	(11.58)
RoundBuyPrice × Gain	1.997***	0.806	3.160***	0.804
	(3.90)	(1.18)	(4.35)	(1.17)
MidHold		-0.512**		
		(-2.22)		
RoundBuyPrice × Gain × MidHold		2.356**		
		(2.42)		
EarlyHold			-1.422***	
			(-3.78)	
RoundBuyPrice × Gain × EarlyHold			-1.335	
			(-0.72)	
LateHold			1.715***	
			(6.33)	
RoundBuyPrice × Gain × LateHold			-2.547**	
			(-2.50)	
Hold[15,90]				-0.621**
				(-2.32)
RoundBuyPrice × Gain × Hold[15,90]				1.937*
				(1.76)
Hold[91,180]				-0.335
				(-1.32)
RoundBuyPrice × Gain × Hold[91,180]				2.989**
				(2.35)
Controls	Yes	Yes	Yes	Yes
Investor FE	Yes	Yes	Yes	Yes
Stock × SellDate FE	Yes	Yes	Yes	Yes
Observations	1,393,400	1,393,400	1,393,400	1,393,400
Adjusted R^2	0.190	0.190	0.191	0.190

t-statistics in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table IA.14: **Propensity to Sell Gains and Buy Price Roundness**

This table replicates the tests from Table 3 but distinguishes round buy prices by whether they end in 5 (Panel A) or 0 (Panel B) to examine whether the propensity to sell at gains differs across these two types of roundness. In Panel A, *BuyPriceEnd*5 is a dummy variable equal to 1 if the purchase price has no fractional part (cents = 00) and the integer amount ends in 5; otherwise, it is 0. In Panel B, *BuyPriceEnd*0 is a dummy variable equal to 1 if the purchase price has no fractional part (cents = 00) and the integer amount ends in 0; otherwise, it is 0. Other variable definitions are provided in the caption of Table 3. All coefficients are multiplied by 100. t-statistics in parentheses; standard errors are double-clustered at the investor and day level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A: Round Buy Price Ends in 5

	(1)	(2)	(3)	(4)	(5)
BuyPriceEnd5	-0.710**	-0.851*	-1.773***	-1.622**	-1.375***
	(-2.29)	(-1.91)	(-3.82)	(-2.26)	(-3.62)
Gain	7.655***	3.959***	3.885***	4.142***	4.786***
	(109.70)	(16.88)	(16.38)	(15.57)	(15.57)
BuyPriceEnd5 × Gain	1.080**	1.127*	2.460***	1.054	1.577**
	(2.42)	(1.79)	(3.49)	(1.47)	(2.57)
Controls	No	Yes	Yes	Yes	Yes
Investor FE	No	No	Yes	No	No
SellDate FE	No	No	No	Yes	No
Stock FE	No	No	No	No	Yes
Investor × SellDate FE	No	No	No	No	Yes
Investor × Stock FE	No	No	No	Yes	No
Stock × SellDate FE	No	No	Yes	No	No
Observations	1,393,400	1,393,400	1,393,400	1,393,400	1,393,400
Adjusted R^2	0.009	0.017	0.190	0.297	0.077

t-statistics in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

II.8 Round Prices Ending in 0 versus 5

Panel B: Round Buy Price Ends in 0

	(1)	(2)	(3)	(4)	(5)
BuyPriceEnd0	-0.029	-0.340	-0.526	-0.322	-0.414
	(-0.09)	(-0.66)	(-1.11)	(-0.41)	(-1.00)
Gain	7.655***	3.959***	3.885***	4.142***	4.786***
	(109.70)	(16.88)	(16.38)	(15.57)	(15.57)
BuyPriceEnd0 × Gain	1.250***	1.799**	1.551**	1.426**	1.766***
	(2.71)	(2.46)	(2.25)	(1.99)	(2.85)
Controls	No	Yes	Yes	Yes	Yes
Investor FE	No	No	Yes	No	No
SellDate FE	No	No	No	Yes	No
Stock FE	No	No	No	No	Yes
Investor × SellDate FE	No	No	No	No	Yes
Investor × Stock FE	No	No	No	Yes	No
Stock × SellDate FE	No	No	Yes	No	No
Observations	1,393,400	1,393,400	1,393,400	1,393,400	1,393,400
Adjusted R^2	0.009	0.017	0.190	0.297	0.077

t-statistics in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

III Stock Splits and Memory Disruption

Table IA.15: **Covariate Balance in the Stock-Split Sample**

This table reports means and standard deviations (in parentheses) for investor-stock holding spells in which the underlying security experiences a within-spell forward stock split between January 1991 and December 1996. *Always* denotes spells whose purchase price is round both before and after applying the split ratio. *Breaks* denotes spells whose purchase price is round before the split but non-round after applying the split ratio. *Never* denotes the remaining spells. Holding days are the number of calendar days in the spell. Position and portfolio values are in dollars. Share 2-for-1 splits is the fraction of spells with a 2-for-1 split ratio.

	Always	Breaks	Never
Holding days	624.0	766.8	810.2
	(573.3)	(590.3)	(612.3)
$\sqrt{\text{TimeOwned}}$	12.925	14.294	14.763
	(6.654)	(6.335)	(6.567)
Return since purchase	0.317	0.181	0.266
	(0.679)	(0.348)	(0.559)
Volatility (252d)	0.029	0.026	0.026
	(0.010)	(0.011)	(0.011)
Position value ($)	11404.9	9845.2	8608.4
	(18519.7)	(41042.4)	(28264.2)
Portfolio value ($)	113616.5	75776.6	87699.1
	(261888.0)	(144682.3)	(288185.5)
Purchase price	36.88	30.18	31.81
	(20.19)	(15.89)	(19.44)
Share 2-for-1 splits	0.972	0.458	0.604
Split ratio	2.06	17.99	17.75
	(0.33)	(45.97)	(116.47)
Spells	144	301	8,849
Investor-stock-days	61,377	155,333	4,866,097

Table IA.16: **Stock Splits and Round Purchase Prices: Always and Breaks Triple Interactions**

This table reports investor-stock-day estimates for holdings with a within-spell forward stock split between January 1991 and December 1996. The dependent variable equals 1 if a stock is sold (fully or partially) on a given day and 0 otherwise (scaled to percentage points). *RoundBuyPrice_Always* equals one if the purchase price is round both before and after applying the split ratio. *RoundBuyPrice_Breaks* equals one if the purchase price is round before the split but non-round after applying the split ratio. *Split* equals one if a within-spell split ex-date falls in the prior 30 calendar days and the position was purchased at least seven days before that ex-date. Column (1) interacts *RoundBuyPrice_Always*, Gain, and *Split*. Column (2) adds log purchase price, return components, time owned, and volatility controls. Column (3) interacts *RoundBuyPrice_Breaks*, Gain, and *Split*. Columns (4) and (5) include both Always and Breaks triple interactions, without and with controls. All specifications include stock-by-sell-date and stock-by-investor fixed effects. t-statistics are in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)
RoundBuyPrice_Always	0.120	-0.248		0.117	-0.248
	(0.38)	(-0.77)		(0.37)	(-0.77)
Gain	0.113***	0.105***	0.115***	0.112***	0.104***
	(12.15)	(5.44)	(12.24)	(11.88)	(5.34)
Split	0.378***	0.370***	0.376***	0.379***	0.372***
	(5.74)	(5.58)	(5.70)	(5.75)	(5.59)
RoundBuyPrice_Always × Gain	0.258***	0.254***		0.260***	0.256***
	(3.90)	(3.80)		(3.93)	(3.83)
RoundBuyPrice_Always × Split	-0.497**	-0.491**		-0.499**	-0.492**
	(-2.37)	(-2.32)		(-2.38)	(-2.33)
Gain × Split	-0.081***	-0.087***	-0.082***	-0.086***	-0.092***
	(-2.88)	(-3.03)	(-2.86)	(-3.01)	(-3.17)
RoundBuyPrice_Always × Gain × Split	0.483**	0.478**		0.487**	0.483**
	(2.07)	(2.03)		(2.09)	(2.05)
RoundBuyPrice_Breaks			0.582	0.592*	0.470
			(1.62)	(1.65)	(1.30)
RoundBuyPrice_Breaks × Gain			0.027	0.030	0.038
			(0.72)	(0.79)	(1.00)
RoundBuyPrice_Breaks × Split			-0.037	-0.040	-0.050
			(-0.38)	(-0.42)	(-0.52)
RoundBuyPrice_Breaks × Gain × Split			0.140	0.143	0.155
			(1.21)	(1.23)	(1.32)
Controls	No	Yes	No	No	Yes
Stock × SellDate FE	Yes	Yes	Yes	Yes	Yes
Stock × Investor FE	Yes	Yes	Yes	Yes	Yes
Observations	4,847,890	4,769,710	4,847,890	4,847,890	4,769,710
Adjusted R^2	0.024	0.024	0.024	0.024	0.024

t-statistics in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table IA.17: **Stock Split and Memory Disruption: Additional Specifications**

This table reports estimation results at the investor-stock-day level for holdings with a within-spell forward stock split between January 1991 and December 1996. The dependent variable equals 1 if a stock is sold (fully or partially) on a given day and 0 otherwise (scaled to percentage points). *RoundBuyPrice_Always* equals one if the purchase price is round both before and after applying the split ratio. *RoundBuyPrice_Breaks* equals one if the purchase price is round before the split but non-round after applying the split ratio. Column (1) retains post-split days only. Column (2) excludes observations for which a within-spell split ex-date falls in the prior 30 calendar days and the position was purchased at least seven days before that ex-date. Column (3) excludes the split ex-date and a $\pm$5 calendar-day window around it. Always$\times$Gain $-$ Breaks$\times$Gain is the difference between the two interaction coefficients. Other variable definitions follow Table 6. All coefficients are multiplied by 100. t-statistics in parentheses; standard errors are double-clustered at the investor and stock level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	Post-split only (1)	Excl. 30d ex-date window (2)	Excl. $\pm$5d window (3)
RoundBuyPrice_Always	-0.504	-0.392	-0.462
	(-0.53)	(-0.74)	(-1.01)
RoundBuyPrice_Breaks	0.814	1.010***	0.346
	(1.34)	(2.68)	(0.75)
Gain	0.006	0.063***	0.052**
	(0.13)	(2.92)	(2.05)
RoundBuyPrice_Always $\times$ Gain	0.372***	0.279***	0.303***
	(2.66)	(2.92)	(2.96)
RoundBuyPrice_Breaks $\times$ Gain	0.047	0.017	0.041
	(0.81)	(0.45)	(1.02)
Always$\times$Gain $-$ Breaks$\times$Gain	0.324**	0.262***	0.262**
	(2.29)	(2.62)	(2.54)
Controls	Yes	Yes	Yes
Stock $\times$ SellDate FE	Yes	Yes	Yes
Stock $\times$ Investor FE	Yes	Yes	Yes
Observations	3,582,657	4,679,329	4,933,868
Adjusted R^2	0.046	0.028	0.024

t-statistics in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table IA.18: **Stock-Split Sample: Spells, Investors, Stocks, and Split Events**

This table reports the number of holding spells, investors, stocks, split events (a stock and its split date), and investor-stock-days in the estimation sample of Table 6, Column (4), separately for the Always, Breaks, and Never groups. The lower panel reports, for Always and Breaks, the share of spells that belong to the largest stocks, the largest split events, and the largest investors.

	Always	Breaks	Never
Spells	144	301	8,849
Investors	138	292	5,995
Stocks	90	168	752
Split events (stock × split date)	99	193	1,138
Investor-stock-days	61,377	155,333	4,866,097
Days per spell	426.2	516.1	549.9
Share of spells in the largest stock	5.6%	3.3%	–
Share of spells in the 5 largest stocks	19.4%	14.3%	–
Share of spells in the 10 largest stocks	29.9%	23.6%	–
Share of spells in the 5 largest split events	12.5%	11.6%	–
Share of spells held by the 5 largest investors	6.9%	4.0%	–

Table IA.19: **Stock-Split Estimates under Alternative Clustering**

This table re-estimates the specification of Table 6, Column (4), and changes only how standard errors are clustered. The dependent variable is the sell indicator in percentage points. The point estimates are identical across rows. A spell is an investor-stock holding spell. A split event is a stock and its split date. The column Clusters reports the number of clusters in each dimension. t-statistics are in parentheses. The last column reports the two-sided p-value of the test that the Always $\times$ Gain and Breaks $\times$ Gain coefficients are equal. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

	Clusters	Always×Gain	Breaks×Gain	Difference	p-value
Investor + stock (baseline)	6,209 / 757	0.281***	0.041	0.240**	0.018
		(2.80)	(1.03)	(2.36)	
Investor	6,209	0.281***	0.041	0.240**	0.016
		(3.01)	(1.09)	(2.41)	
Stock	757	0.281***	0.041	0.240**	0.018
		(2.81)	(1.03)	(2.36)	
Spell	9,294	0.281***	0.041	0.240**	0.016
		(3.01)	(1.09)	(2.41)	
Split event	1,151	0.281***	0.041	0.240**	0.030
		(2.64)	(0.93)	(2.17)	
Investor + stock + date	6,209 / 757 / 1,517	0.281***	0.041	0.240**	0.020
		(2.74)	(1.02)	(2.33)	

Table IA.20: **Stock-Split Estimates after Dropping Stocks with Many Always or Breaks Spells**

This table re-estimates the specification of Table 6, Column (4), after dropping stocks ranked by their number of Always and Breaks spells combined. The first row repeats the full-sample estimate. Standard errors are double-clustered by investor and stock. t-statistics are in parentheses. The last two columns report the number of Always and Breaks spells that remain. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

	Always×Gain	Breaks×Gain	Difference	# Always	# Breaks
None (full sample)	0.281***	0.041	0.240**	144	301
	(2.80)	(1.03)	(2.36)		
Drop stock ranked 1	0.281***	0.040	0.241**	136	294
	(2.68)	(1.00)	(2.26)		
Drop stock ranked 2	0.279***	0.042	0.237**	144	291
	(2.78)	(1.01)	(2.31)		
Drop stock ranked 3	0.273***	0.043	0.230**	136	299
	(2.63)	(1.09)	(2.20)		
Drop stock ranked 4	0.280***	0.049	0.231**	144	291
	(2.79)	(1.20)	(2.26)		
Drop stock ranked 5	0.291***	0.048	0.243**	143	293
	(2.86)	(1.14)	(2.33)		
Drop stocks ranked 1–5	0.280**	0.062	0.217*	127	264
	(2.54)	(1.32)	(1.93)		

IV Market-Level Trading Intensity

IV.1 Covariate Balance and Trading Robustness

Table IA.21: **Round versus Non-round Covariate Balance**

This table reports within-firm adjacent-fiscal-quarter differences in firm-quarter covariates. The sample retains consecutive fiscal-quarter earnings announcements for the same PERMNO in which *Round* differs between the two quarters. For each pair, Round (Non-round) is the covariate value in the quarter with $Round = 1$ ($Round = 0$). Diff. is the within-pair Round minus Non-round difference. The *t*-statistic is for the mean Diff., clustered by firm. $Round = 1$ when the median IBES consensus, rounded to two decimals, is a nonzero multiple of \$0.05. Size is the natural log of Compustat market capitalization. B/M decile is the within-calendar-quarter decile of book-to-market. Institutional ownership is lagged 13F holdings over shares outstanding. *N* analysts is the IBES number of estimates. σ_{est} is the IBES standard deviation of estimates. Reporting lag is announcement date minus fiscal period end, in days. Earnings volatility is the standard deviation of quarterly EPS over the prior eight quarters. Sample construction matches Table 11.

Variable	Round	Non-round	Diff.	*t*-stat	*N* pairs
ln(Size)	13.841	13.840	0.000	0.03	102,358
B/M decile	5.527	5.517	0.010	3.12	102,327
Institutional ownership	0.264	0.264	0.000	0.84	100,945
N analysts	7.686	7.684	0.002	0.46	103,206
$\ln(1+\sigma_{\text{est}})$	0.051	0.052	-0.001	-3.12	103,206
Reporting lag (days)	31.131	31.157	-0.026	-0.52	103,206
Earnings volatility	0.429	0.430	-0.001	-1.57	101,168
Switch pairs (all covariates)	103,206				
Firms	8,983				

Within-pair Round minus Non-round differences; *t*-statistics clustered by firm. *N* pairs is the non-missing count for that covariate.

Table IA.22: **Announcement-Window Trading: Firm Fixed Effects**

This table reports OLS estimates of retail trade flow and log(Turnover) averaged over trading days $[0, +1]$ on *Round*, standardized absolute percent surprise SR^{abs}, and *Round* $\times$ SR^{abs}. Columns (1) and (3) include firm fixed effects. Columns (2) and (4) include firm-by-year fixed effects. All columns include the spillover control block used in Table 10 columns (2) and (4). *Round* $= 1$ when the median IBES consensus, rounded to two decimals, is a nonzero multiple of \$0.05. Sample construction matches Table 10. Standard errors are clustered by announcement date. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

	Retail trade flow		log(Turnover)	
	Firm FE	Firm×Year FE	Firm FE	Firm×Year FE
	(1)	(2)	(3)	(4)
Round	0.013**	0.008	0.006**	0.004
	(2.07)	(1.48)	(2.00)	(1.43)
SR^{abs}	0.021***	0.028***	0.013***	0.019***
	(5.77)	(8.69)	(8.06)	(12.66)
Round $\times$ SR^{abs}	0.023**	0.019**	0.034***	0.023***
	(2.15)	(2.02)	(7.24)	(5.41)
Controls	Yes	Yes	Yes	Yes
Observations	87,843	86,729	345,861	339,553
Adj. R^2	0.174	0.043	0.374	0.023

t-statistics in parentheses; clustered by announcement date.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table IA.23: **Announcement-Window Trading: Abnormal Retail Flow and Turnover**

This table reports OLS estimates of abnormal retail trade flow and abnormal log(Turnover) on *Round*, standardized absolute percent surprise SR^{abs}, and *Round* $\times$ SR^{abs}. For each pre-announcement baseline $W \in \{[-21,-2],[-30,-11],[-40,-11]\}$, abnormal retail trade flow equals the average of daily retail trade flow over trading days $[0,+1]$ minus the average over trading days in W, and abnormal log(Turnover) equals the analogous difference for daily log(VOL/SHROUT). Each abnormal series is winsorized at the 1st and 99th percentiles. Columns (1)–(3) use abnormal retail trade flow; columns (4)–(6) use abnormal log(Turnover). Within each outcome, the three columns use baselines $[-21,-2]$, $[-30,-11]$, and $[-40,-11]$, respectively. All columns include the spillover control block used in Table 10 columns (2) and (4). *Round* $= 1$ when the median IBES consensus, rounded to two decimals, is a nonzero multiple of \$0.05. Sample construction matches Table 10. Standard errors are clustered by announcement date. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

	Abnormal retail trade flow			Abnormal log(Turnover)		
	$[-21,-2]$	$[-30,-11]$	$[-40,-11]$	$[-21,-2]$	$[-30,-11]$	$[-40,-11]$
	(1)	(2)	(3)	(4)	(5)	(6)
Round	-0.015**	-0.011*	-0.010	-0.006**	-0.004	-0.004
	(-2.51)	(-1.82)	(-1.60)	(-2.44)	(-1.51)	(-1.47)
SR^{abs}	0.040***	0.041***	0.039***	0.022***	0.024***	0.024***
	(12.08)	(11.61)	(11.40)	(15.90)	(15.77)	(15.83)
Round $\times$ SR^{abs}	0.024**	0.033***	0.031***	0.024***	0.025***	0.024***
	(2.51)	(3.21)	(2.94)	(5.96)	(5.88)	(5.58)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	86,752	86,499	86,404	345,681	345,671	345,654
Adj. R^2	0.179	0.183	0.187	0.107	0.104	0.104

t-statistics in parentheses; clustered by announcement date.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table IA.24: **Announcement-Window Trading: Alternative Surprise Scalings**

This table reports OLS estimates of retail trade flow and log(Turnover) averaged over trading days $[0, +1]$ on *Round*, absolute-surprise decile SR^{abs}, and *Round* $\times$ SR^{abs}. Columns (1)–(3) use retail trade flow; columns (4)–(6) use log(Turnover). In columns (1) and (4), signed surprise equals $(Actual - Consensus)/|P|$, where P is the last CRSP closing price strictly before the announcement trading day. In columns (2) and (5), signed surprise equals $(Actual - Consensus)/\sigma_{\text{EPS}}$, where σ_{EPS} is the trailing Compustat quarterly EPS standard deviation. In columns (3) and (6), signed surprise equals $Actual - Consensus$ in dollars. SR^{abs} is the within-calendar-quarter decile of the absolute value of that signed surprise (1 = smallest, 10 = largest), after winsorizing the signed series at the 1st and 99th percentiles. All columns include the spillover control block used in Table 10 columns (2) and (4), with FE_q defined as the within-calendar-quarter decile of the same signed surprise. *Round* $= 1$ when the median IBES consensus, rounded to two decimals, is a nonzero multiple of \$0.05. Sample construction matches Table 10. Standard errors are clustered by announcement date. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

	Retail trade flow			log(Turnover)		
	$\lvert S\rvert/P$	$\lvert S\rvert/\sigma_{\text{EPS}}$	$\lvert S\rvert_{\$}$	$\lvert S\rvert/P$	$\lvert S\rvert/\sigma_{\text{EPS}}$	$\lvert S\rvert_{\$}$
	(1)	(2)	(3)	(4)	(5)	(6)
Round	0.020	-0.028*	-0.008	0.010	-0.003	0.015**
	(1.26)	(-1.78)	(-0.48)	(1.56)	(-0.49)	(2.11)
SR^{abs}	0.072***	0.017***	0.030***	0.037***	0.020***	0.031***
	(39.86)	(9.97)	(18.18)	(50.17)	(29.48)	(46.90)
Round $\times$ SR^{abs}	0.001	0.012***	0.006**	-0.003**	0.001	-0.004***
	(0.32)	(4.20)	(2.35)	(-2.37)	(0.62)	(-3.59)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	88,033	88,033	88,033	346,313	346,310	346,313
Adj. R^2	0.527	0.515	0.518	0.598	0.595	0.598

t-statistics in parentheses; clustered by announcement date.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

V Market-Level Announcement Returns

V.1 Windows, Definitions, and Matching

Table IA.25: **Pre-Announcement Cumulative Abnormal Returns**

This table reports OLS estimates of pre-announcement cumulative abnormal returns on *Round*, standardized earnings surprise, and their interaction. Dependent variables are market-adjusted CAR over trading-day windows $[-5,-1]$ and $[-10,-1]$, winsorized at the 1% and 99% levels and scaled to percent. Panel A restricts to Surprise > 0 and includes *Round*, standardized SR^+, and *Round* $\times$ SR^+. Panel B restricts to Surprise ≤ 0 and includes *Round*, standardized SR^-, and *Round* $\times$ SR^-. $SR^+ = \max(Surprise, 0)$ and $SR^- = \min(Surprise, 0)$, each standardized to mean zero and unit variance within the corresponding surprise-sign subsample. Percent surprise is $(Actual - Consensus)/|Consensus|$ from IBES, winsorized at the 1% and 99% levels. All columns include the firm-level control block used in Table 11 with controls. *Round* $= 1$ when the median IBES consensus, rounded to two decimals, is a nonzero multiple of \$0.05. Sample construction matches that table. Standard errors are clustered by announcement date. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

	CAR$[-5,-1]$ (1)	CAR$[-10,-1]$ (2)
Panel A: Surprise > 0		
Round	-0.059*	-0.039
	(-1.88)	(-0.88)
SR^+	0.131***	0.203***
	(7.19)	(8.18)
Round $\times$ SR^+	0.044	0.078
	(1.01)	(1.32)
Observations	191,166	191,166
Adj. R^2	0.014	0.019
Panel B: Surprise ≤ 0		
Round	-0.083**	-0.219***
	(-2.17)	(-4.07)
SR^-	0.205***	0.379***
	(8.40)	(11.02)
Round $\times$ SR^-	0.018	0.129*
	(0.34)	(1.76)
Observations	151,829	151,829
Adj. R^2	0.005	0.008
Firm controls	Yes	Yes

t-statistics in parentheses; clustered by announcement date.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table IA.26: **Round Consensus Definitions**

This table reports OLS estimates of CAR$(0,+1)$ and CAR$(2,+75)$ on *Round*, signed standardized percent surprise, and *Round*$\times$ surprise. Panel A restricts to Surprise $>$ 0 and reports *Round* $\times$ SR^{+}. Panel B restricts to Surprise $\leq$ 0 and reports *Round* $\times$ SR^{-}. Specifications include firm-level controls, calendar indicators, and Fama–French 10 industry indicators. SR^{+} is standardized max$(Surprise, 0)$ and SR^{-} is standardized min$(Surprise, 0)$ within the corresponding surprise-sign sample. Round definitions: *Medest*, \$0.05 (\$0.10) equals one when the median consensus is a nonzero multiple of \$0.05 (\$0.10); *Medest*, *integer* equals one when the median consensus is a nonzero integer; *NearRound* equals one in the non-\$0.05 subsample with $|Medest| \geq$ \$0.10 and second decimal digit in $\{1,4,6,9\}$; *Placebo Round* is the mean *Round*$\times$ surprise coefficient from 100 random reassignments of the \$0.05 Round label within the surprise-sign sample that preserve the Round share, with t equal to that mean divided by the cross-reassignment standard deviation. Sample construction matches Table 11. Standard errors are clustered by announcement date, except for Placebo Round as noted above. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

	CAR$(0,+1)$		CAR$(2,+75)$	
Round definition	Coeff.	N	Coeff.	N
Panel A: Surprise $>$ 0; Round$\times SR^{+}$				
Medest, \$0.05	0.208***	191,166	−0.346**	191,166
	(3.61)		(−2.07)	
Medest, \$0.10	0.236***	191,166	−0.724***	191,166
	(2.82)		(−2.93)	
Medest, integer	0.479*	191,166	−2.156***	191,166
	(1.72)		(−3.16)	
NearRound	0.005	116,309	0.037	116,309
	(0.08)		(0.25)	
Placebo Round	0.008	191,166	0.019	191,166
	(0.16)		(0.16)	
Panel B: Surprise $\leq$ 0; Round$\times SR^{-}$				
Medest, \$0.05	0.050	151,829	0.099	151,829
	(0.78)		(0.48)	
Medest, \$0.10	0.008	151,829	0.481*	151,829
	(0.09)		(1.74)	
Medest, integer	0.221	151,829	1.730**	151,829
	(1.03)		(2.09)	
NearRound	−0.126*	84,592	−0.197	84,592
	(−1.74)		(−0.79)	
Placebo Round	0.004	151,829	0.011	151,829
	(0.07)		(0.05)	

t-statistics in parentheses; clustered by announcement date (Placebo Round: mean/SD across 100 reassignments). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table IA.27: **Cumulative Abnormal Returns by Surprise Sign: Alternative Windows and Adjustments**

This table reports OLS estimates of cumulative abnormal returns on *Round*, signed standardized percent surprise, and *Round*× surprise. Panel A restricts to Surprise > 0 and uses SR^+. Panel B restricts to Surprise ≤ 0 and uses SR^-. Columns (1)–(3) use market-adjusted CARs; columns (4)–(6) use Fama–French 10 industry-adjusted CARs. Dependent variables are CAR(0) in cols. (1) and (4), CAR$(-1,+1)$ in cols. (2) and (5), and CAR$(2,+90)$ in cols. (3) and (6), winsorized at the 1% and 99% levels and scaled to percent. Market-adjusted returns subtract the CRSP value-weighted market return. Industry-adjusted returns subtract the Kenneth French value-weighted FF10 industry portfolio return matched to Compustat SIC. All columns include size and book-to-market deciles; $\log(1+N_{ann,t})$; $\log(1+N_{est})$; reporting lag, its square, and its cube; institutional-ownership decile; earnings volatility and persistence; share turnover; and year, month, day-of-week, and Fama–French 10 industry indicators. *Round* $= 1$ when the median IBES consensus, rounded to two decimals, is a nonzero multiple of \$0.05. Sample construction matches Table 11. Standard errors are clustered by announcement date. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

	Market-adjusted			Industry-adjusted		
	CAR(0) (1)	CAR$(-1,+1)$ (2)	CAR$(2,+90)$ (3)	CAR(0) (4)	CAR$(-1,+1)$ (5)	CAR$(2,+90)$ (6)
Panel A: Surprise > 0						
Round	-0.0100	-0.1173***	-1.0856***	-0.0121	-0.1196***	-1.0799***
	(-0.40)	(-2.77)	(-8.20)	(-0.48)	(-2.83)	(-8.05)
SR^+	0.3101***	0.8683***	0.4127***	0.3093***	0.8659***	0.4388***
	(20.70)	(32.99)	(5.34)	(20.61)	(32.48)	(5.70)
Round $\times$ SR^+	0.1277***	0.2643***	-0.3964**	0.1206***	0.2565***	-0.3200*
	(3.50)	(4.29)	(-2.12)	(3.31)	(4.15)	(-1.71)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	191,166	191,166	191,166	191,166	191,166	191,166
Adj. R^2	0.014	0.026	0.031	0.013	0.026	0.029
Panel B: Surprise ≤ 0						
Round	-0.0382	-0.1760***	-1.3240***	-0.0409	-0.1886***	-1.3296***
	(-1.31)	(-3.61)	(-7.85)	(-1.40)	(-3.85)	(-7.94)
SR^-	0.2293***	0.6563***	0.1850*	0.2287***	0.6556***	0.1383
	(11.90)	(20.55)	(1.73)	(11.88)	(20.51)	(1.31)
Round $\times$ SR^-	0.0615	0.0153	0.3744*	0.0589	0.0289	0.4213*
	(1.47)	(0.22)	(1.66)	(1.41)	(0.42)	(1.86)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	151,829	151,829	151,829	151,829	151,829	151,829
Adj. R^2	0.011	0.025	0.035	0.011	0.025	0.033

Table IA.28: **Round Consensus after Matching and Entropy Balancing**

This table reports estimates of cumulative abnormal returns on *Round*, standardized positive percent surprise SR^+, and *Round* $\times$ SR^+ in the Surprise > 0 sample. Panel A uses CAR$(0, +1)$; Panel B uses CAR$(2, +75)$. Column (1) uses the unmatched sample. Column (2) uses a one-to-one exact-matched sample that pairs each *Round* $= 1$ observation with one *Round* $= 0$ observation within cells of Fama–French 10 industry, size decile, analyst-coverage decile, and institutional-ownership decile. Column (3) uses one-to-one nearest-neighbor propensity-score matching: the propensity score is the fitted probability from a logit of *Round* on size decile, $\log(1 + N_{est})$, institutional-ownership decile, and book-to-market decile; each *Round* $= 1$ observation is paired to its nearest *Round* $= 0$ neighbor on the propensity score within Fama–French 10 industry by calendar-year cells, without replacement, subject to a caliper of 0.2 times the sample standard deviation of the propensity score. Column (4) uses entropy-balancing ATT weights that reweight *Round* $= 0$ observations so that the weighted means of size decile, $\log(1 + N_{est})$, institutional-ownership decile, and Fama–French 10 industry indicators match the *Round* $= 1$ means. All columns include the firm-level control block used in Table 11 with controls. *Round* $= 1$ when the median IBES consensus, rounded to two decimals, is a nonzero multiple of \$0.05. Sample construction matches that table. Standard errors are clustered by announcement date. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

	Unmatched (1)	Exact match (2)	PS match (3)	Entropy balance (4)
Panel A: CAR$(0, +1)$, *Surprise* > 0				
Round	-0.073*	-0.074	-0.071	-0.069*
	(-1.82)	(-1.49)	(-1.47)	(-1.73)
SR^+	0.838***	0.883***	0.871***	0.841***
	(33.36)	(20.43)	(20.09)	(33.10)
Round $\times$ SR^+	0.208***	0.192***	0.204***	0.223***
	(3.61)	(2.84)	(3.04)	(3.86)
Observations	191,166	90,476	91,638	191,166
Adj. R^2	0.025	0.026	0.025	0.025
Panel B: CAR$(2, +75)$, *Surprise* > 0				
Round	-0.947***	-0.910***	-0.942***	-0.954***
	(-7.71)	(-6.06)	(-6.31)	(-7.75)
SR^+	0.430***	0.473***	0.563***	0.468***
	(6.11)	(3.95)	(4.55)	(6.35)
Round $\times$ SR^+	-0.346**	-0.359*	-0.412**	-0.338**
	(-2.07)	(-1.82)	(-2.14)	(-2.02)
Observations	191,166	90,476	91,638	191,166
Adj. R^2	0.027	0.028	0.027	0.027
Firm controls	Yes	Yes	Yes	Yes

t-statistics in parentheses; clustered by announcement date.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

V.2 Guidance, Information Environment, and Miss-Sample Heterogeneity

Table IA.29: **Round Consensus, Management Guidance, and Just-Beat**

This table reports OLS estimates of cumulative abnormal returns on *Round*, standardized positive percent surprise SR^+, and *Round* $\times$ SR^+ in the Surprise $>$ 0 sample. Panel A uses CAR$(0,+1)$; Panel B uses CAR$(2,+75)$. Column (1) is the baseline specification with firm-level controls. Column (2) adds *HasGuidance*. Column (3) retains firm-quarters with *HasGuidance* $=$ 0. Column (4) adds *JustBeat*. Column (5) adds *JustBeat* and *Round* $\times$ SR^+ $\times$ *JustBeat*. *HasGuidance* $=$ 1 when the firm has at least one IBES quarterly EPS management guidance announcement for the same fiscal period end (year and month of *fpedats*) with guidance announcement date strictly before the earnings announcement date. *JustBeat* $=$ 1 when *Actual* $-$ *Consensus* lies in $(0, 0.01]$ dollars. *Round* $=$ 1 when the median IBES consensus, rounded to two decimals, is a nonzero multiple of \$0.05. Sample construction and the firm-level control block match Table 11 with controls. Standard errors are clustered by announcement date. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

	Baseline (1)	+Guidance (2)	No guidance (3)	+JustBeat (4)	×JustBeat (5)
Panel A: CAR$(0,+1)$, Surprise $>$ 0					
Round	-0.073*	-0.071*	-0.084**	-0.155***	-0.202***
	(-1.82)	(-1.76)	(-1.96)	(-3.86)	(-4.55)
SR^+	0.838***	0.832***	0.821***	0.732***	0.727***
	(33.36)	(33.12)	(30.90)	(29.03)	(28.70)
Round $\times$ SR^+	0.208***	0.207***	0.201***	0.194***	0.221***
	(3.61)	(3.60)	(3.31)	(3.39)	(3.75)
HasGuidance	—	-0.517***	—	—	—
	(—)	(-9.54)	(—)	(—)	(—)
JustBeat	—	—	—	-1.418***	-1.473***
	(—)	(—)	(—)	(-33.58)	(-31.12)
Round $\times$ SR^+ $\times$ *JustBeat*	—	—	—	—	-0.564***
	(—)	(—)	(—)	(—)	(-2.92)
Observations	191,166	191,166	157,033	191,166	191,166
Adj. R^2	0.025	0.026	0.029	0.031	0.031
Panel B: CAR$(2,+75)$, Surprise $>$ 0					
Round	-0.947***	-0.948***	-1.024***	-0.917***	-0.980***
	(-7.71)	(-7.72)	(-7.47)	(-7.46)	(-7.19)
SR^+	0.430***	0.433***	0.449***	0.469***	0.463***
	(6.11)	(6.14)	(5.91)	(6.63)	(6.52)
Round $\times$ SR^+	-0.346**	-0.345**	-0.327*	-0.341**	-0.305*
	(-2.07)	(-2.07)	(-1.84)	(-2.04)	(-1.78)
HasGuidance	—	0.215	—	—	—
	(—)	(1.55)	(—)	(—)	(—)
JustBeat	—	—	—	0.527***	0.454***
	(—)	(—)	(—)	(4.26)	(3.34)
Round $\times$ SR^+ $\times$ *JustBeat*	—	—	—	—	-0.754
	(—)	(—)	(—)	(—)	(-1.32)
Observations	191,166	191,166	157,033	191,166	191,166
Adj. R^2	0.027	0.027	0.027	0.027	0.027
Firm controls	Yes	Yes	Yes	Yes	Yes

t-statistics in parentheses; clustered by announcement date.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table IA.30: **Miss-Sample Round Interaction with Short-Sale and Attention Moderators**

This table reports OLS estimates for earnings announcements with Surprise ≤ 0. The dependent variables are CAR$(0,+1)$ and CAR$(2,+75)$, market-adjusted, winsorized at the 1% and 99% levels, and scaled to percent. Each specification includes *Round*, standardized SR^-, *Round* $\times$ SR^-, a moderator M, *Round* $\times$ M, $SR^- \times M$, and *Round* $\times$ $SR^- \times M$. SR^- is min$(Surprise, 0)$ standardized within the miss sample. The table reports *Round* $\times$ SR^-, the triple interaction *Round* $\times$ $SR^- \times M$, and the net coefficient when $M = 1$ (the sum of *Round* $\times$ SR^- and the triple). Moderators equal one when the underlying measure is above (or, for short interest, below) its within-calendar-quarter median: institutional-ownership decile; log market capitalization; lagged share turnover; residual institutional-ownership decile from a within-quarter regression on size and book-to-market deciles; Compustat short interest scaled by market capitalization (as-of before the announcement); $\log(1+N_{est})$; and $\log(1+N_{ann,t})$. The baseline row omits M. All columns include size and book-to-market deciles; $\log(1+N_{ann,t})$; $\log(1+N_{est})$; reporting lag, its square, and its cube; institutional-ownership decile; earnings volatility and persistence; share turnover; and year, month, day-of-week, and Fama–French 10 industry indicators. *Round* $= 1$ when the median IBES consensus, rounded to two decimals, is a nonzero multiple of \$0.05. Sample construction matches Table 11 Panel B. Standard errors are clustered by announcement date. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

	CAR$(0,+1)$			CAR$(2,+75)$			
	Round $\times$ SR^-	$\times M$	Net$(M=1)$	*Round* $\times$ SR^-	$\times M$	Net$(M=1)$	N
Baseline	0.0498	—	0.0498	0.0995	—	0.0995	151,829
	(0.78)		(0.78)	(0.48)		(0.48)	
High IO	0.0352	0.0361	0.0712	-0.0780	0.3502	0.2722	151,829
	(0.46)	(0.27)		(-0.31)	(0.78)		
Large size	0.0446	0.0232	0.0679	-0.1046	0.3409	0.2364	151,829
	(0.58)	(0.17)		(-0.42)	(0.76)		
High turnover	0.1243	-0.1446	-0.0202	0.6403**	-1.0005**	-0.3602	151,829
	(1.46)	(-1.17)		(2.25)	(-2.37)		
High residual IO	0.0081	0.1085	0.1167	-0.0142	0.1836	0.1694	151,829
	(0.10)	(0.84)		(-0.05)	(0.41)		
Low short interest	0.0013	0.1110	0.1123	0.1390	-0.4628	-0.3237	75,141
	(0.01)	(0.62)		(0.36)	(-0.83)		
High analyst coverage	0.0220	0.1100	0.1320	0.0476	-0.0883	-0.0407	151,829
	(0.30)	(0.73)		(0.20)	(-0.19)		
Busy announcement day	0.0061	0.1008	0.1069	-0.0022	0.2534	0.2512	151,829
	(0.07)	(0.77)		(-0.01)	(0.62)		
Controls				Yes			

Table IA.31: **Round Consensus, Analyst Coverage, Forecast Dispersion, and Announcement Outcomes**

This table reports OLS estimates of announcement returns and announcement-window trading intensity on *Round*, surprise, *Round*×surprise, within-FE_q standardized analyst coverage and forecast dispersion, and *Round* interactions with those standardized measures. Columns (1)–(2) restrict to Surprise > 0. The dependent variables are CAR$(0,+1)$ and CAR$(2,+75)$, market-adjusted, winsorized at the 1% and 99% levels, and scaled to percent. Surprise enters as standardized SR^{+}. Columns (3)–(4) use retail trade flow and log(Turnover) averaged over trading days $[0,+1]$. Surprise enters as standardized absolute percent surprise SR^{abs}. FE_q is the within-calendar-quarter decile of percent earnings surprise (1 = lowest, 10 = highest). $Numest^{z}$ is the IBES number of estimates, standardized to mean zero and unit variance within FE_q. $Disp^{z}$ is IBES forecast dispersion (*stdev*), with missing dispersion set to zero when the number of estimates is at most one, standardized to mean zero and unit variance within FE_q. $Round = 1$ when the median IBES consensus, rounded to two decimals, is a nonzero multiple of \$0.05. Columns (1)–(2) include size and book-to-market deciles; $\log(1+N_{ann,t})$; reporting lag, its square, and its cube; institutional-ownership decile; earnings volatility and persistence; share turnover; and year, month, day-of-week, and Fama-French 10 industry indicators. Columns (3)–(4) include the spillover control block used in Table 10 columns (2) and (4). Standard errors are clustered by announcement date. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

	Surprise > 0		Announcement trading	
	CAR$(0,+1)$	CAR$(2,+75)$	Retail flow	log(Turnover)
	(1)	(2)	(3)	(4)
Round	-0.0654	-0.8853***	0.0596***	0.0206***
	(-1.62)	(-7.24)	(7.08)	(6.03)
SR^{+} / SR^{abs}	0.7919***	0.3648***	0.0330***	0.0216***
	(31.45)	(5.11)	(7.14)	(11.37)
Round × SR^{+} / *Round* × SR^{abs}	0.2023***	-0.3468**	0.0290**	0.0373***
	(3.52)	(-2.08)	(2.28)	(6.95)
$Numest^{z}$	0.3152***	0.3860***	0.2475***	0.1284***
	(11.41)	(4.73)	(52.65)	(56.79)
$Disp^{z}$	0.0020	-0.3914**	0.0089	0.0094**
	(0.05)	(-2.05)	(0.78)	(2.53)
Round × $Numest^{z}$	0.0633	0.2291**	-0.0210***	0.0018
	(1.58)	(2.01)	(-3.03)	(0.57)
Round × $Disp^{z}$	-0.0821	-0.0567	0.0025	-0.0084**
	(-1.60)	(-0.25)	(0.20)	(-1.97)
Constant	-3.5222	-13.2366	8.9893***	-0.6773***
	(-0.44)	(-0.79)	(56.58)	(-3.26)
Controls	Yes	Yes	Yes	Yes
Observations	191,166	191,166	88,033	346,313
Adj. R^2	0.026	0.028	0.514	0.589

VI Attention Crowding

The main text reports the trading and return evidence for round-target firms. This section tests whether round-target announcers also affect same-day non-round peers. Hirshleifer et al. (2009) show that extraneous earnings announcements on the same day distract investors and lead to underreaction to other firms' news. If attention is a limited resource and many firms with round forecast targets announce on the same day within a given surprise group, attention may tilt toward them and away from peers with non-round targets. Those non-round firms may then under-react at announcement and catch up later as drift. To test this, I keep only non-round firms ($Round = 0$) and estimate

$$\begin{aligned} CAR_{i,t} = \alpha + \beta_1 FE_{q,i} + \beta_2 FracRank_t \\ + \beta_3 FE_{q,i} \times FracRank_t + X\gamma + \varepsilon_{i,t}, \end{aligned} \tag{12}$$

where FE_q is the within-calendar-quarter decile of percent surprise (1 = lowest, 10 = highest), and $FracRank_t$ is the within-quarter decile of the daily fraction of $FE_q = k$ announcers with $Round = 1$, matched to firm i's own surprise decile and announcement date. Controls parallel the fully specified trading-intensity and return specifications and include control$\times FE_q$ interactions. In columns that focus on extreme good news, I replace FE_q with an indicator for $FE_q = 10$ and interact that indicator with $FracRank_t$. If round-target crowding diverts attention from non-round peers, the sensitivity of returns to a firm's own surprise rank should fall at the announcement and rise subsequently. The spillover hypothesis therefore predicts $\beta_3 < 0$ for CAR$(0, +1)$ and $\beta_3 > 0$ for CAR$(2, +75)$.

Table IA.32 lines up with that prediction. Across all FE_q deciles with controls, $FE_q \times FracRank_t$ equals -0.018 for CAR$(0, +1)$ and $+0.022$ for CAR$(2, +75)$. Because both FE_q and $FracRank_t$ are decile ranks, a one-decile increase in round-target crowding flattens the announcement-window surprise slope by 0.018 percentage

points and steepens the post-announcement slope by 0.022 percentage points. Moving from the least to the most crowded day ($FracRank_t$ from 1 to 10) therefore reduces the return associated with a one-decile higher surprise by about 0.16 percentage points at announcement and raises it by about 0.20 percentage points over CAR$(2, +75)$. When more round-target peers announce in the same surprise decile on the same day, non-round firms' announcement returns respond less to their own surprise rank, and the missing piece shows up later.[29] Looking only at extreme good news ($FE_q = 10$), $FE_q 10 \times FracRank_t$ equals -0.167 at announcement and $+0.279$ over CAR$(2, +75)$: a one-decile increase in crowding lowers announcement CAR by 0.17 percentage points for top-decile surprises and raises subsequent CAR by 0.28 percentage points.

Taken together, these estimates are consistent with round-target announcers drawing attention away from same-day non-round peers, in the same spirit as the same-day earnings distraction documented by Hirshleifer et al. (2009), although they do not isolate attention from other same-day influences.

[29] Table IA.33 replaces the day-level $FracRank_t$ decile with a continuous leave-one-out round-announcer share $Frac_t^{-i}$. The leave-one-out construction ensures that firm i does not enter its own crowding measure, and the continuous share uses the intensive margin of same-day round peers rather than a coarse rank. $FE_q \times Frac_t^{-i}$ remains negative for CAR$(0, +1)$ and positive for CAR$(2, +75)$. A within-quarter, within-FE_q permutation of day-level fractions breaks the link between firm i and its actual same-day peers and yields null interactions, so the spillover is tied to realized round-target crowding rather than to the unconditional distribution of round shares.

Table IA.32: **Cumulative Abnormal Returns, Earnings Surprise, and Round-Announcer Fraction**

The table reports OLS estimates for non-round firms ($Round = 0$) among roundness-eligible earnings announcements with valid CAR and return data over trading days $[0, +1]$ relative to the quarterly earnings announcement date. The dependent variable is $\text{CAR}(0, +1)$ in cols. (1)–(2) and (5), and $\text{CAR}(2, +75)$ in cols. (3)–(4) and (6); $\text{CAR}(0, +1)$ is the market-adjusted cumulative abnormal return over days 0 and $+1$, winsorized at the 1% and 99% levels and scaled to percent. FE_q is the within-calendar-quarter decile of percent earnings surprise based on median IBES consensus EPS (1 = lowest, 10 = highest). $Round = 1$ when the median IBES consensus EPS forecast, rounded to two decimal places, is a nonzero multiple of \$0.05. $FracRank_t$ is the within-calendar-quarter decile of the daily fraction of $FE_q = k$ announcers with $Round = 1$, matched to each firm's FE_q and announcement date t. Cols. (1) and (3) include FE_q, $FracRank_t$, and $FE_q \times FracRank_t$. Cols. (2) and (4) add size and book-to-market deciles (within calendar quarter, from Compustat); $\log(1 + N_{FEk,t})$; $\log(1 + N_{est})$; reporting lag, its square, and its cube; institutional-ownership decile (within calendar quarter, from Thomson Reuters 13F holdings scaled by CRSP shares outstanding, lagged one quarter); earnings volatility and persistence (rolling eight-quarter Compustat EPS); share turnover (average daily volume divided by shares outstanding, lagged one quarter); indicator variables for year, month, day of week, and Fama–French 10 industry; and each control interacted with FE_q. Cols. (5)–(6) restrict to $FE_q \in \{1, 10\}$; FE_q10 equals one when $FE_q = 10$; controls, calendar and industry indicators, and control$\times FE_q10$ interactions as in cols. (2) and (4). $N_{FEk,t}$ is the number of $FE_q = k$ earnings announcements on date t; N_{est} is the number of IBES EPS forecasts. t-statistics in parentheses; standard errors clustered by announcement date. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	All FE_q deciles				$FE_q = 10$	
	$\text{CAR}(0, +1)$		$\text{CAR}(2, +75)$		$\text{CAR}(0, +1)$	$\text{CAR}(2, +75)$
	(1)	(2)	(3)	(4)	(5)	(6)
FE_q	0.8302***	0.1887**	0.1548***	0.8708***	—	—
	(46.64)	(2.16)	(2.68)	(3.48)		
$FracRank_t$	0.0409**	0.1053***	-0.1564**	-0.1173*	0.0818**	-0.2289**
	(2.19)	(5.23)	(-2.44)	(-1.76)	(2.43)	(-2.12)
$FE_q \times FracRank_t$	-0.0102***	-0.0183***	0.0294***	0.0221**	—	—
	(-3.37)	(-5.58)	(2.96)	(2.10)		
FE_q10	—	—	—	—	3.1989***	6.7920*
					(2.62)	(1.83)
$FE_q10 \times FracRank_t$	—	—	—	—	-0.1667***	0.2792**
					(-3.61)	(1.97)
Constant	-4.4166***	-2.6218	0.8268**	-1.6573	-6.4889	0.0899
	(-40.53)	(-0.64)	(2.21)	(-0.27)	(-0.50)	(0.02)
Controls	No	Yes	No	Yes	Yes	Yes
Observations	267,767	258,234	267,767	258,234	51,125	51,125
Adj. R^2	0.082	0.090	0.002	0.028	0.144	0.041

Table IA.33: **Earnings Surprise, Leave-One-Out Round-Announcer Fraction, and Cumulative Abnormal Returns**

This table reports OLS estimates of CAR$(0,+1)$ and CAR$(2,+75)$ on FE_q, a leave-one-out continuous round-announcer fraction, and their interaction, for non-round firms ($Round = 0$). FE_q is the within-calendar-quarter decile of percent earnings surprise based on median IBES consensus EPS (1 = lowest, 10 = highest). $Frac_t^{-i}$ is the share of other same-day announcers in the same FE_q cell with $Round = 1$. $Frac_t^{\text{plac}}$ is a within-calendar-quarter, within-FE_q permutation of day-level leave-one-out fractions across announcement dates. $Round = 1$ when the median IBES consensus, rounded to two decimals, is a nonzero multiple of \$0.05. Columns (1)–(2) use $Frac_t^{-i}$. Columns (3)–(4) use $Frac_t^{\text{plac}}$. All columns include $\log(1 + N_{FEk,t})$; $\log(1 + N_{ann,t})$; the leave-one-out same-day Fama–French 10 industry round share; size and book-to-market deciles; $\log(1 + N_{est})$; reporting lag, its square, and its cube; institutional-ownership decile; earnings volatility and persistence; share turnover; year, month, day-of-week, and Fama–French 10 industry indicators; and each control interacted with FE_q. CAR$(0,+1)$ and CAR$(2,+75)$ are market-adjusted, winsorized at the 1% and 99% levels, and scaled to percent. $N_{FEk,t}$ is the number of $FE_q = k$ announcements on date t; $N_{ann,t}$ is the number of earnings announcements on date t. Standard errors are clustered by announcement date. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

	Non-round		Non-round, placebo	
	CAR$(0,+1)$	CAR$(2,+75)$	CAR$(0,+1)$	CAR$(2,+75)$
	(1)	(2)	(3)	(4)
FE_q	0.1633*	1.0070***	0.1358	1.0718***
	(1.69)	(3.61)	(1.40)	(3.85)
$Frac_t^{-i}$	0.6978***	-1.2289*	—	—
	(3.30)	(-1.81)		
$Frac_t^{\text{plac}}$	—	—	0.0493	0.0581
			(0.26)	(0.09)
$FE_q \times Frac_t^{-i}$	-0.1255***	0.2127**	—	—
	(-3.67)	(1.96)		
$FE_q \times Frac_t^{\text{plac}}$	—	—	-0.0120	-0.0434
			(-0.39)	(-0.44)
Constant	1.3735	-1.6957	1.5640	-1.9897
	(0.52)	(-0.25)	(0.59)	(-0.29)
Controls	Yes	Yes	Yes	Yes
Observations	238,883	238,883	238,883	238,883
Adj. R^2	0.089	0.028	0.089	0.028